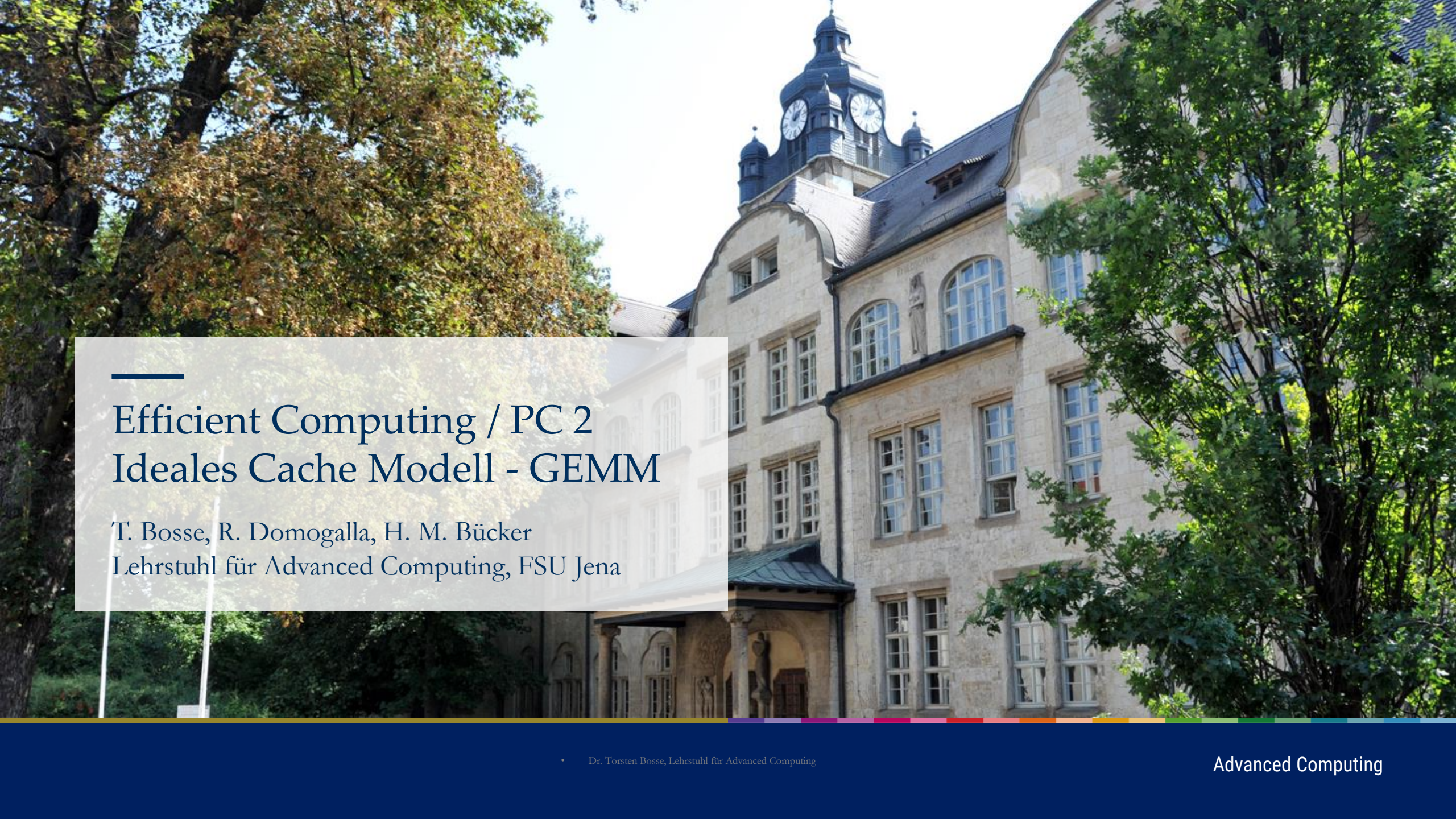


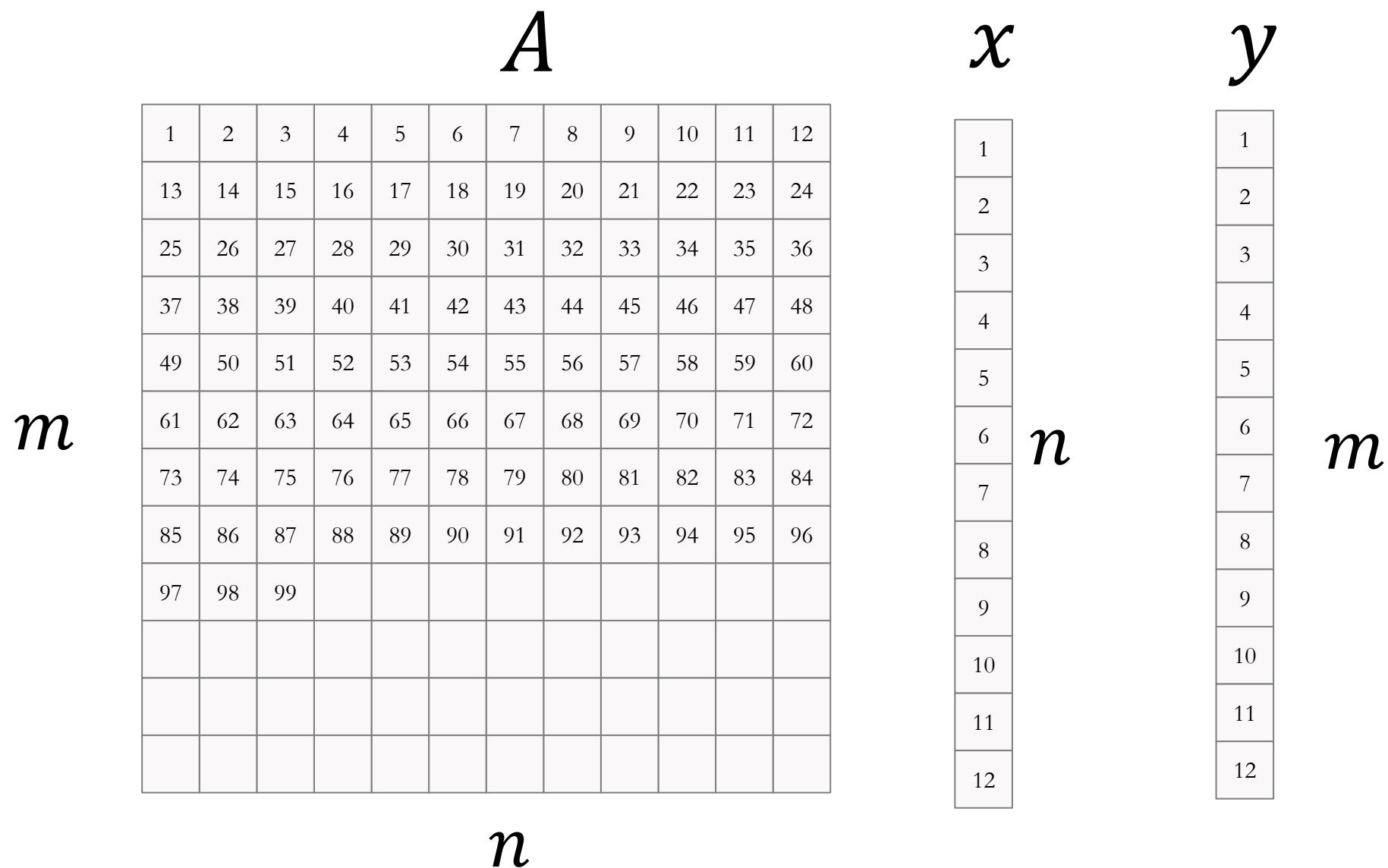


Efficient Computing / PC 2

Ideales Cache Modell - GEMM

T. Bosse, R. Domogalla, H. M. Bucker
Lehrstuhl für Advanced Computing, FSU Jena

Memory Layout

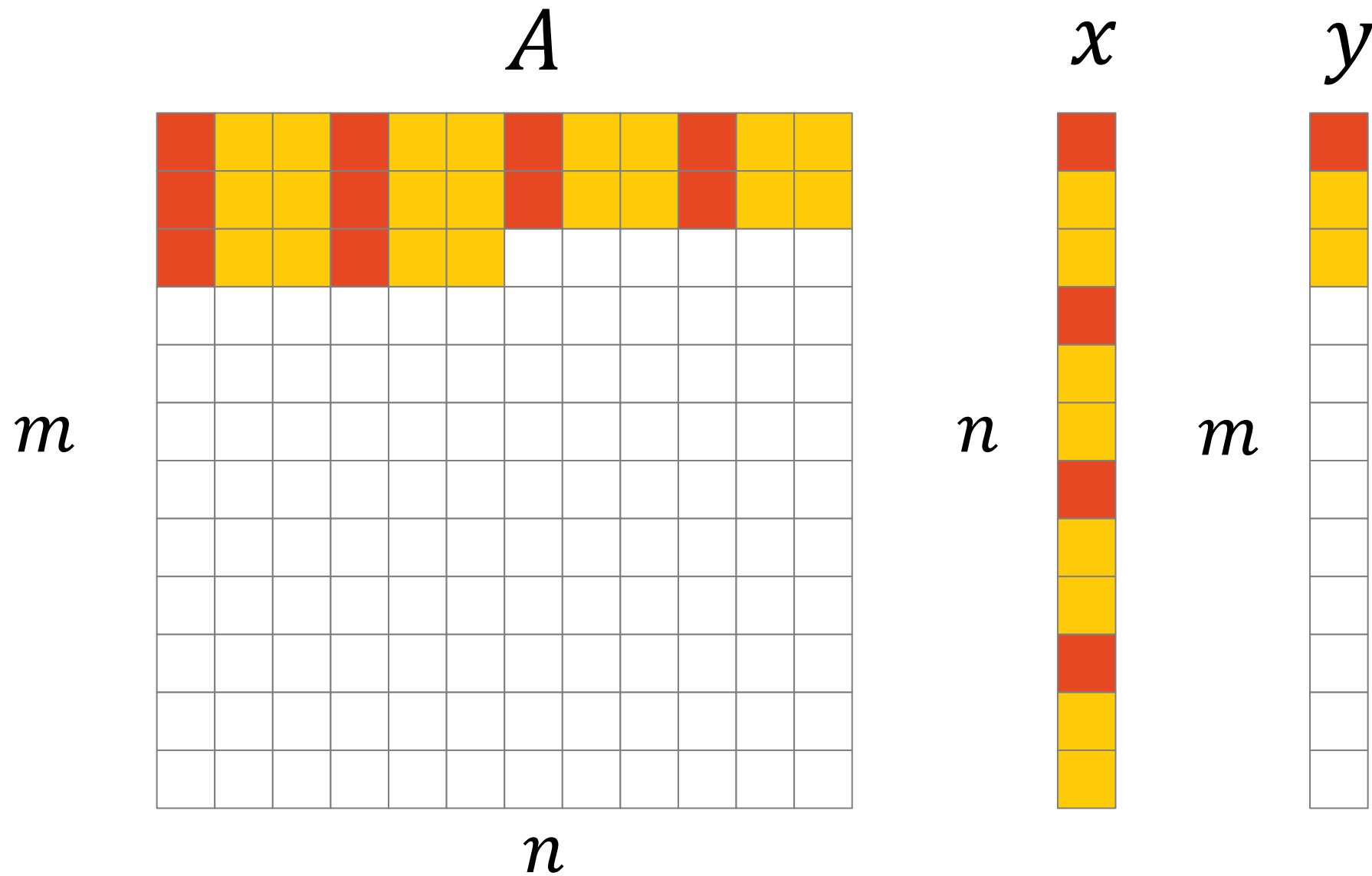


Matrix-Vector Product

```
for(int i=0; i<m; i++){  
    y[i] = 0;  
    for(int j=0; j<n; j++){  
        y[i] = y[i] + A[i][j]*x[j];  
    }  
}
```

Matrix Vector Product

$$mn + m + n \ll M$$



Transposed Matrix Vector Product

```
for(int i=0; i<n; i++){  
    y[i] = 0;  
    for(int j=0; j<m; j++){  
        y[i] = y[i] + A[j][i]*x[j];  
    }  
}
```

Transposed Matrix Vector Product

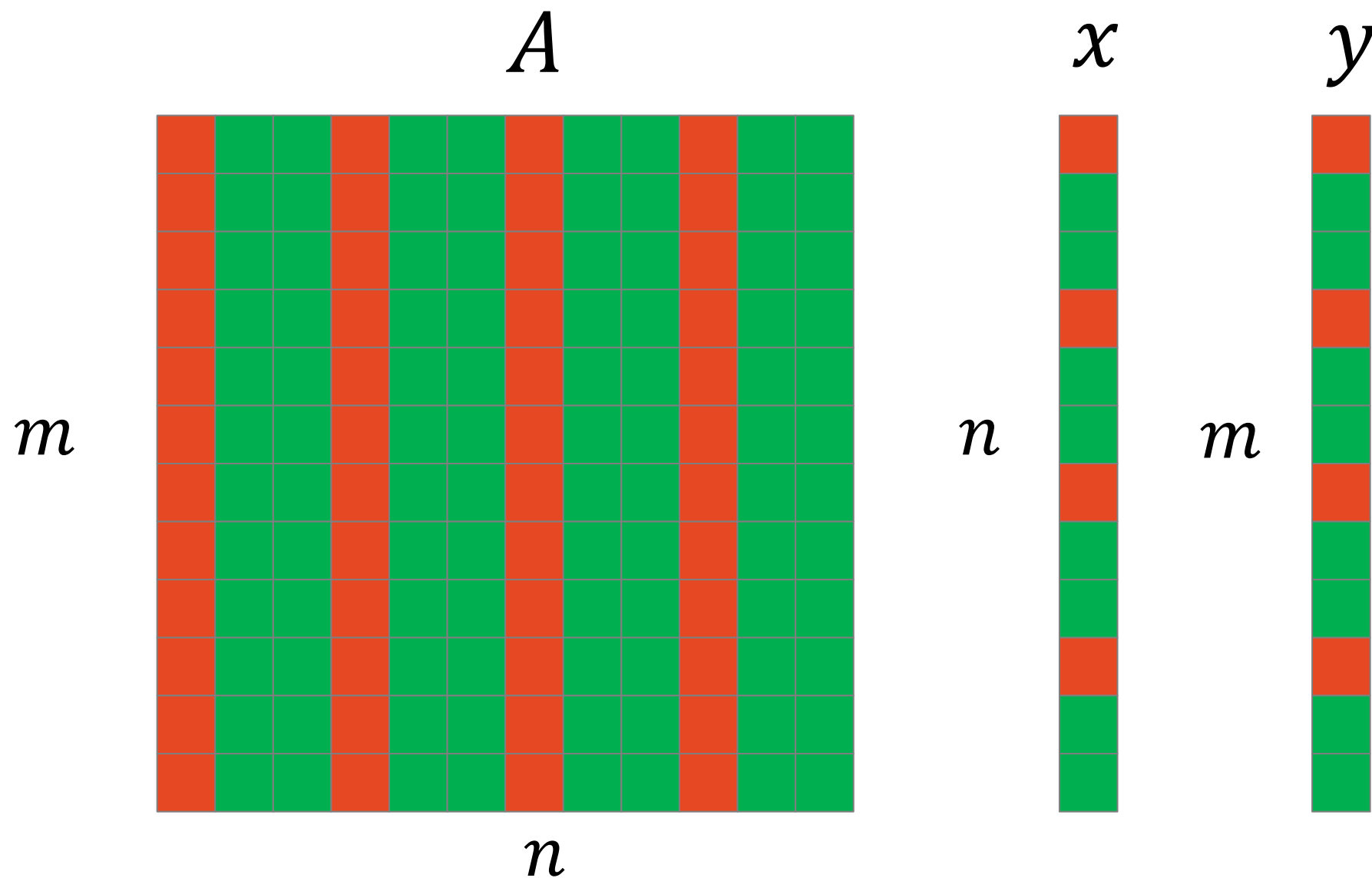
$$mn + m + n \ll M$$

A

 x y
$$m$$
$$m$$
$$n$$
$$n$$

Hits and Misses for (transposed) MVP

$$mn + m + n \ll M$$



Analysis

$$mn + m + n \ll M$$

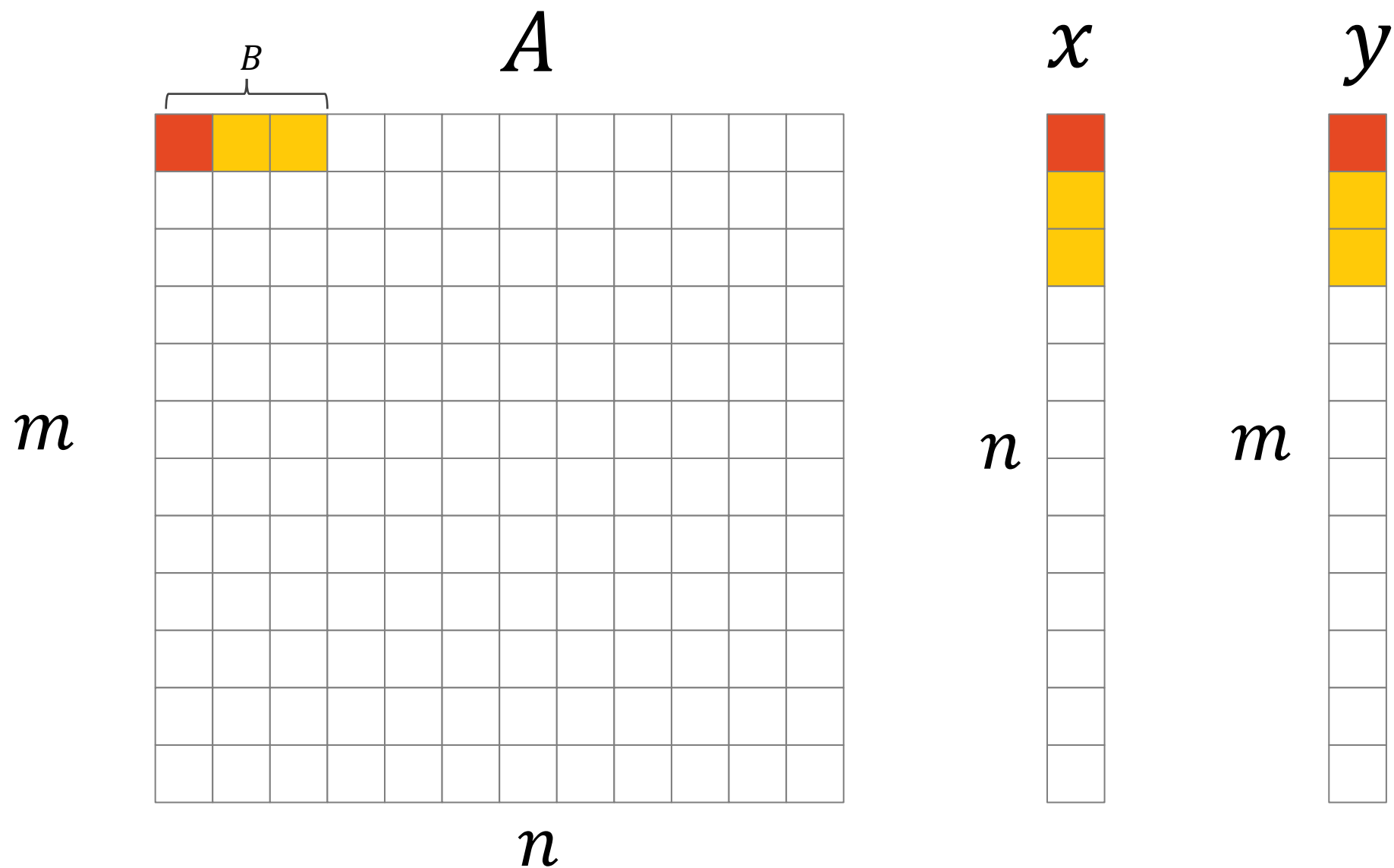
```
for(int i=0; i<m; i++){  
    y[i] = 0;  
    for(int j=0; j<n; j++){  
        y[i] = y[i] + A[i][j]*x[j];  
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```
for(int i=0; i<n; i++){  
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    }  
}
```

$$Q(n, m; B, M) \approx m \lceil n/B \rceil + \lceil n/B \rceil + \lceil m/B \rceil \approx \frac{mn}{B}$$

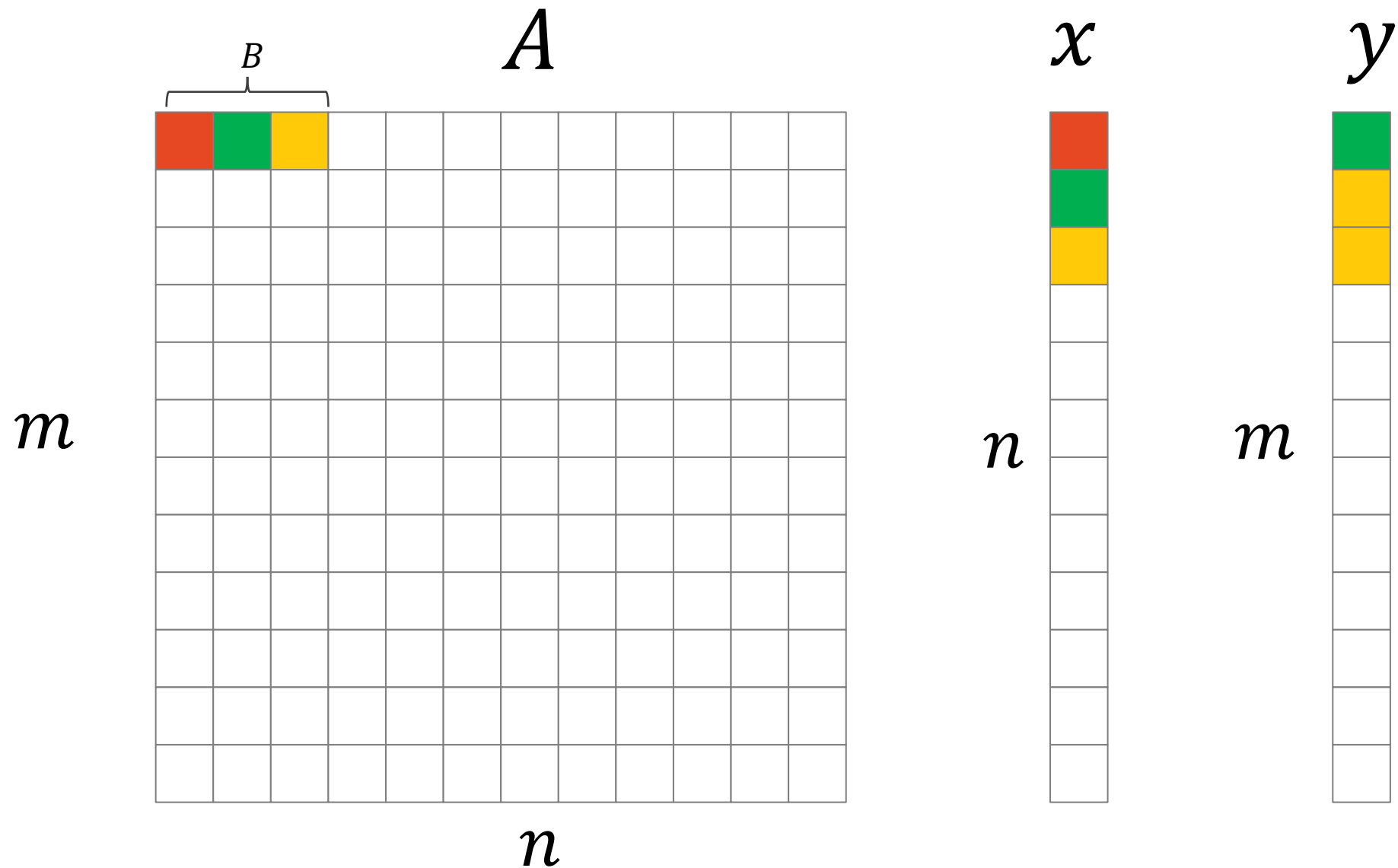
Matrix Vector Product

$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$



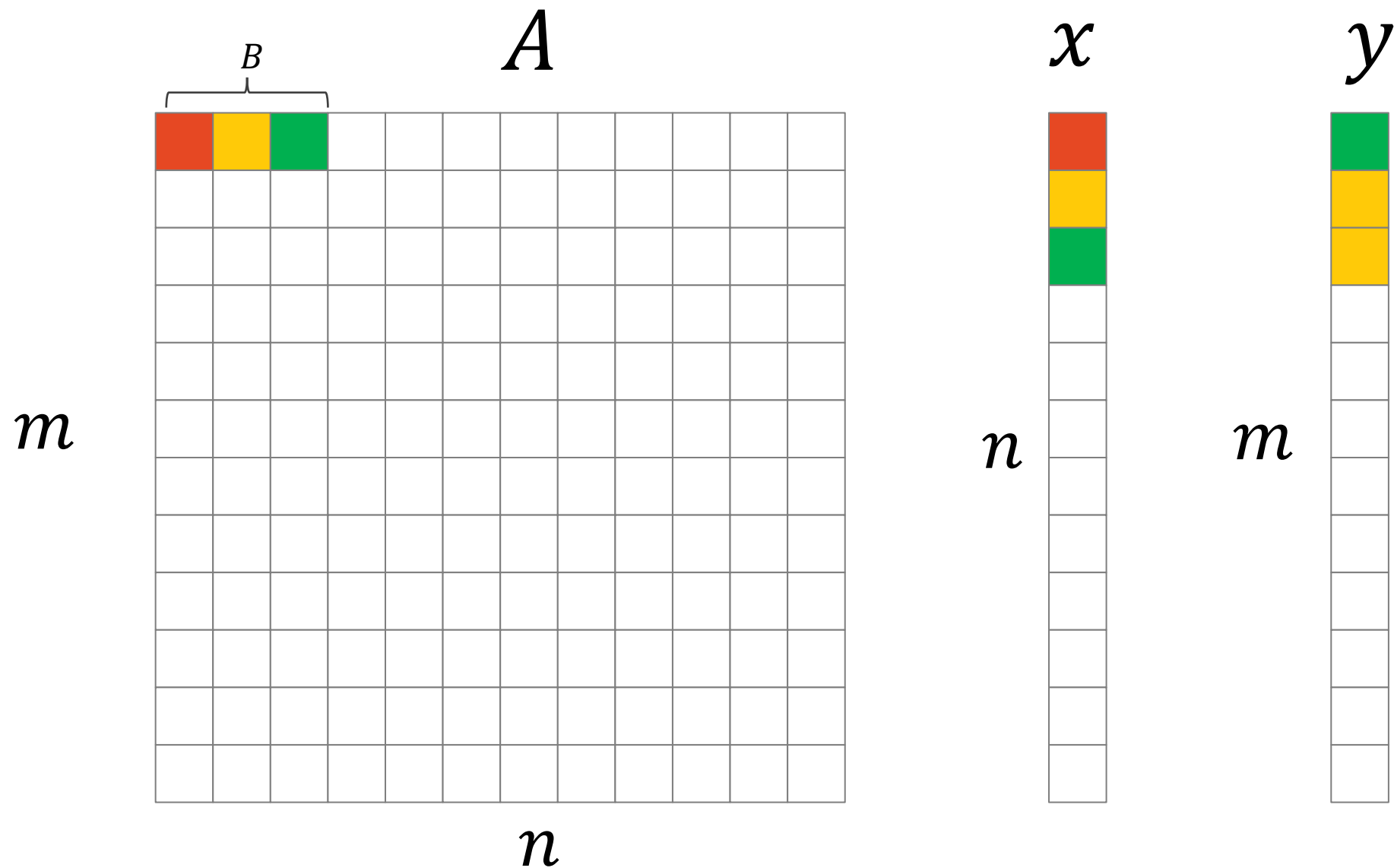
Matrix Vector Product

$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$



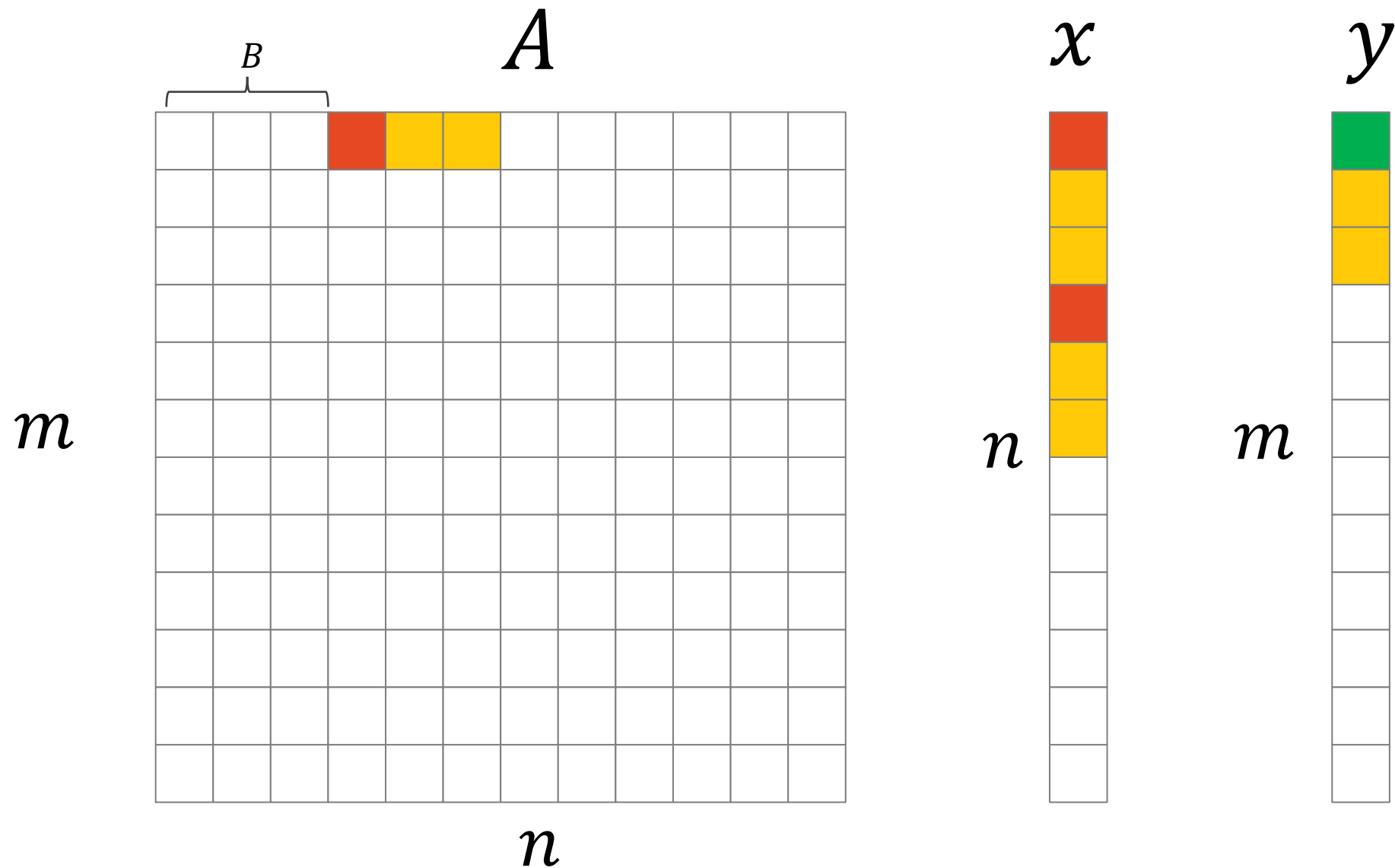
Matrix Vector Product

$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$



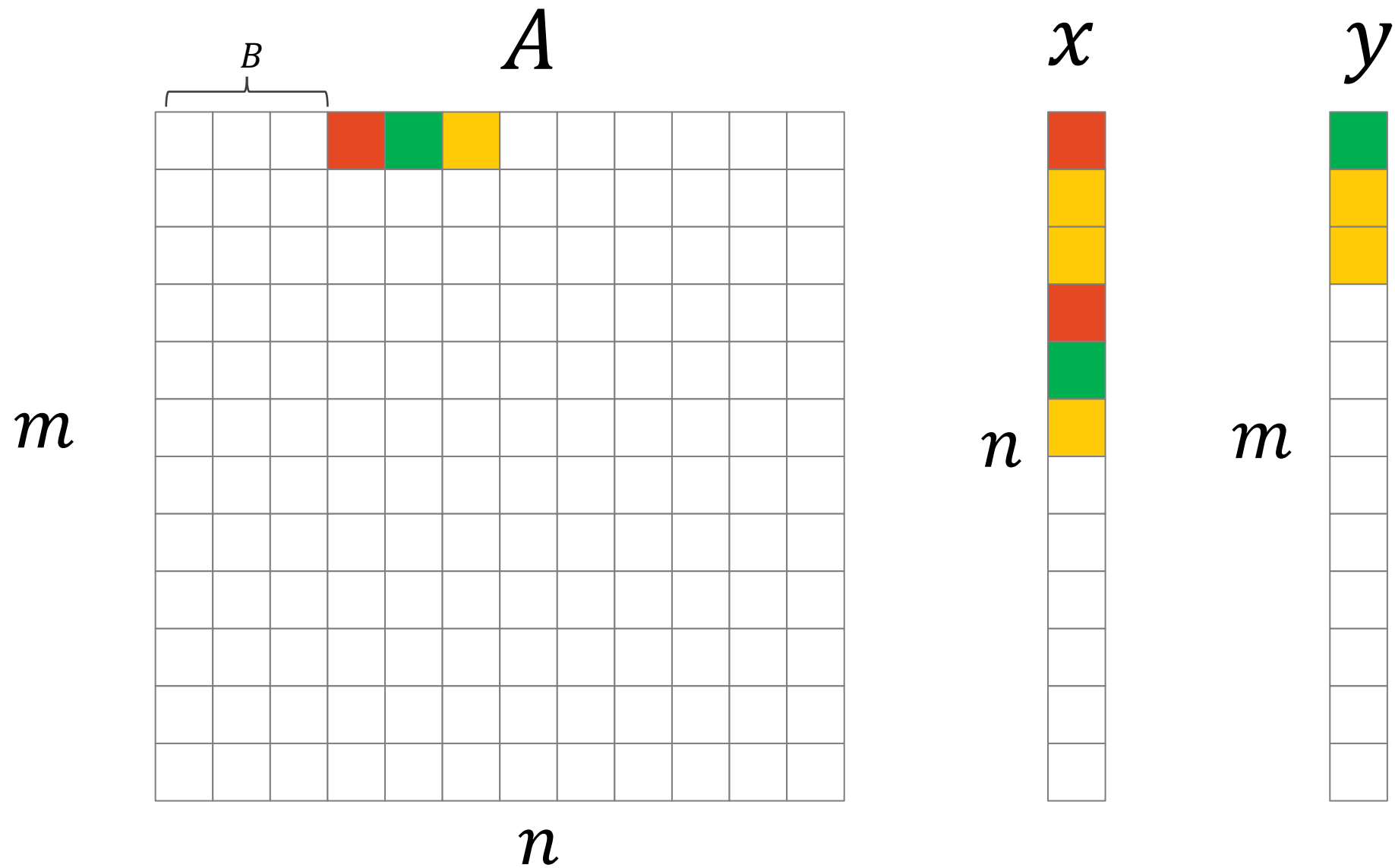
Matrix Vector Product

$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$



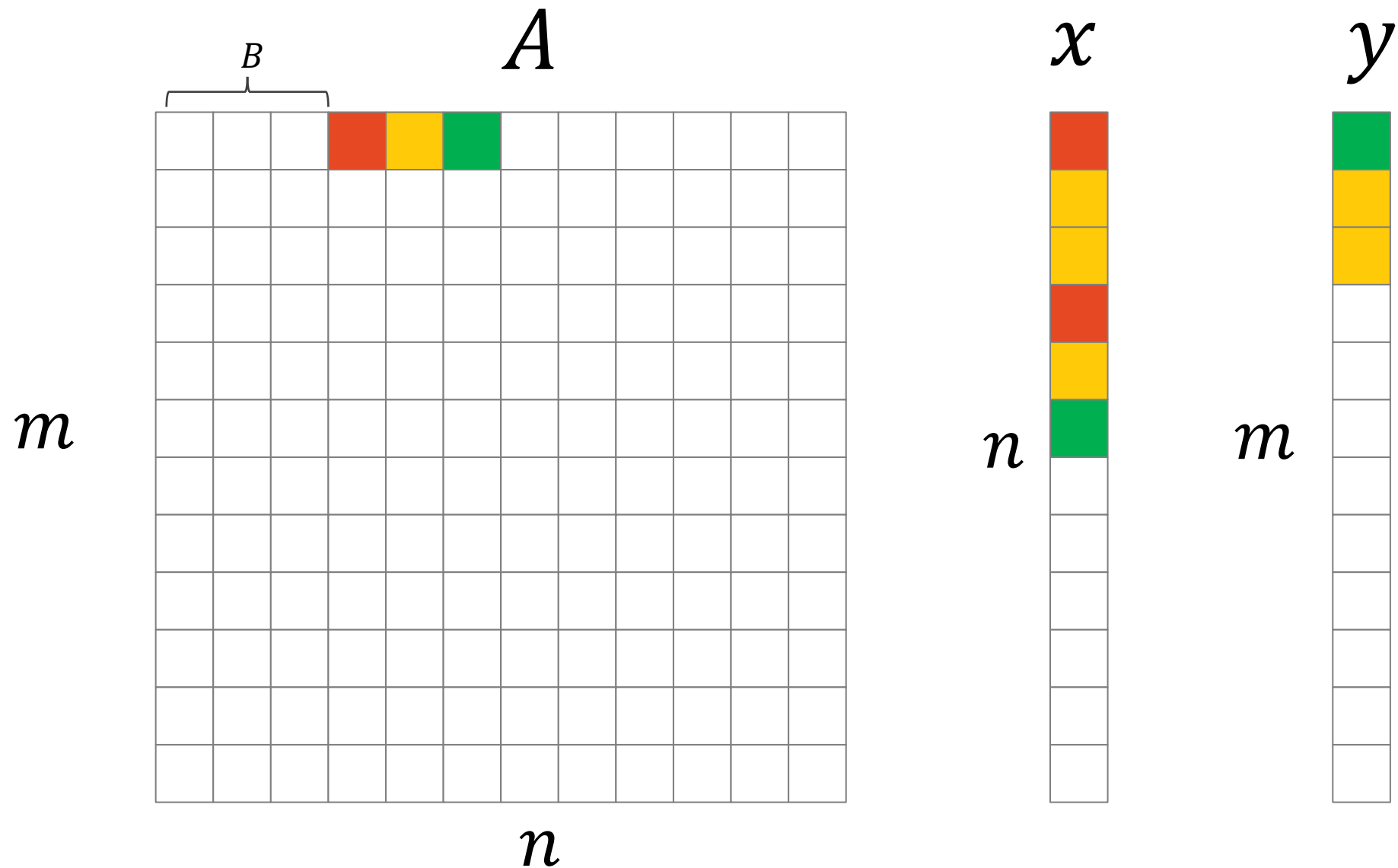
Matrix Vector Product

$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$



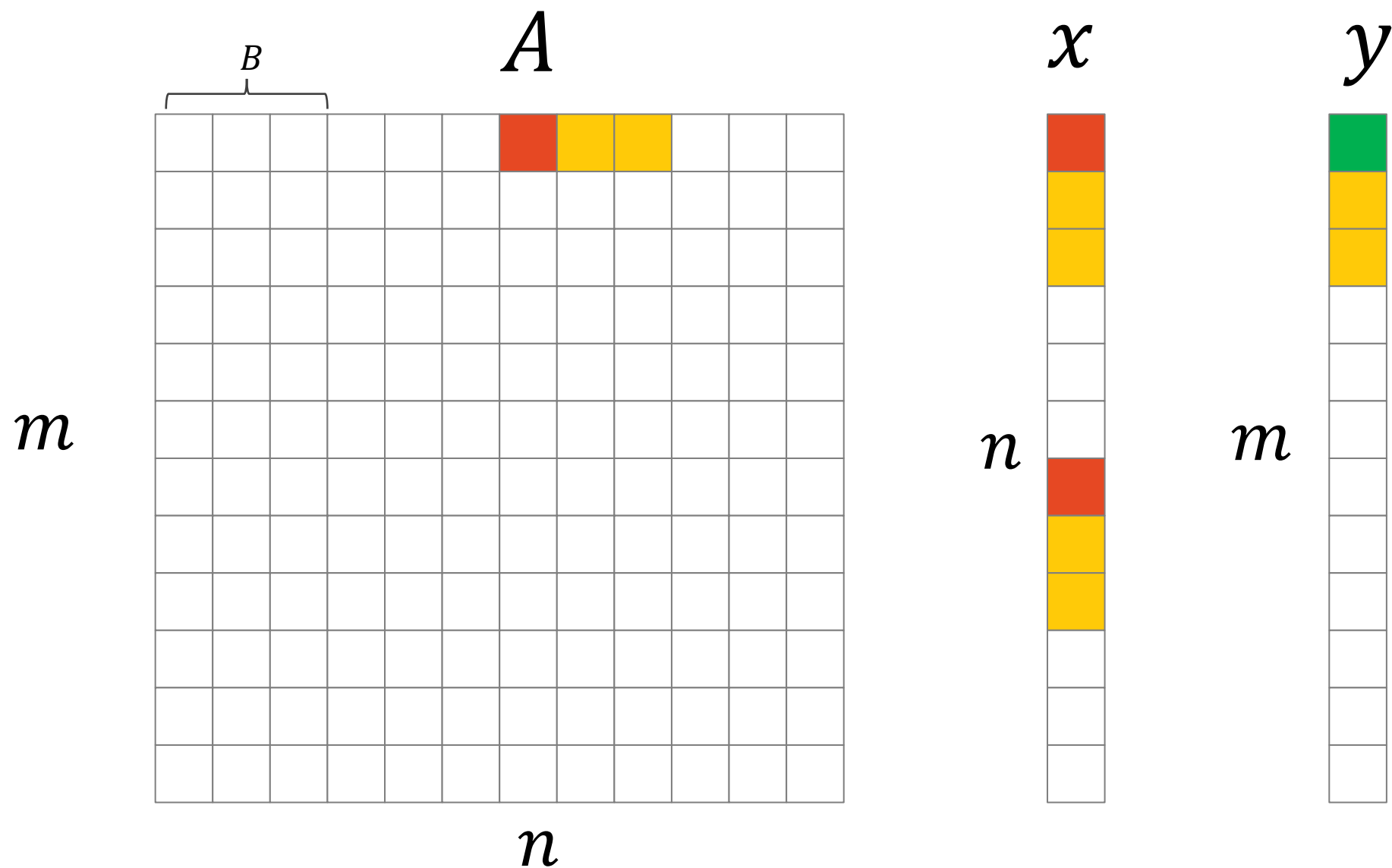
Matrix Vector Product

$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$



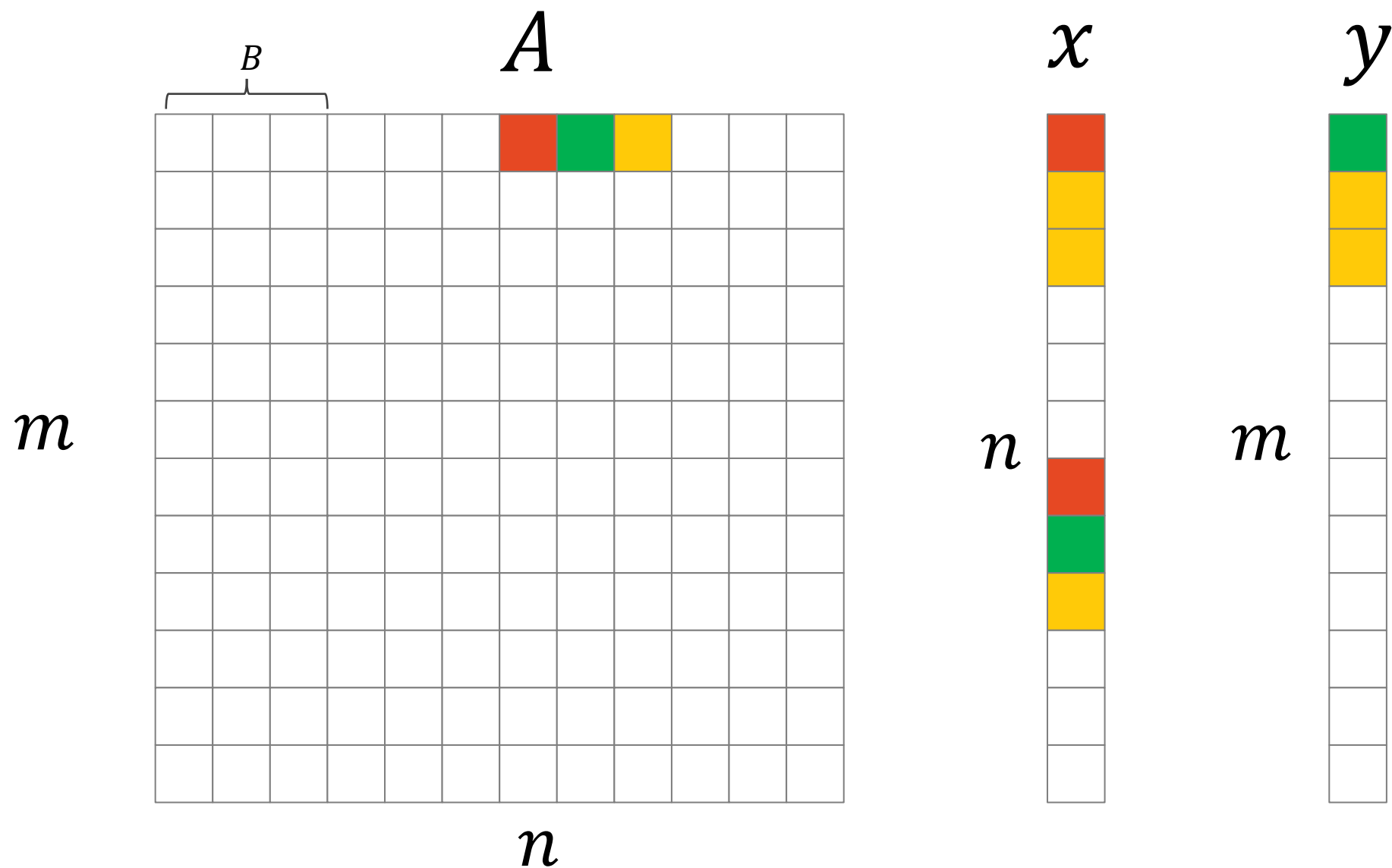
Matrix Vector Product

$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$



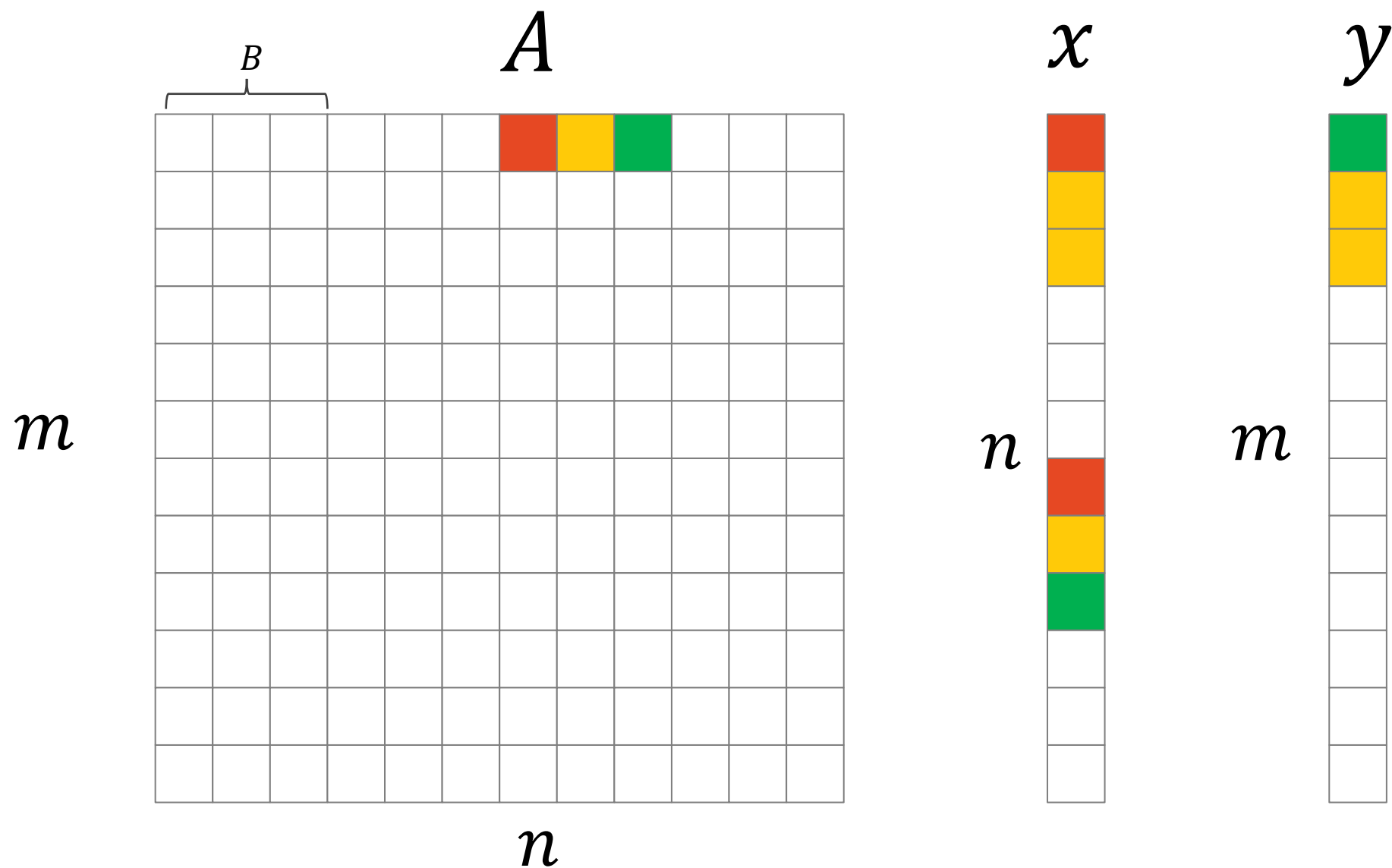
Matrix Vector Product

$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$



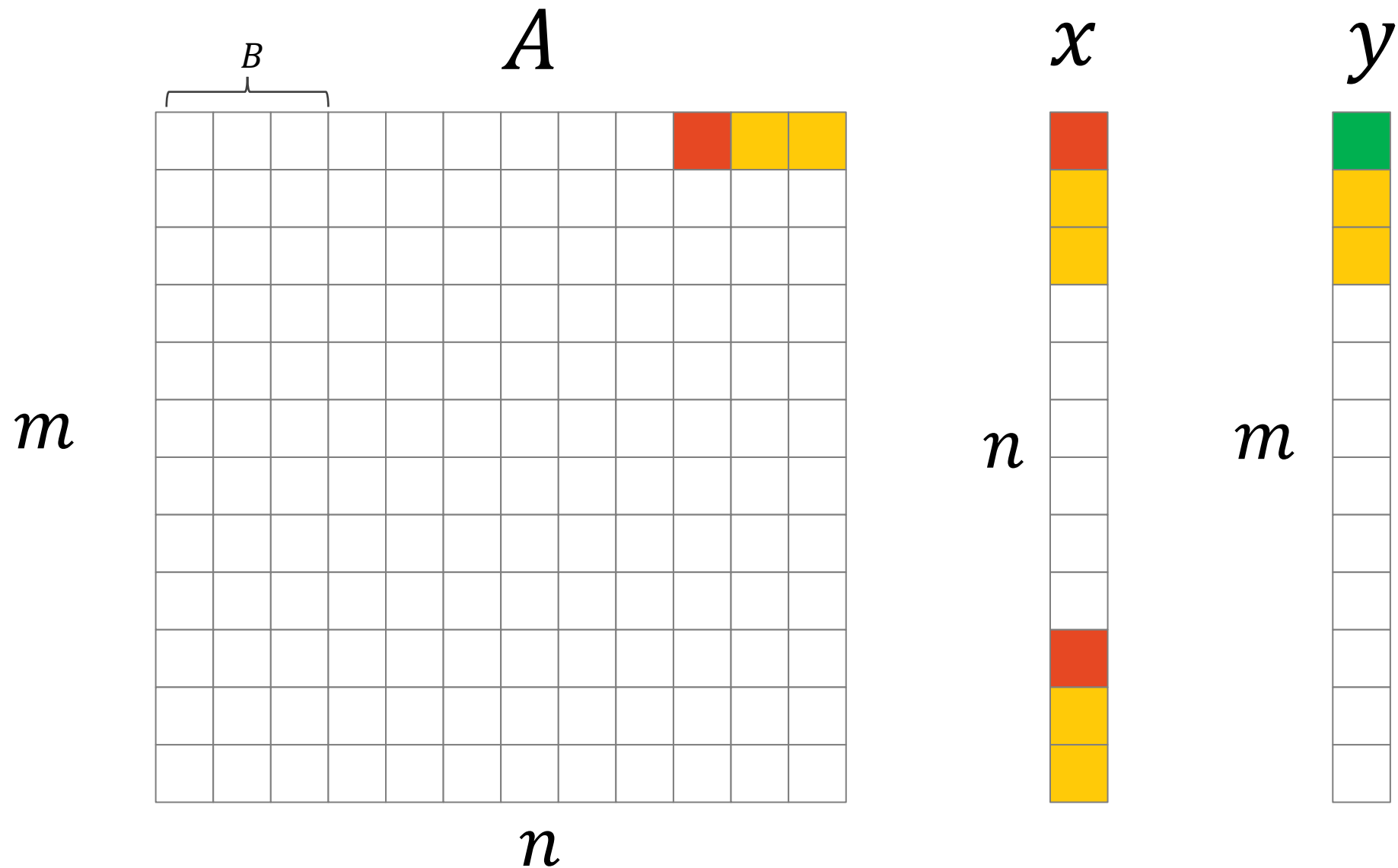
Matrix Vector Product

$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$



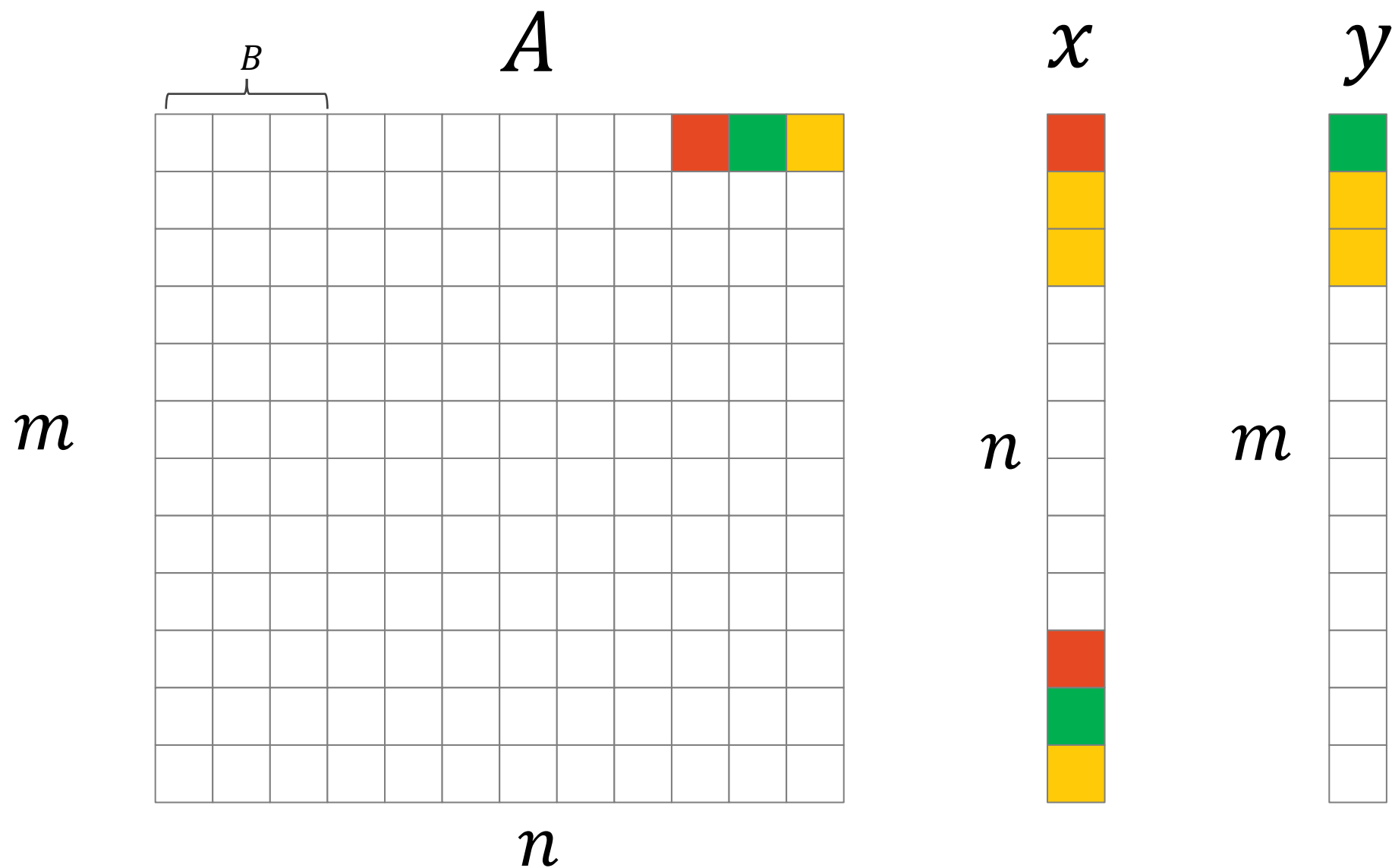
Matrix Vector Product

$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$



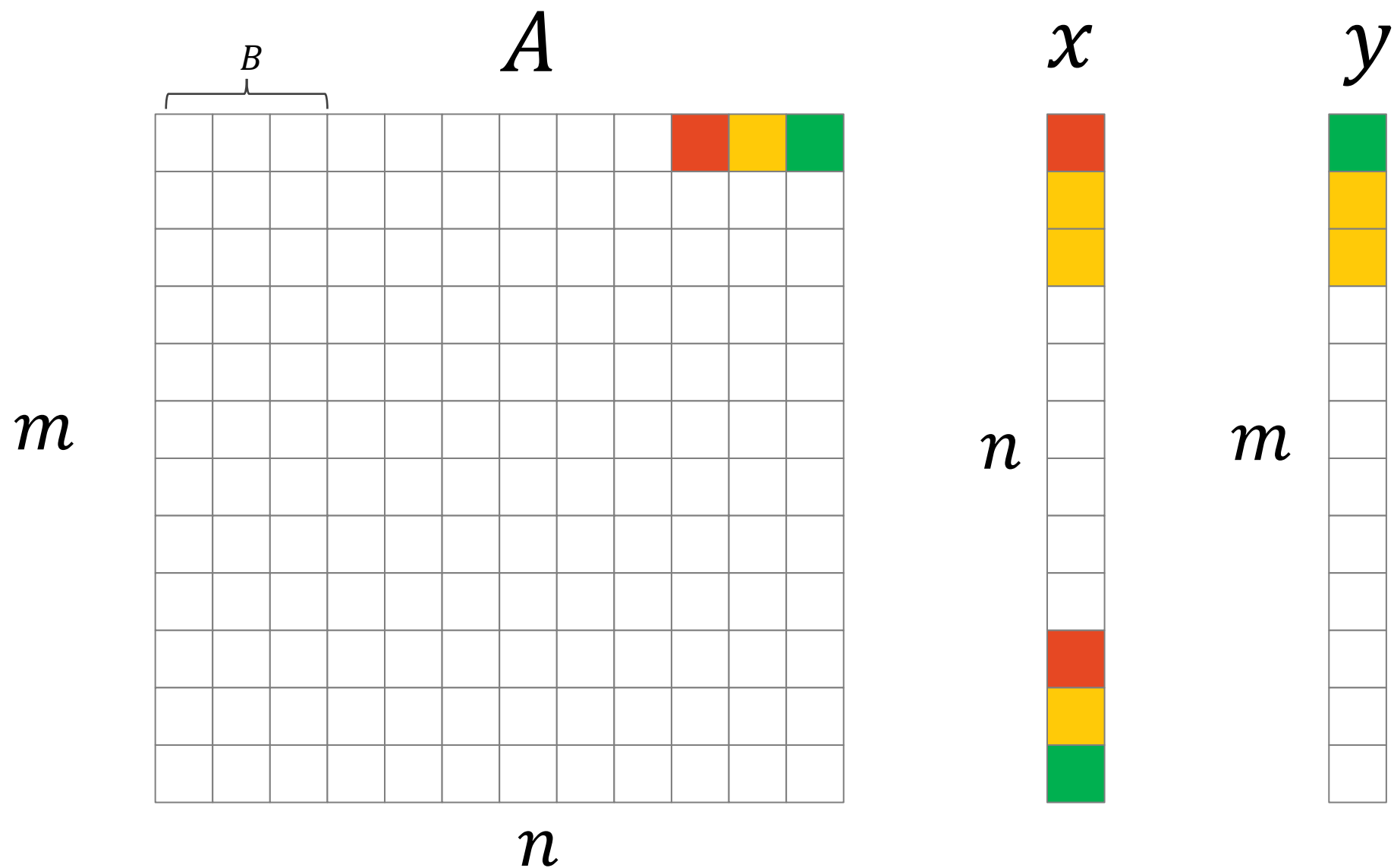
Matrix Vector Product

$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$



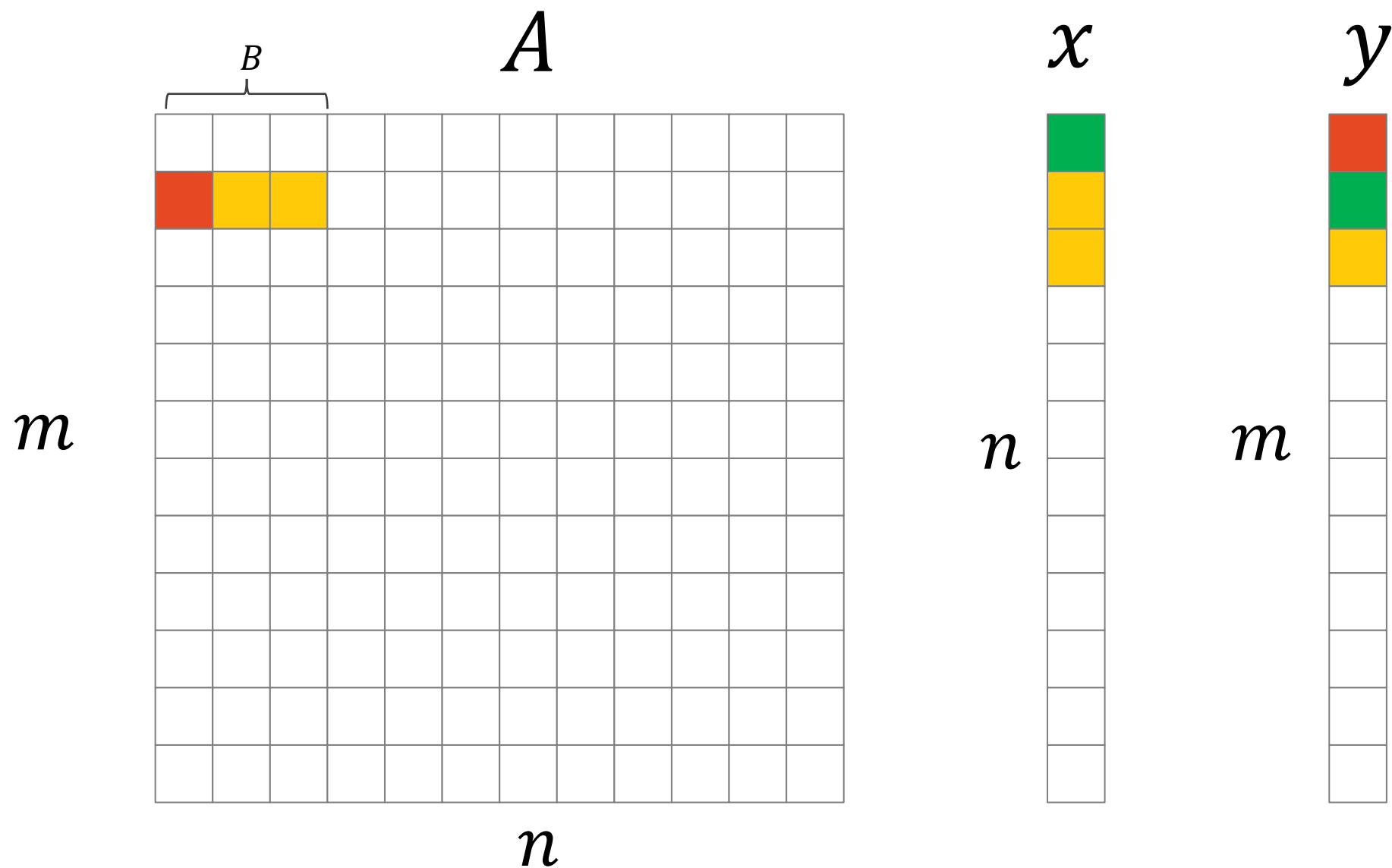
Matrix Vector Product

$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$



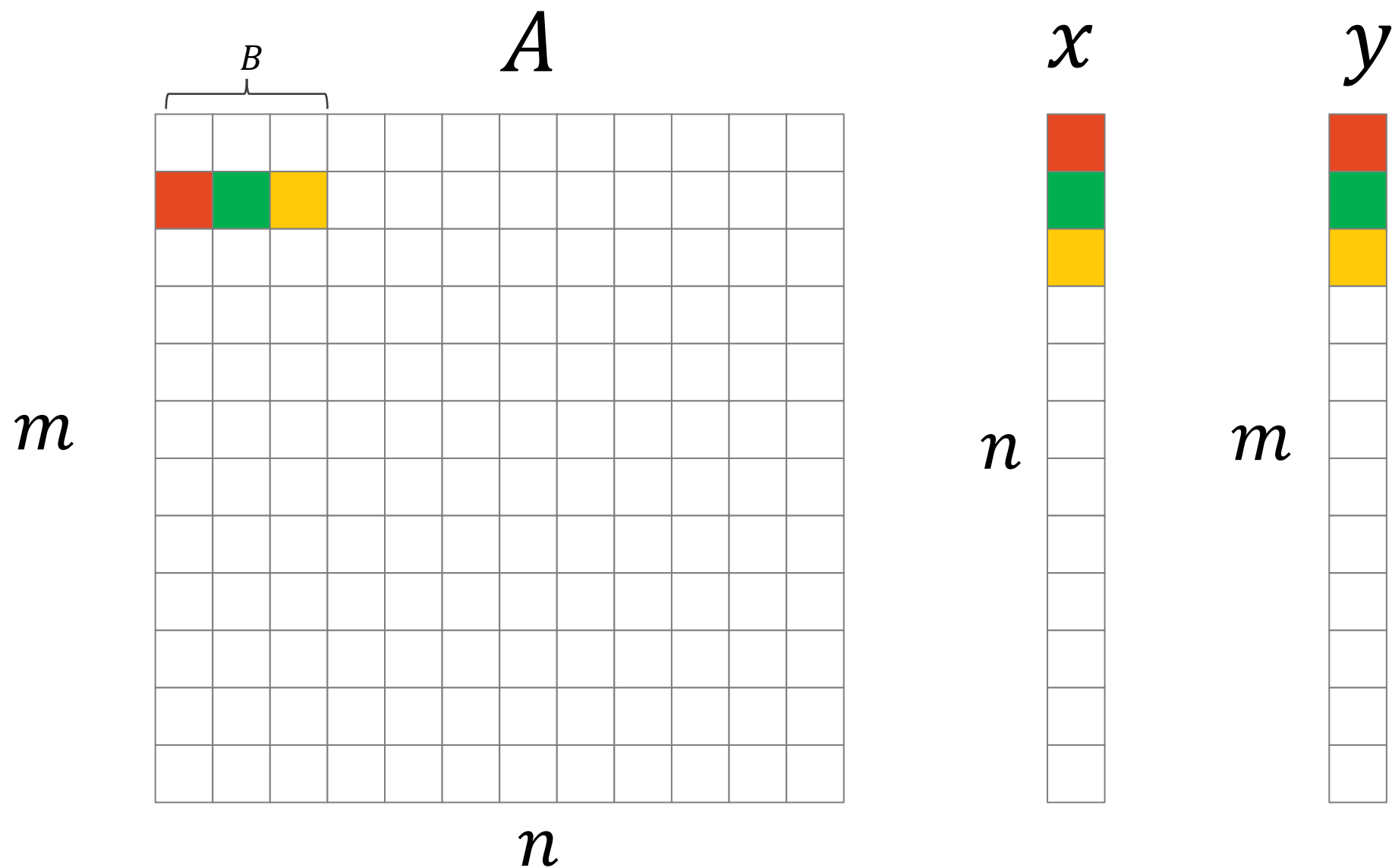
Matrix Vector Product

$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$



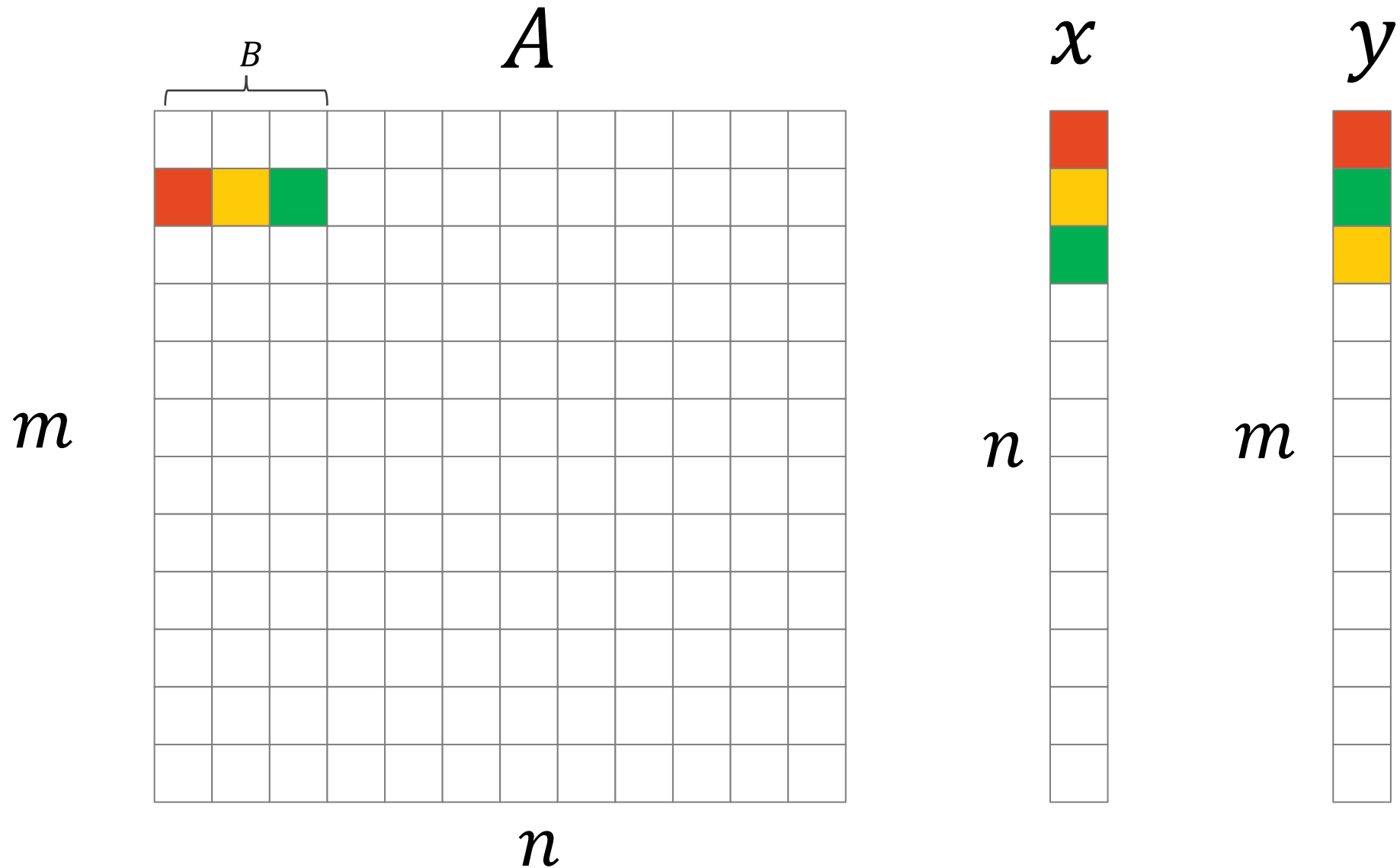
Matrix Vector Product

$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$



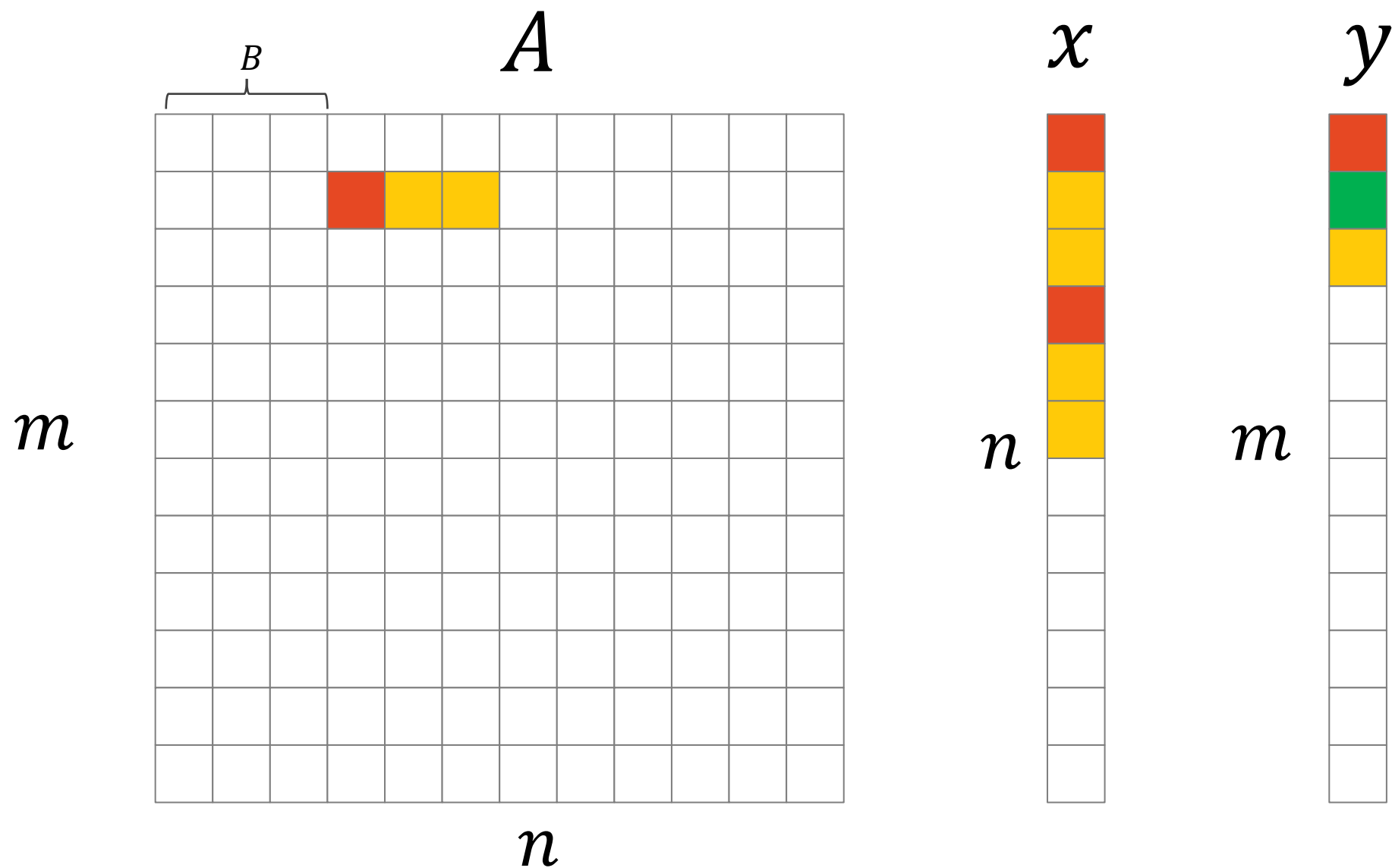
Matrix Vector Product

$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$



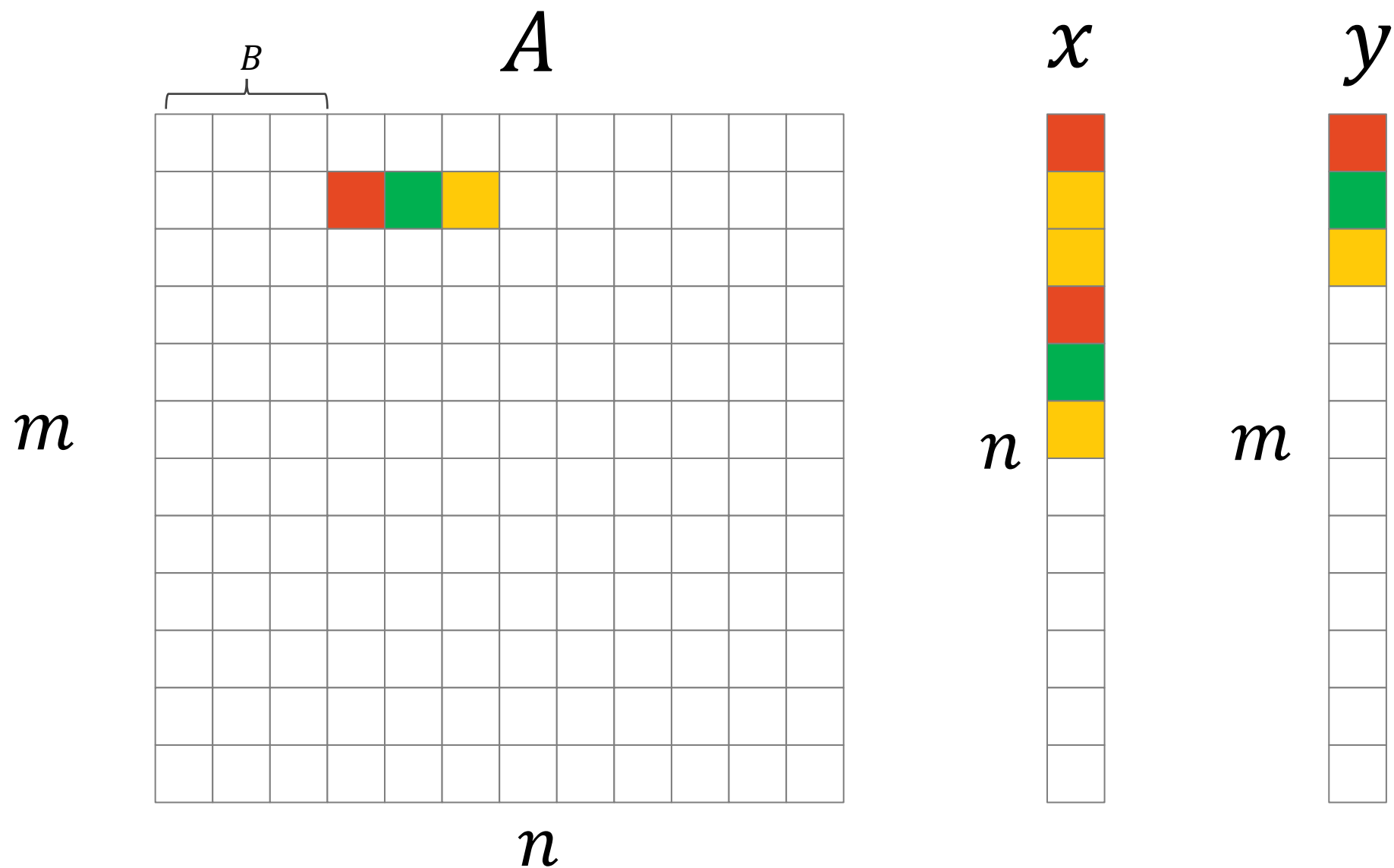
Matrix Vector Product

$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$



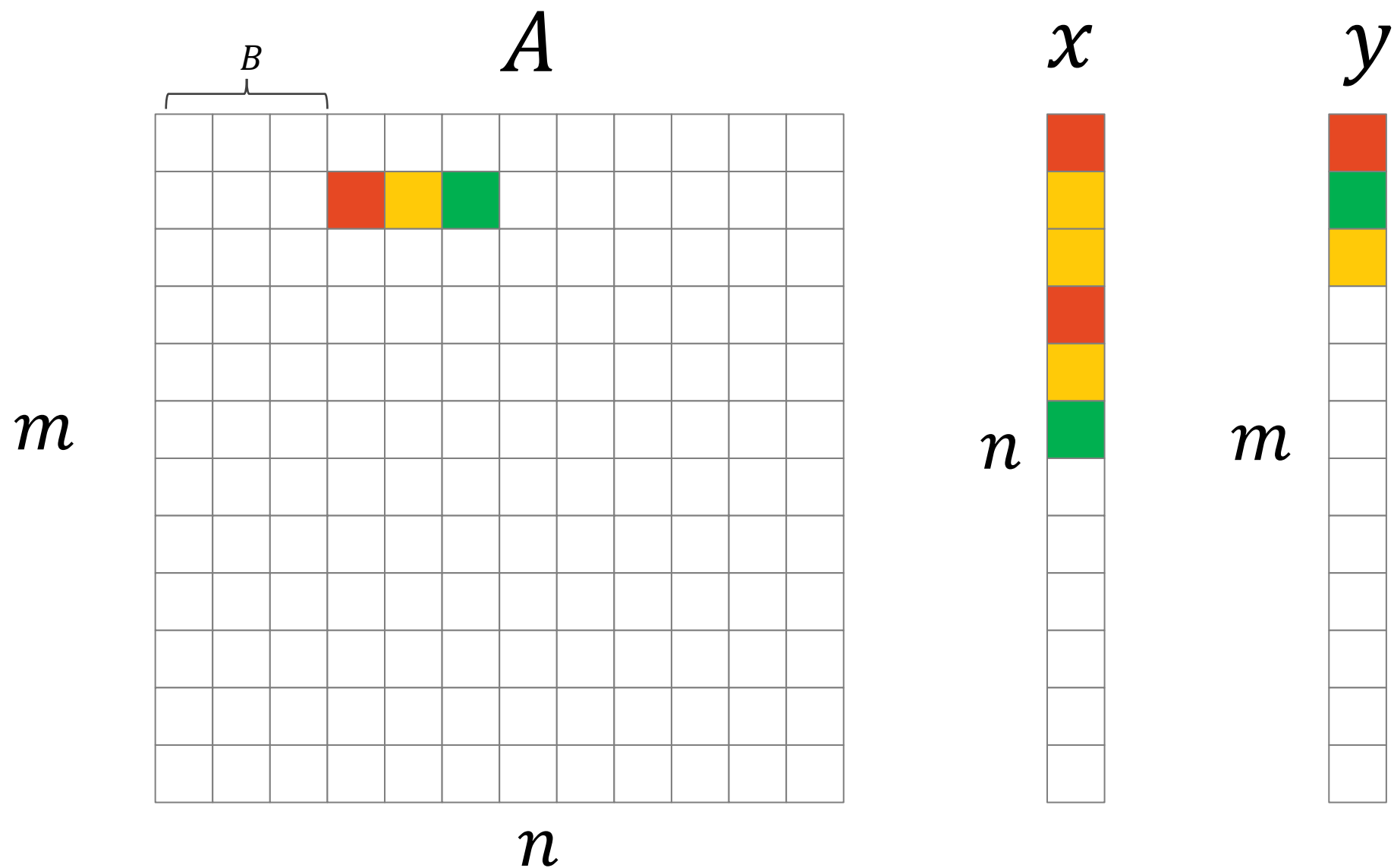
Matrix Vector Product

$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$



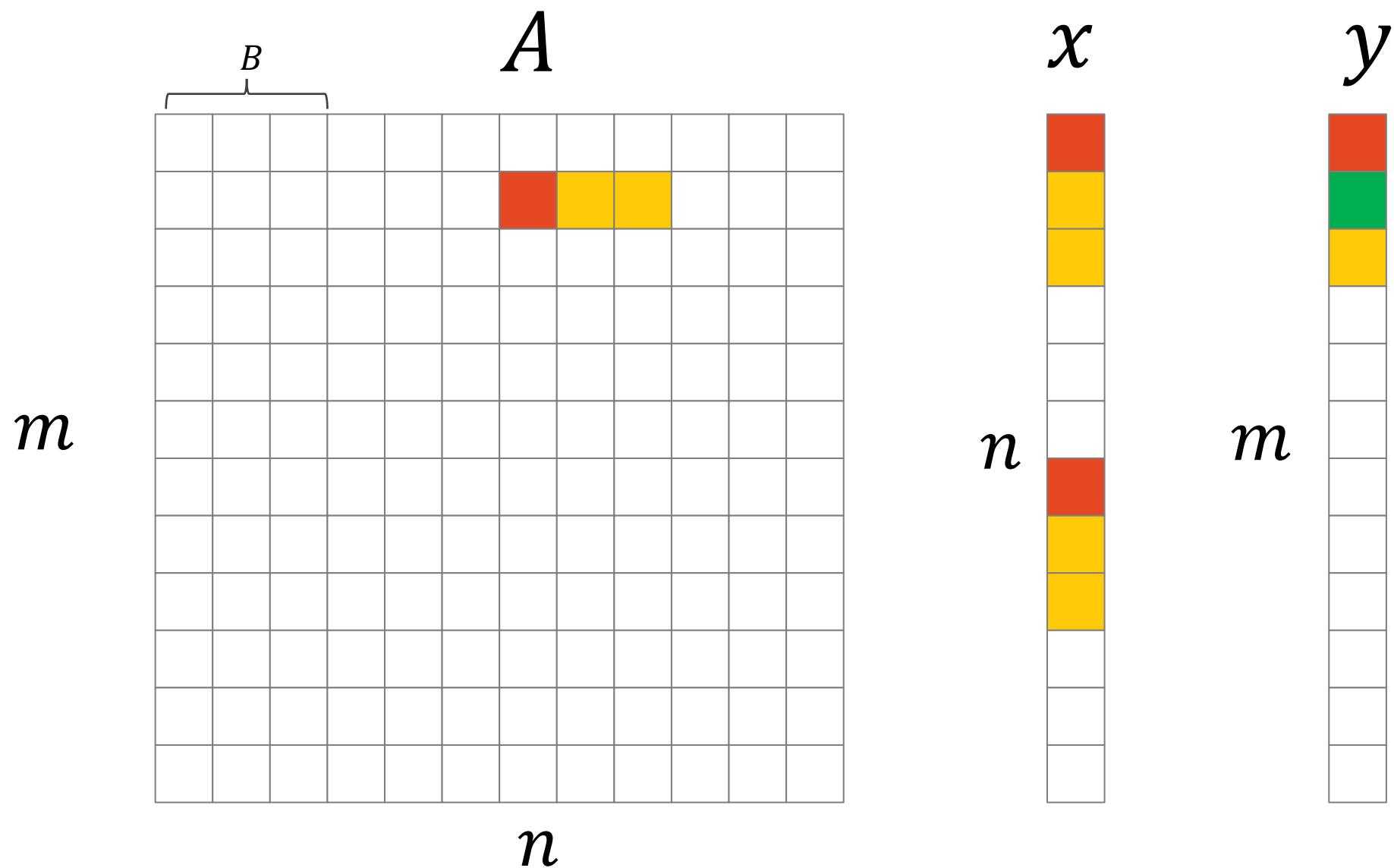
Matrix Vector Product

$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$



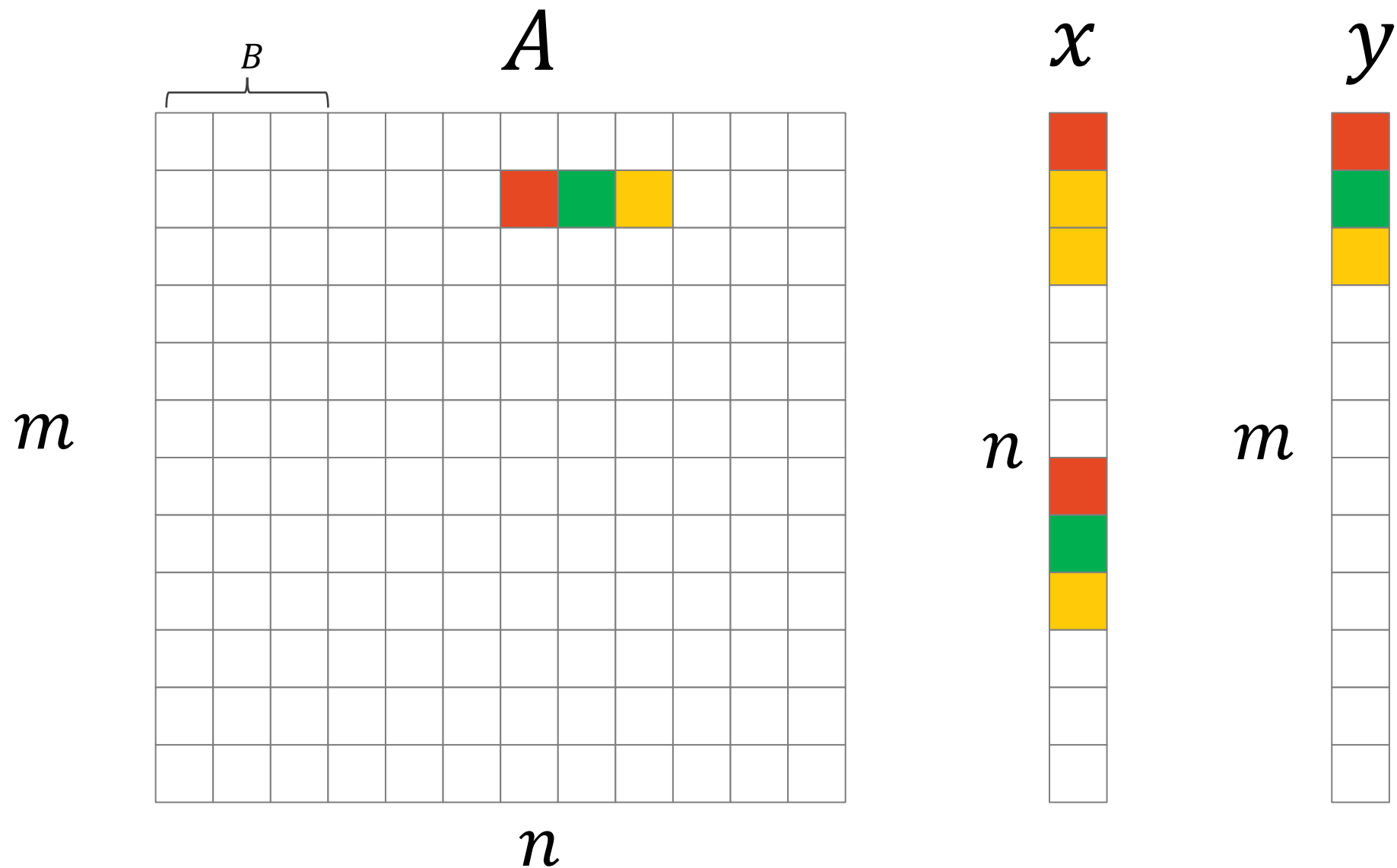
Matrix Vector Product

$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$



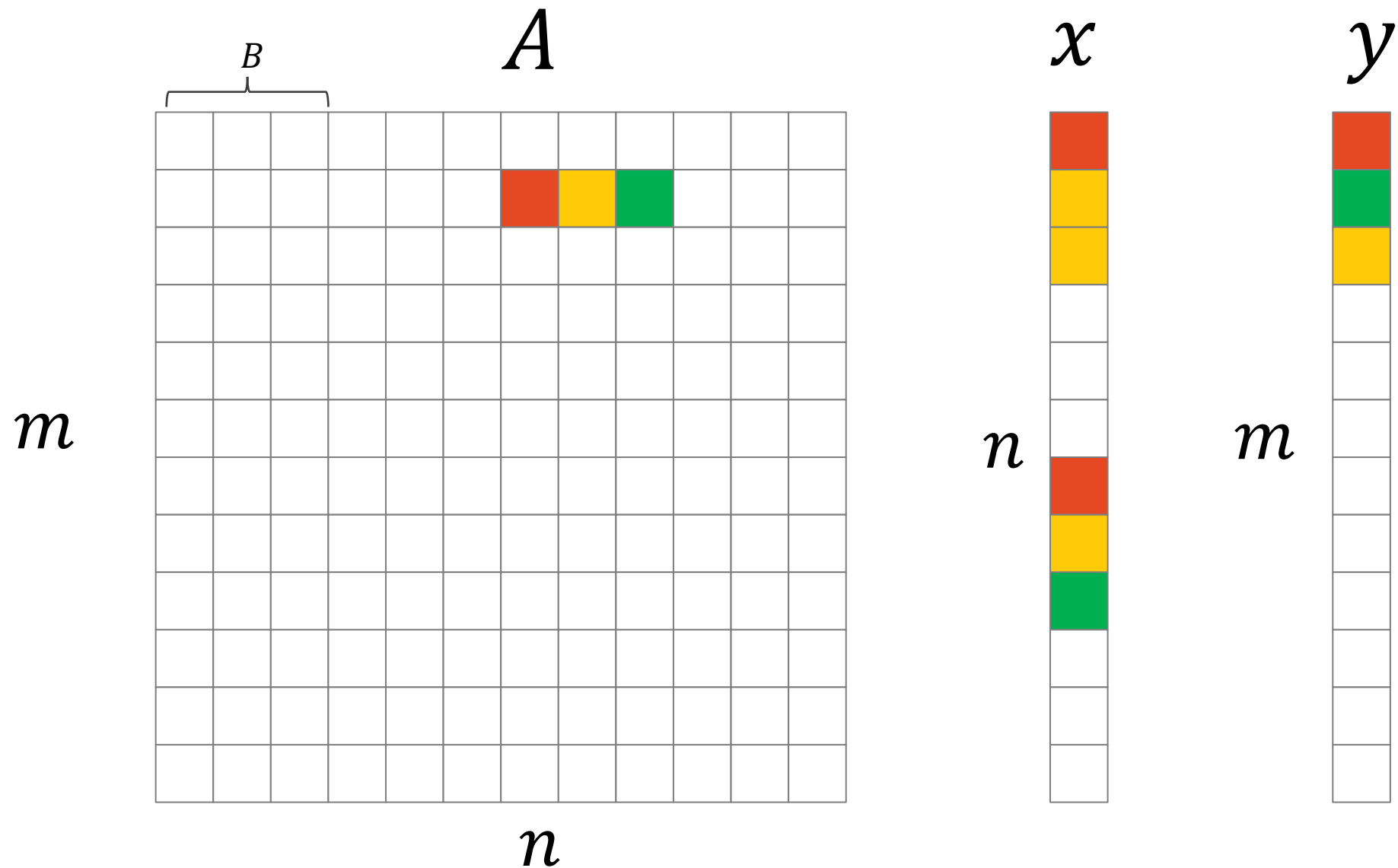
Matrix Vector Product

$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$



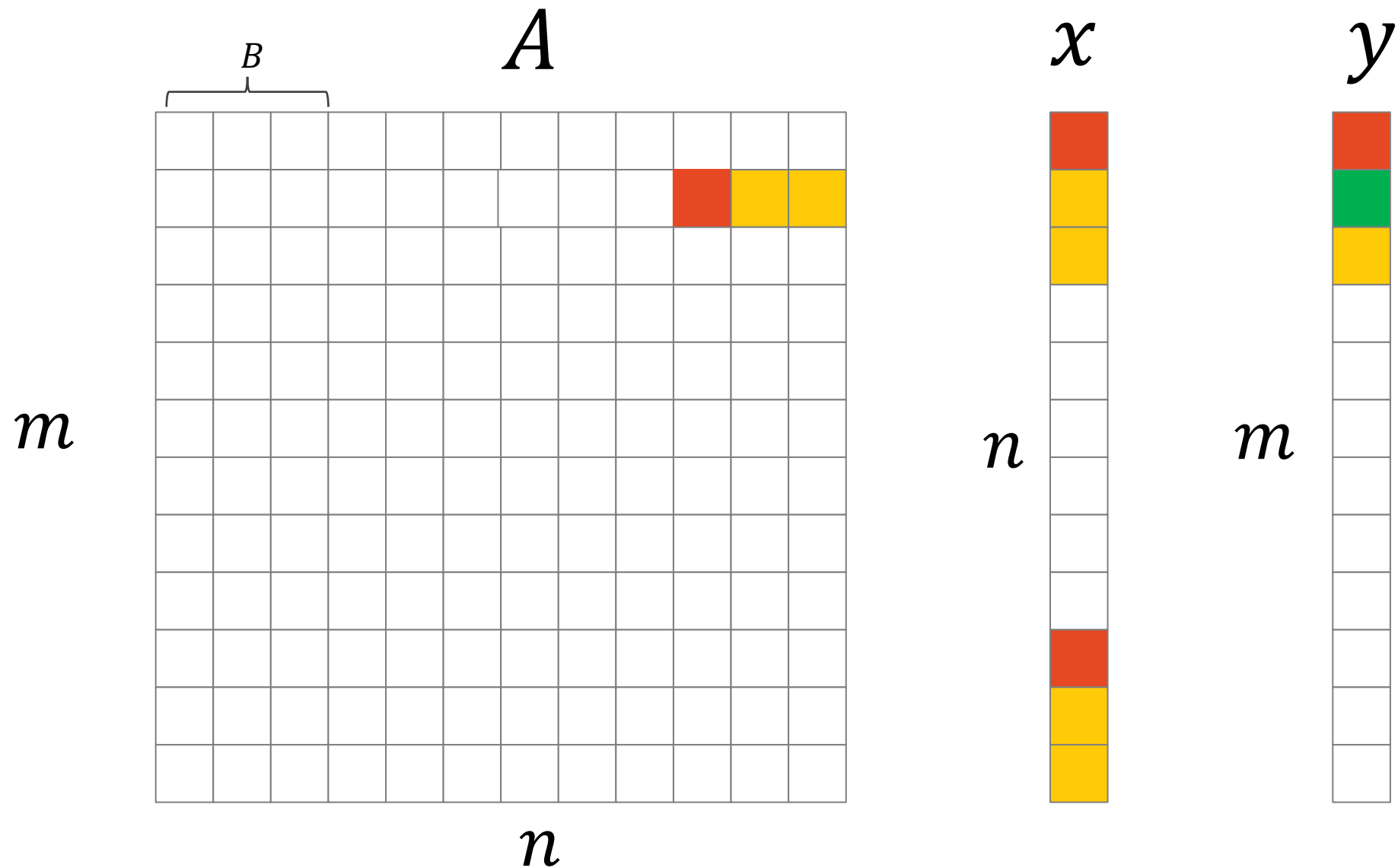
Matrix Vector Product

$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$



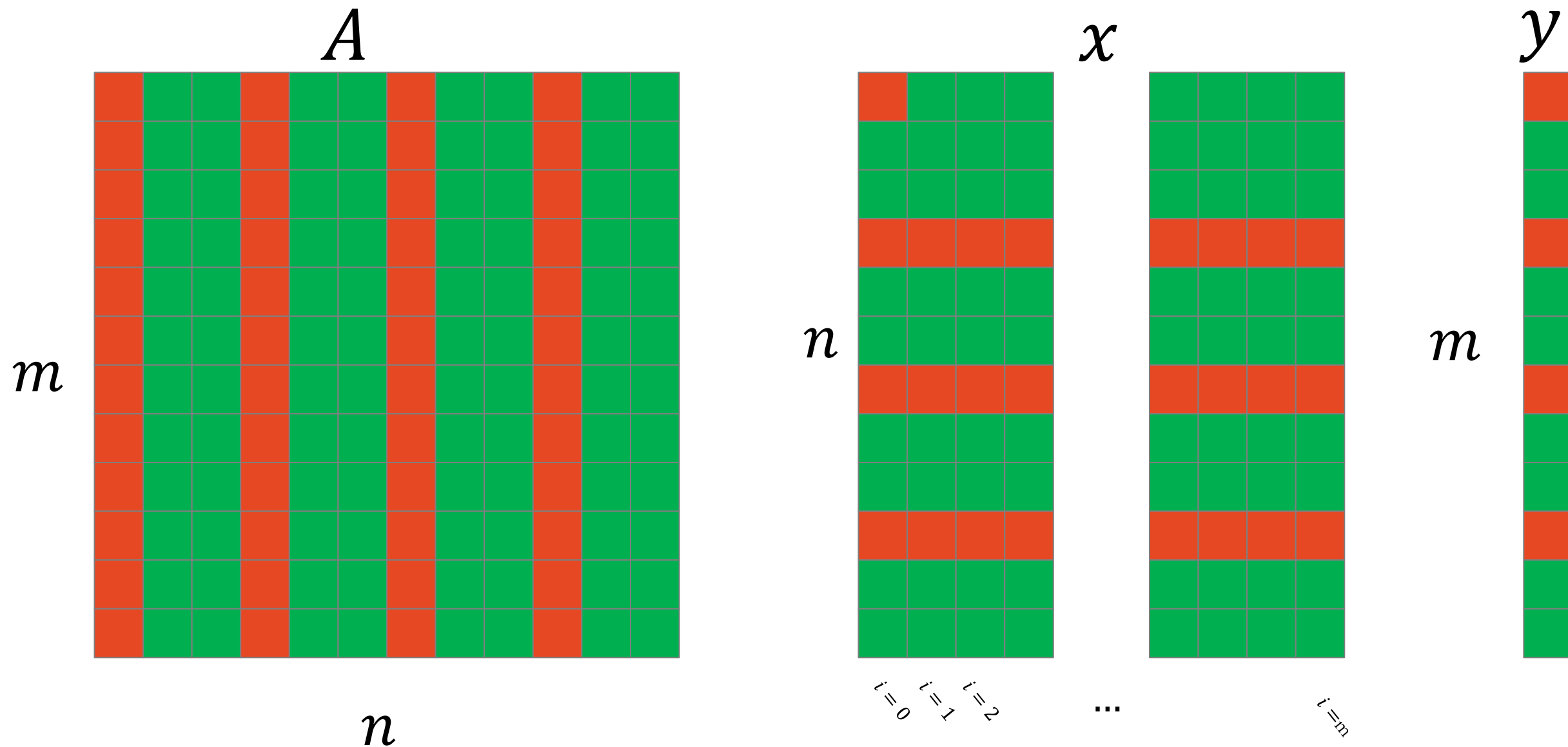
Matrix Vector Product

$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$



Hits and Misses for MVP

$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$



Analysis

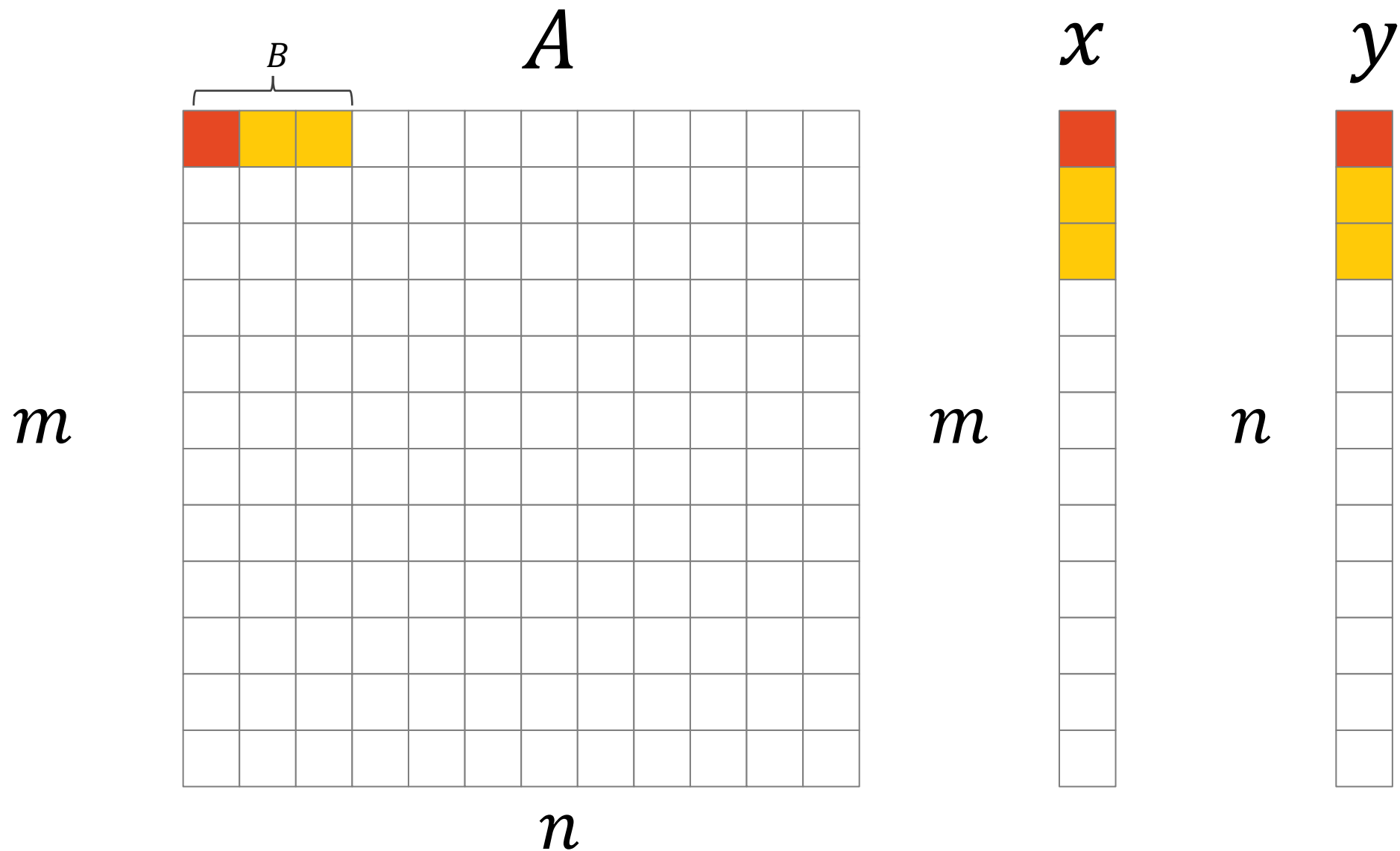
$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$

```
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        y[i] = y[i] + A[i][j]*x[j];  
    }  
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```

$$Q(n, m; B, M) \approx m(\lceil n/B \rceil + \lceil n/B \rceil) \approx \frac{2mn}{B}$$

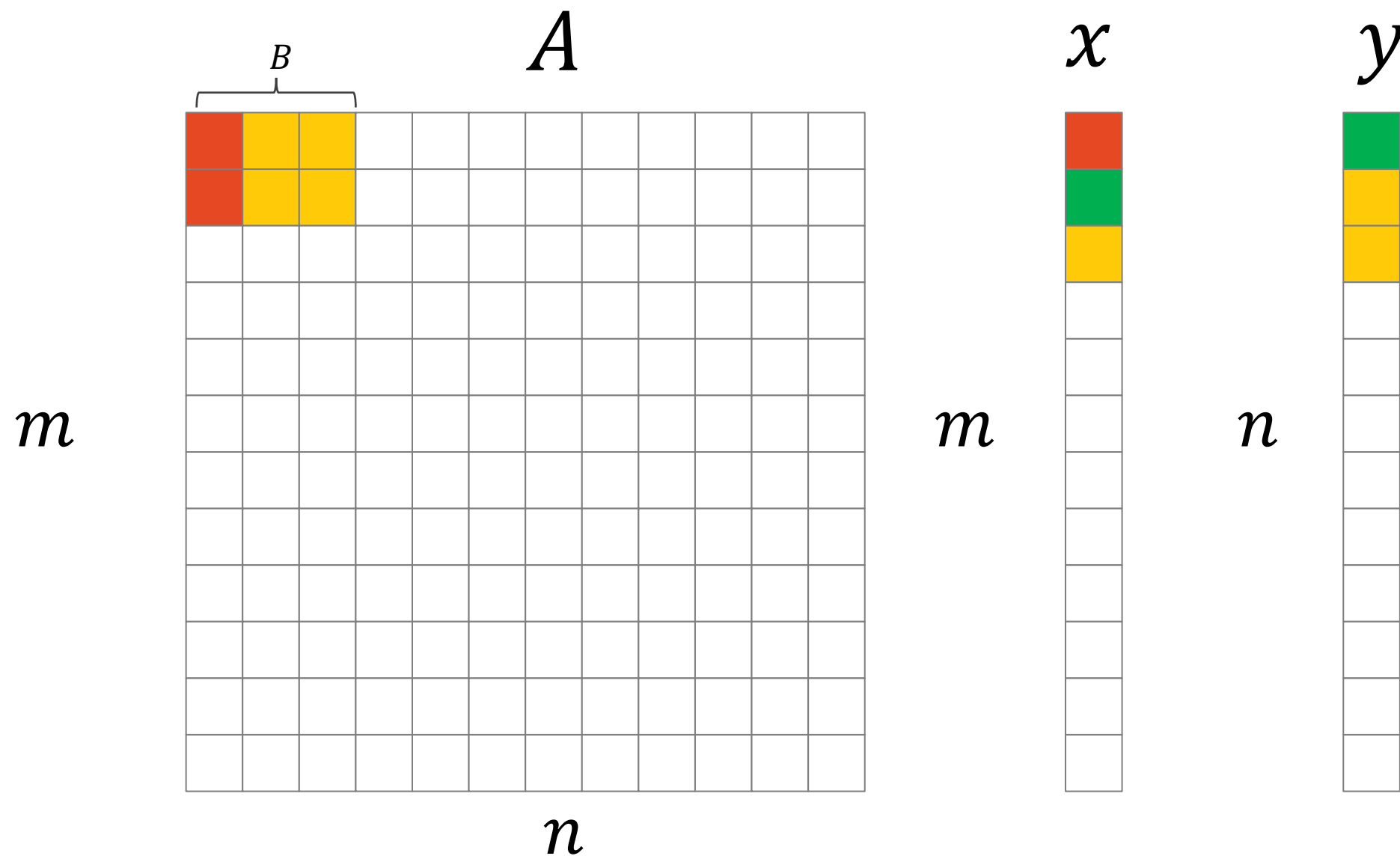
Transpose Matrix Vector Product

$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$



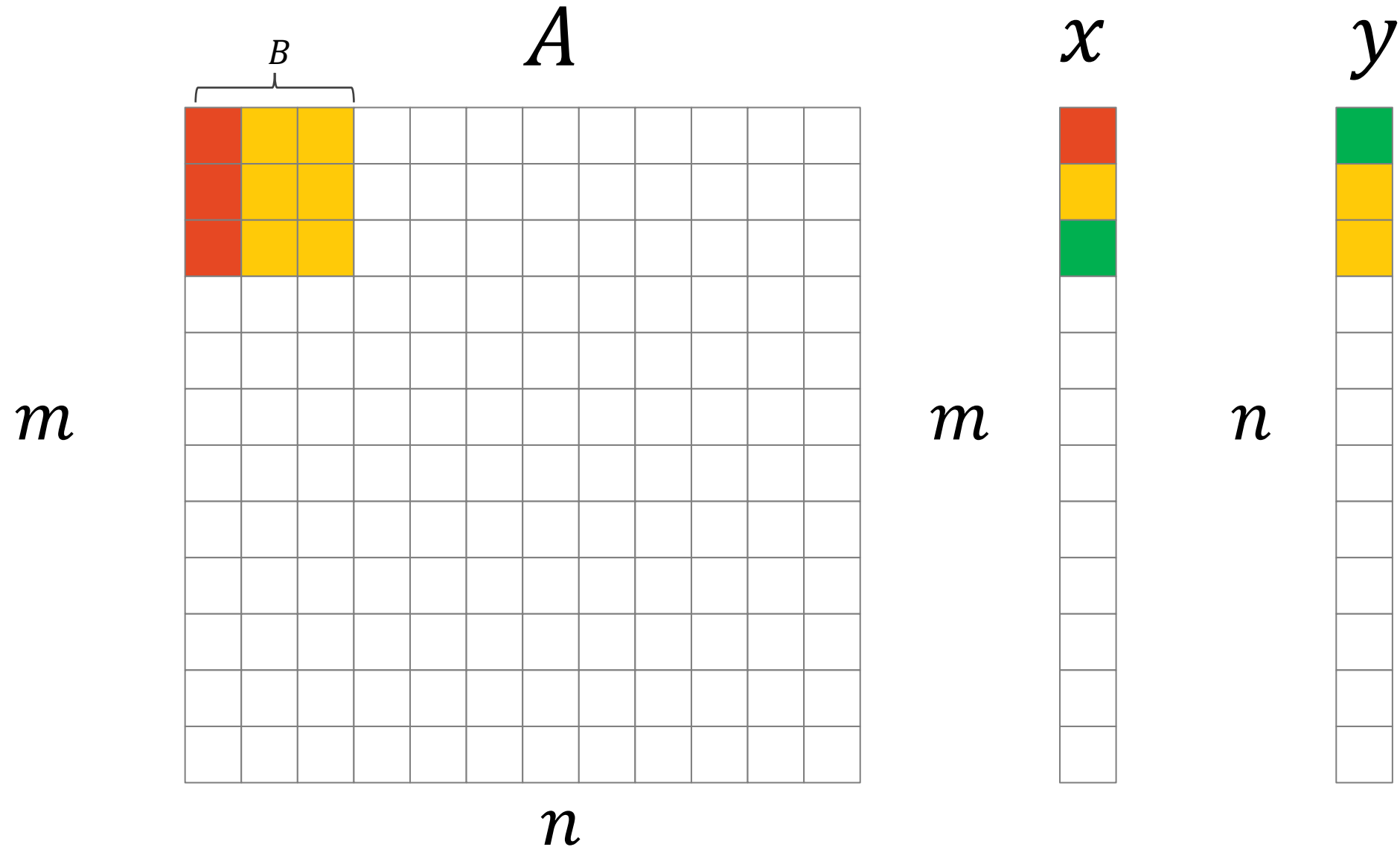
Transpose Matrix Vector Product

$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$



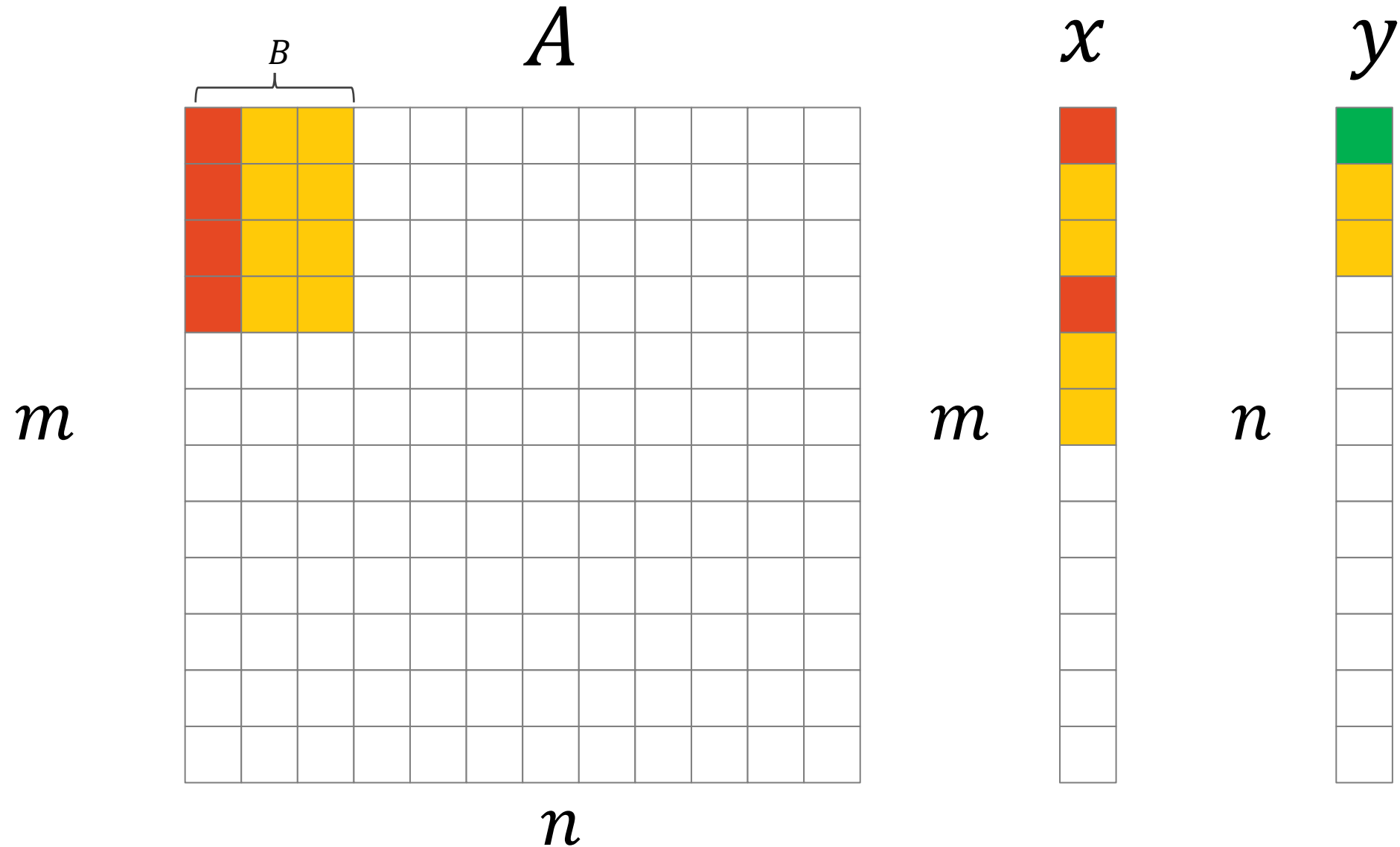
Transpose Matrix Vector Product

$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$



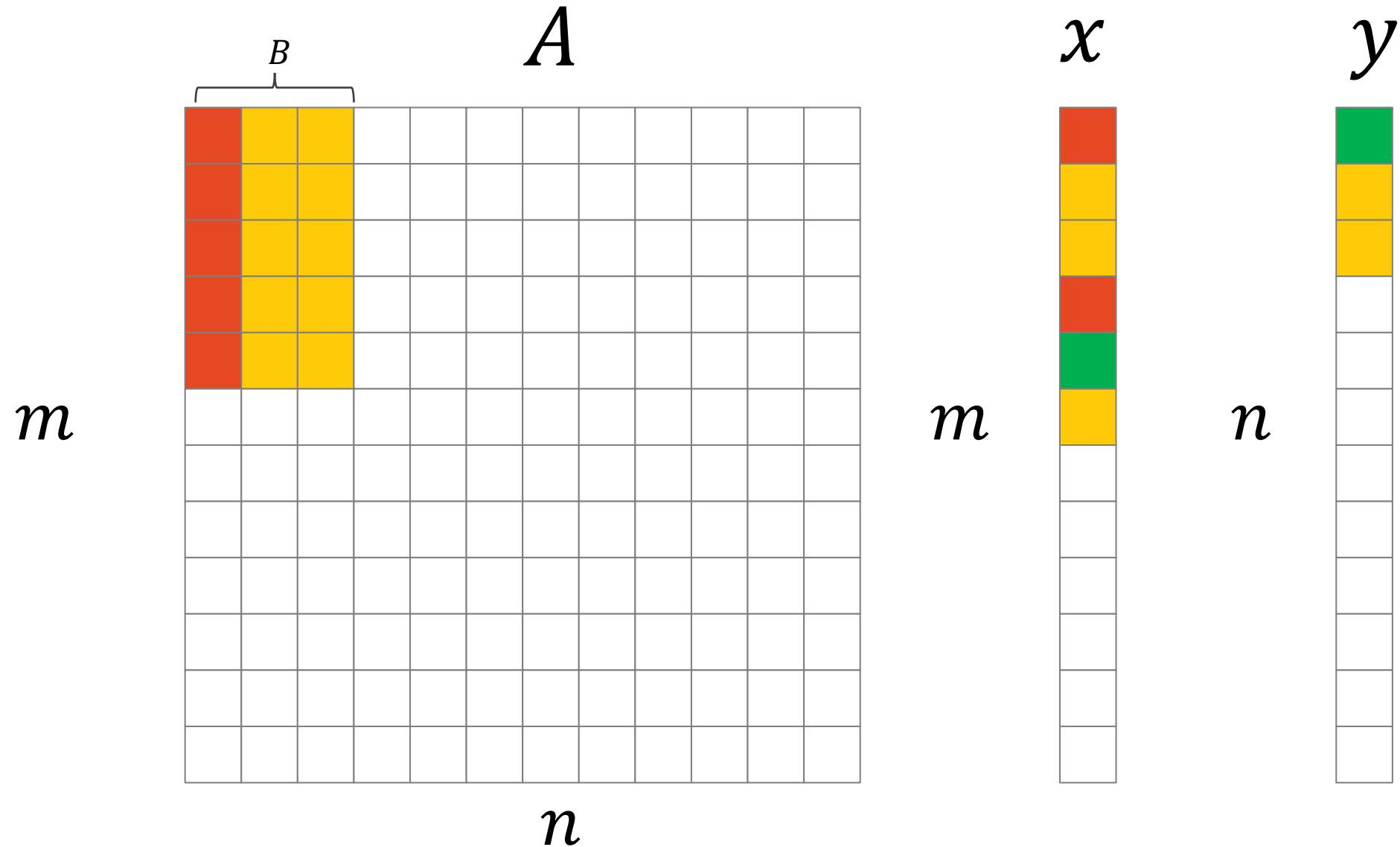
Transpose Matrix Vector Product

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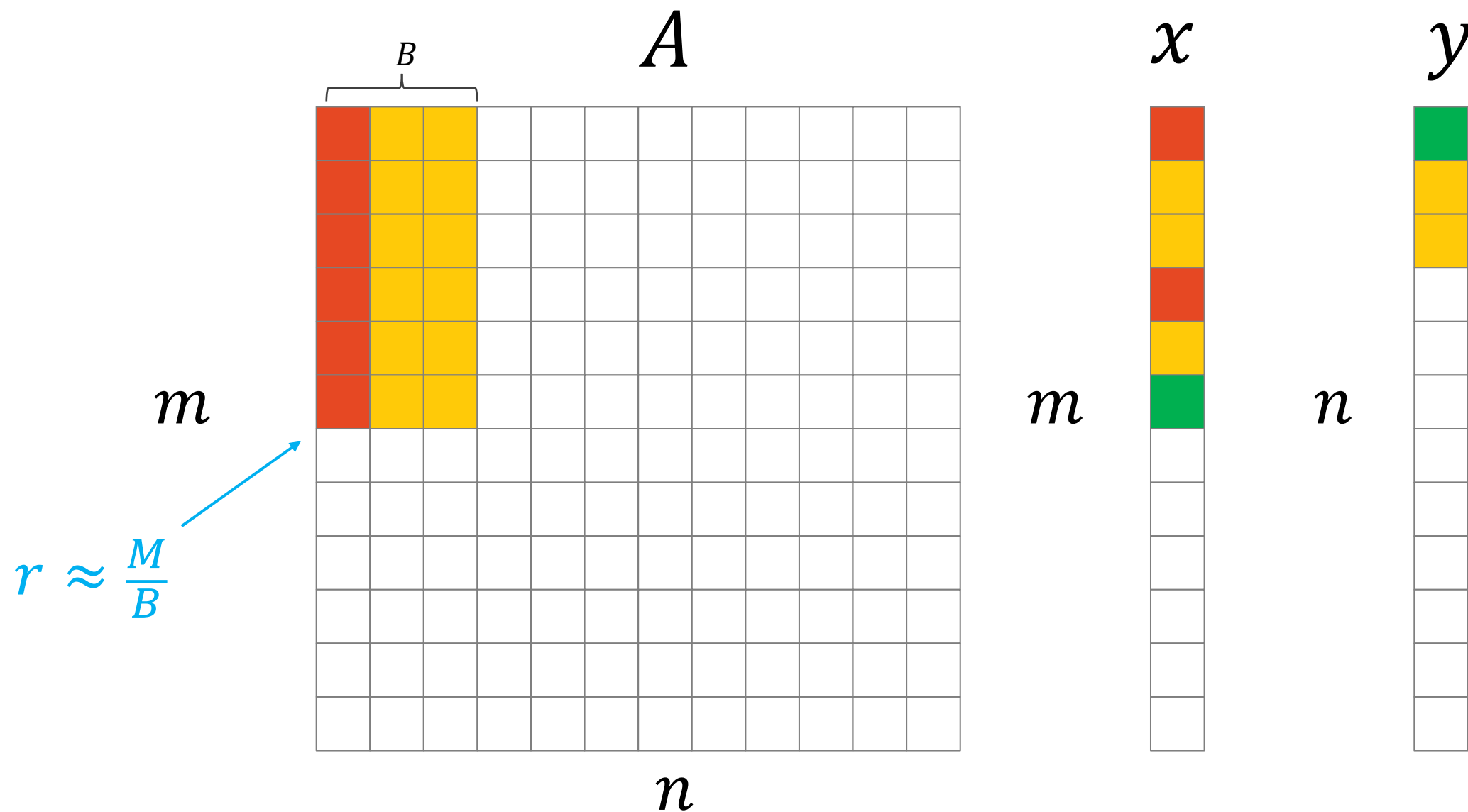
Transpose Matrix Vector Product

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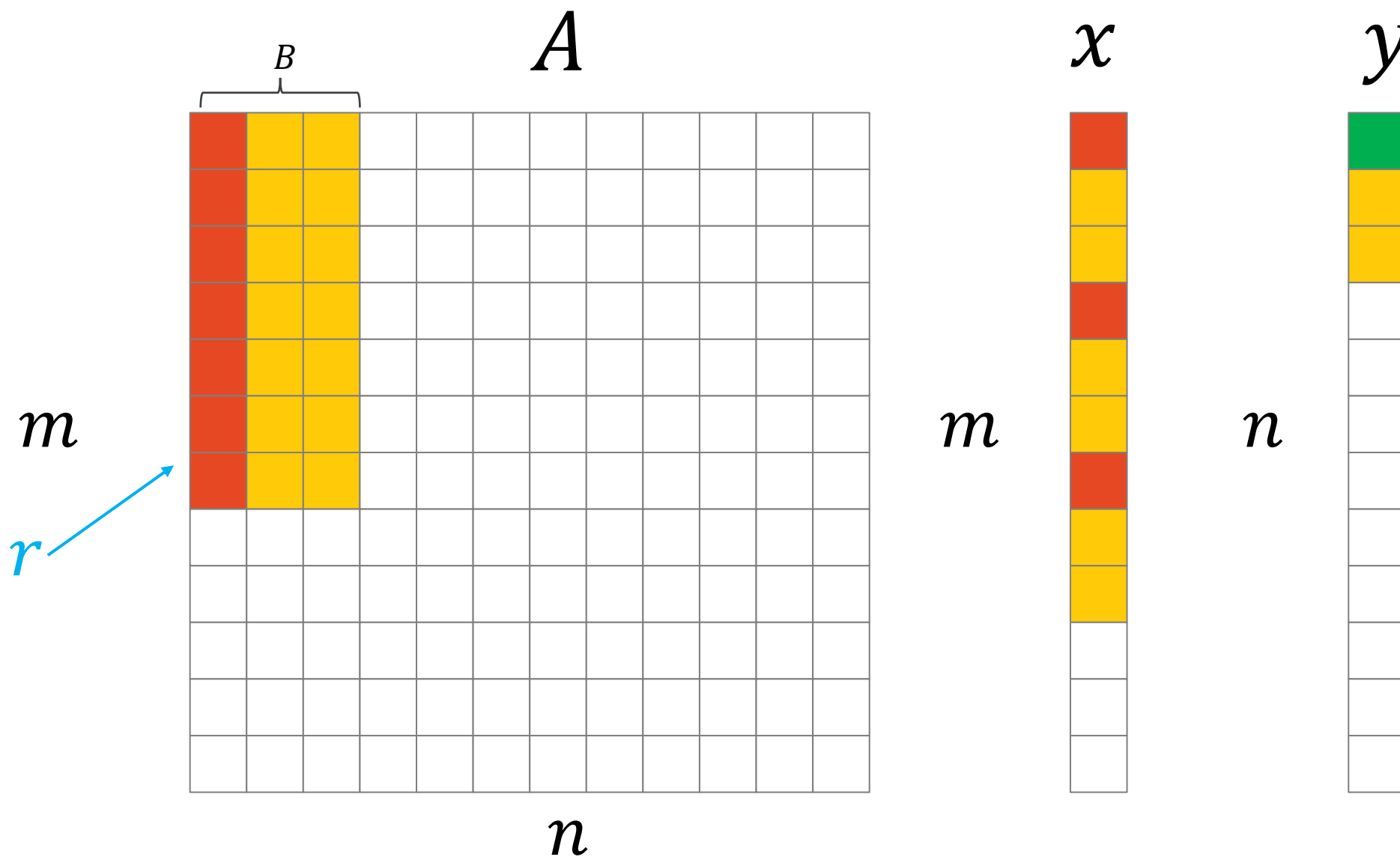
Transpose Matrix Vector Product

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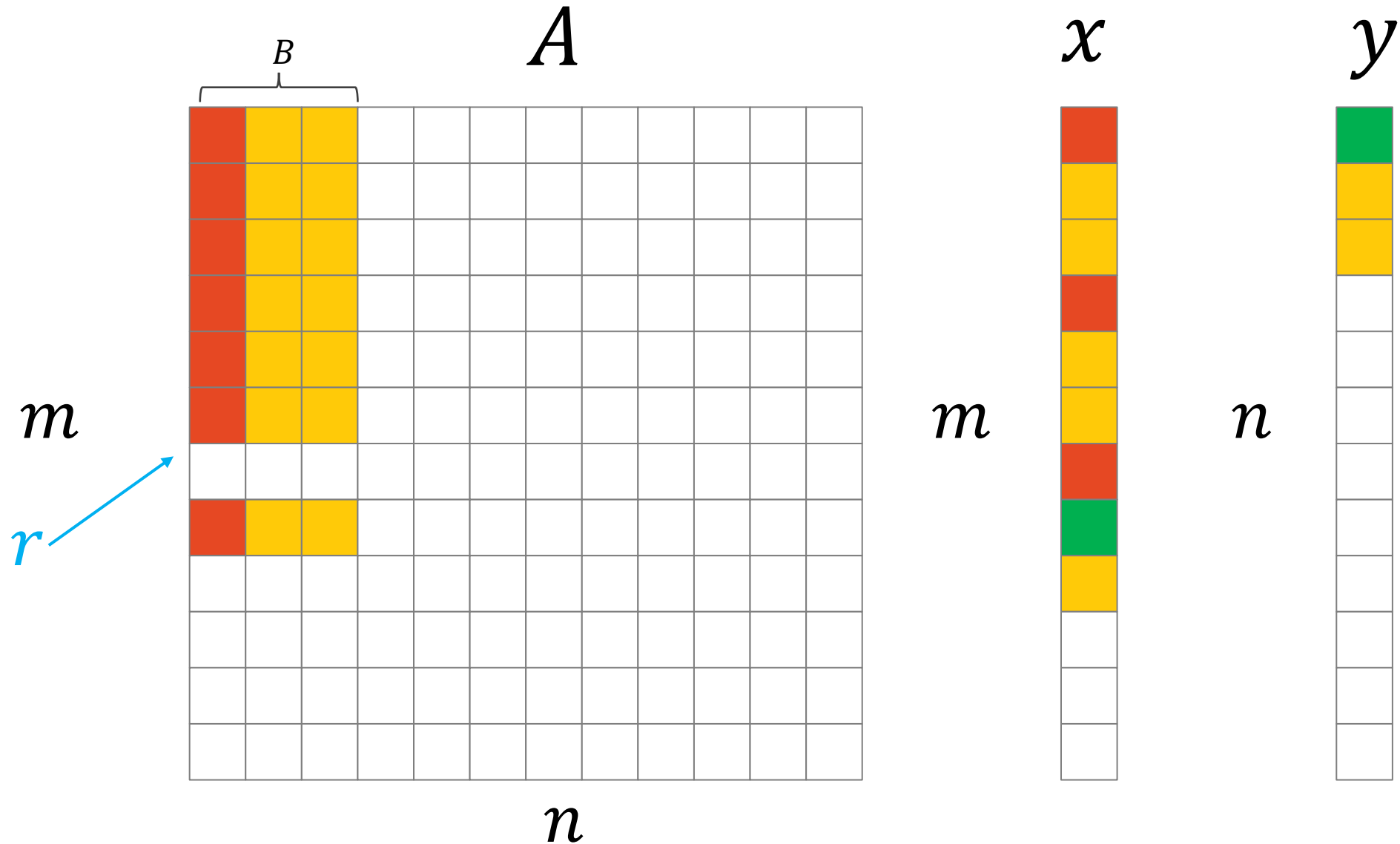
Transpose Matrix Vector Product

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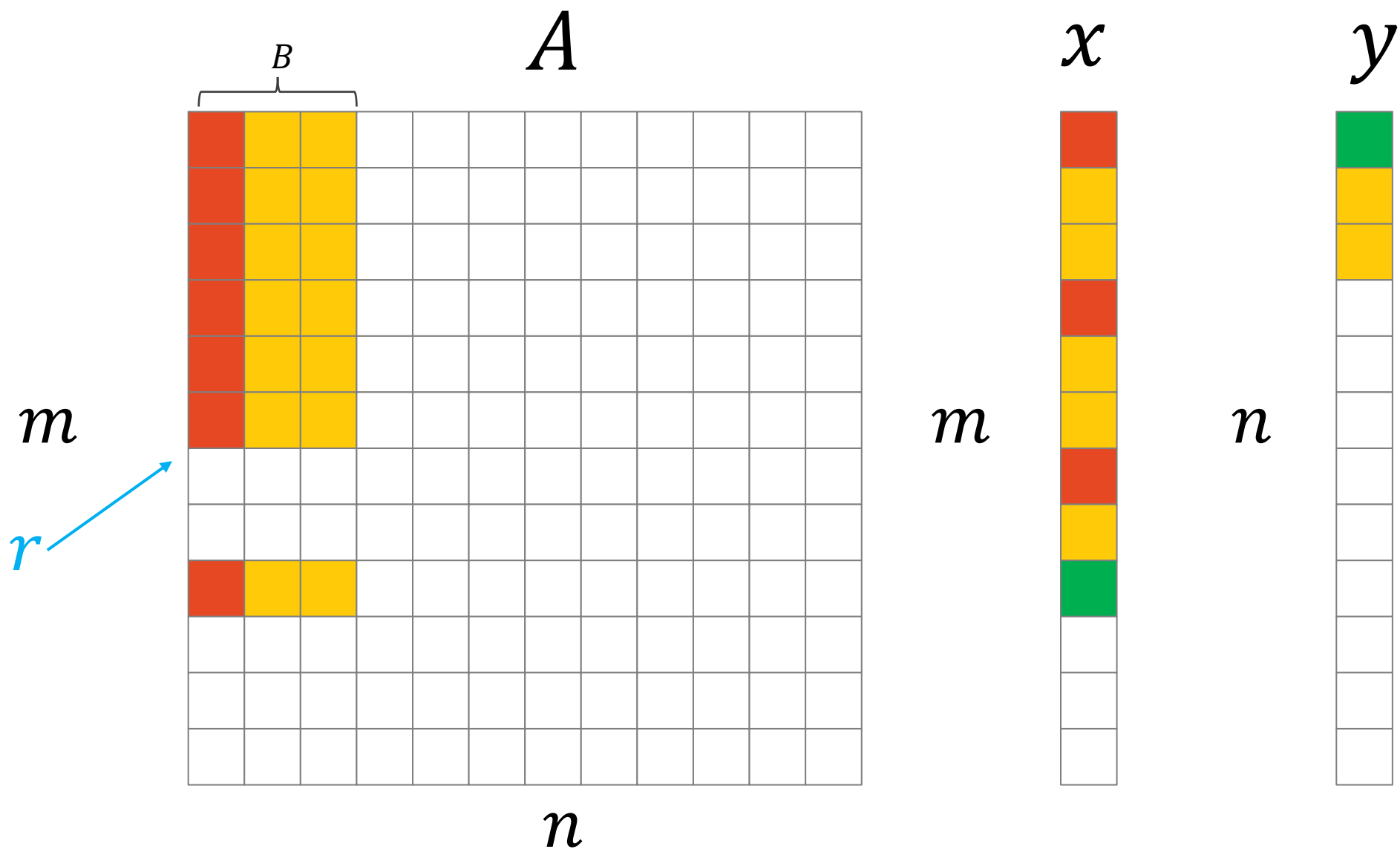
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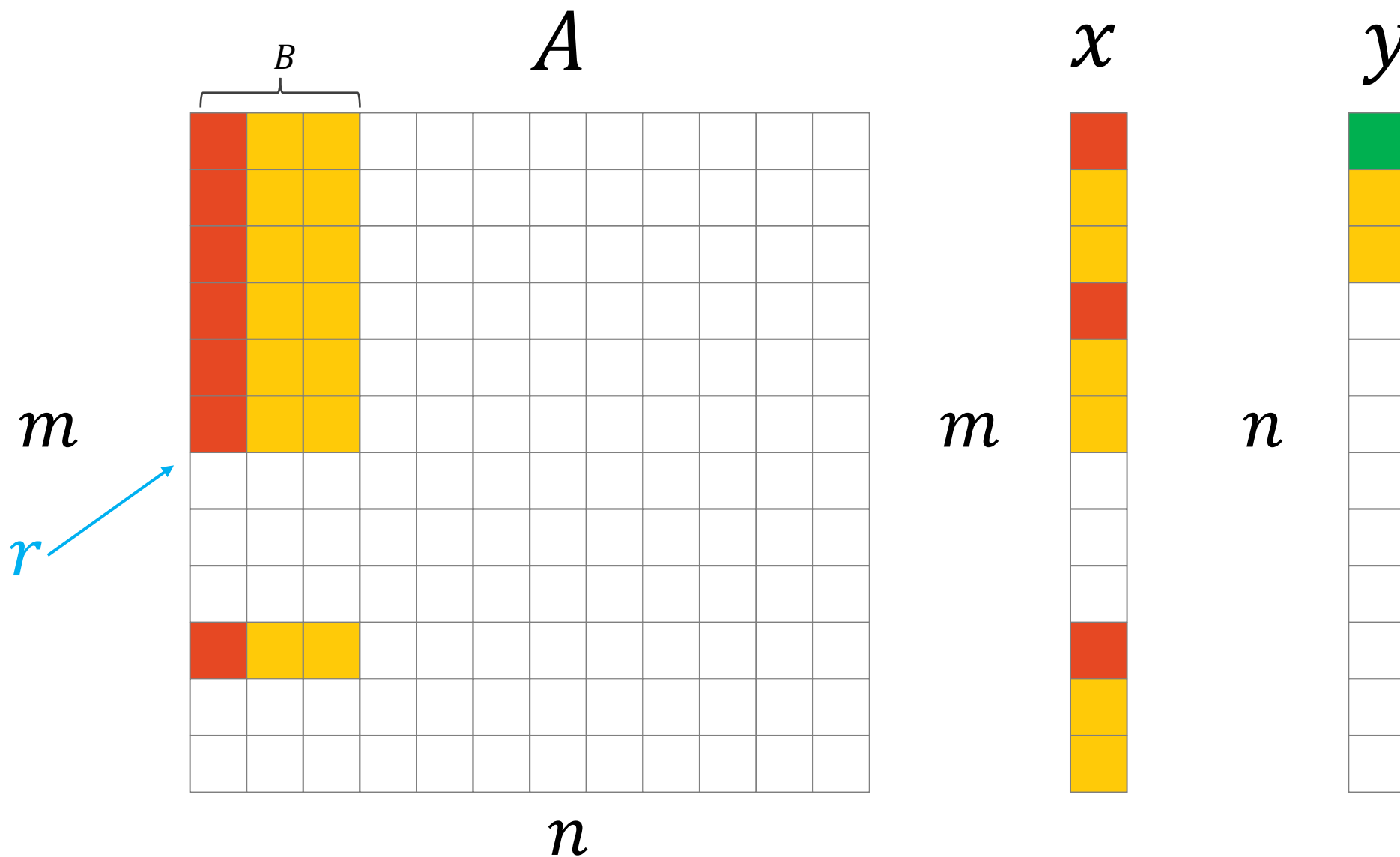
Transpose Matrix Vector Product

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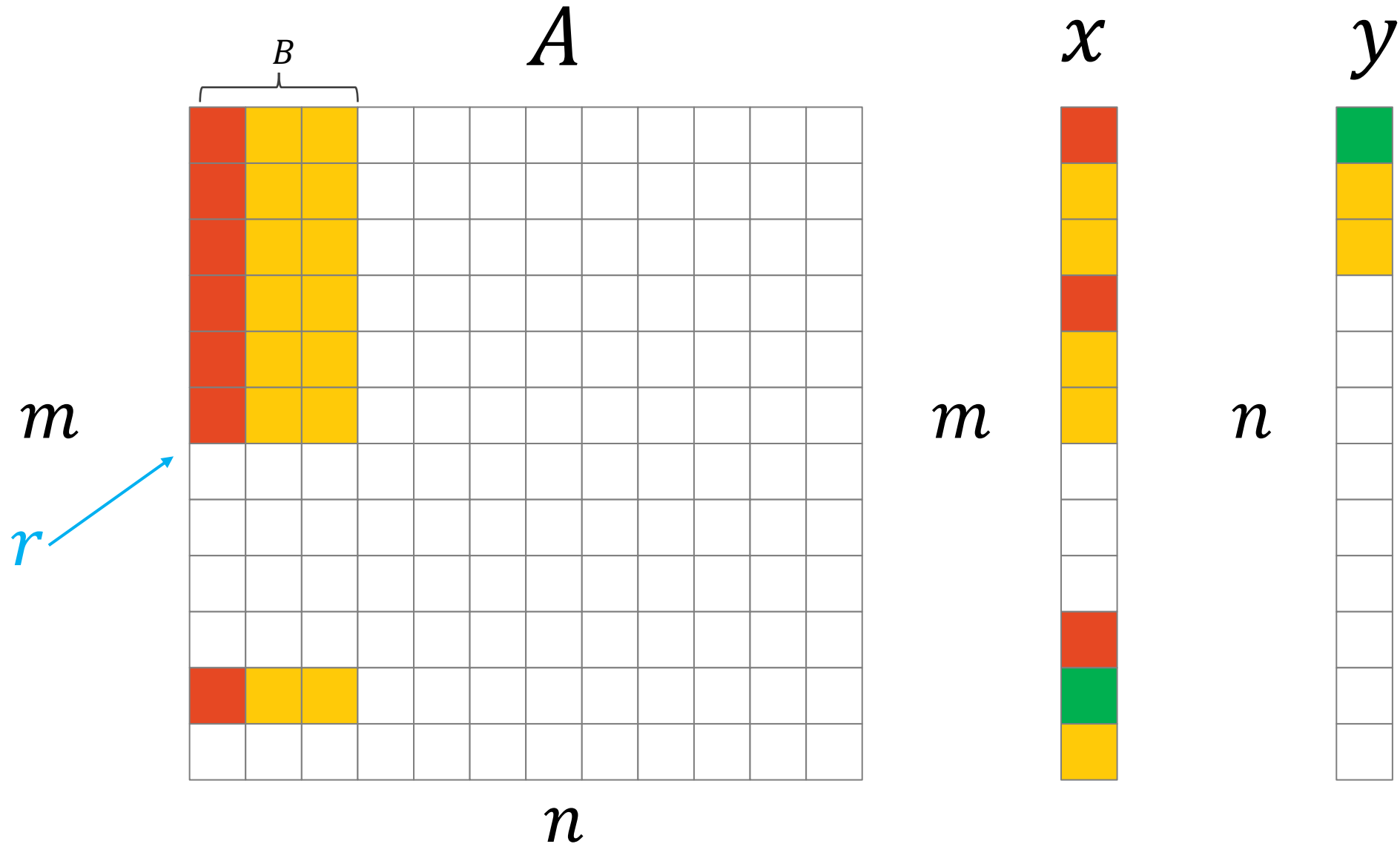
Transpose Matrix Vector Product

$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$



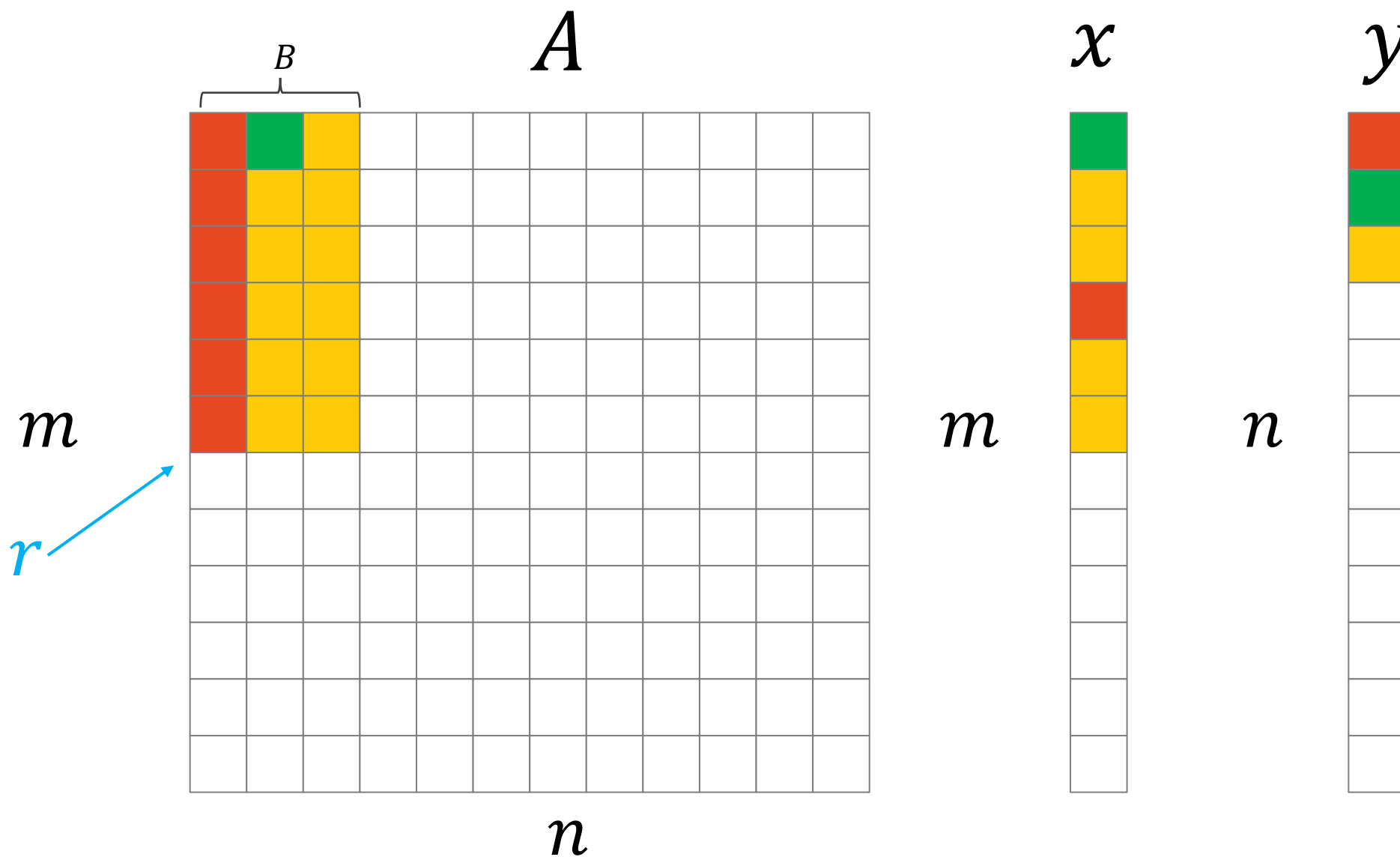
Transpose Matrix Vector Product

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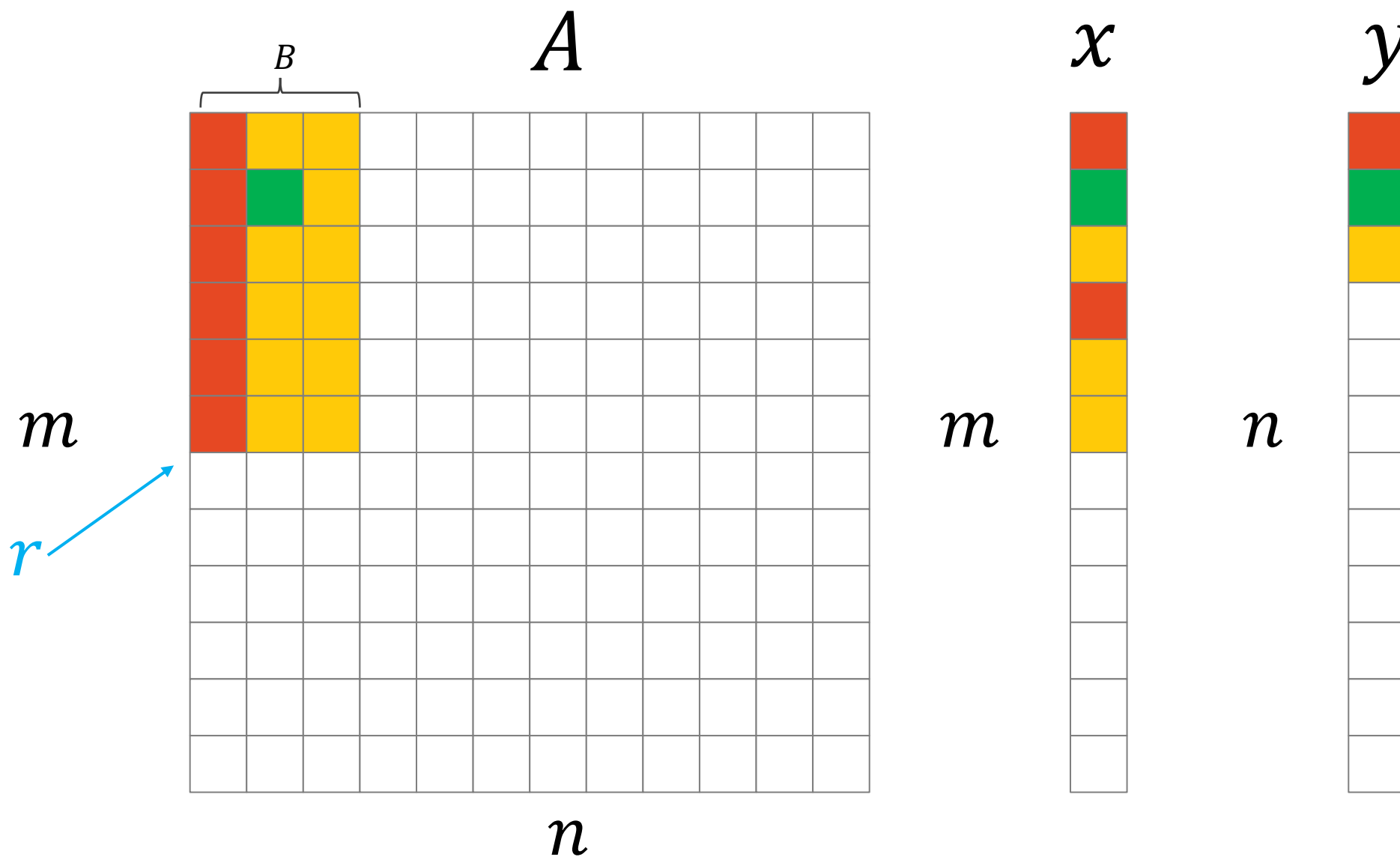
Transpose Matrix Vector Product

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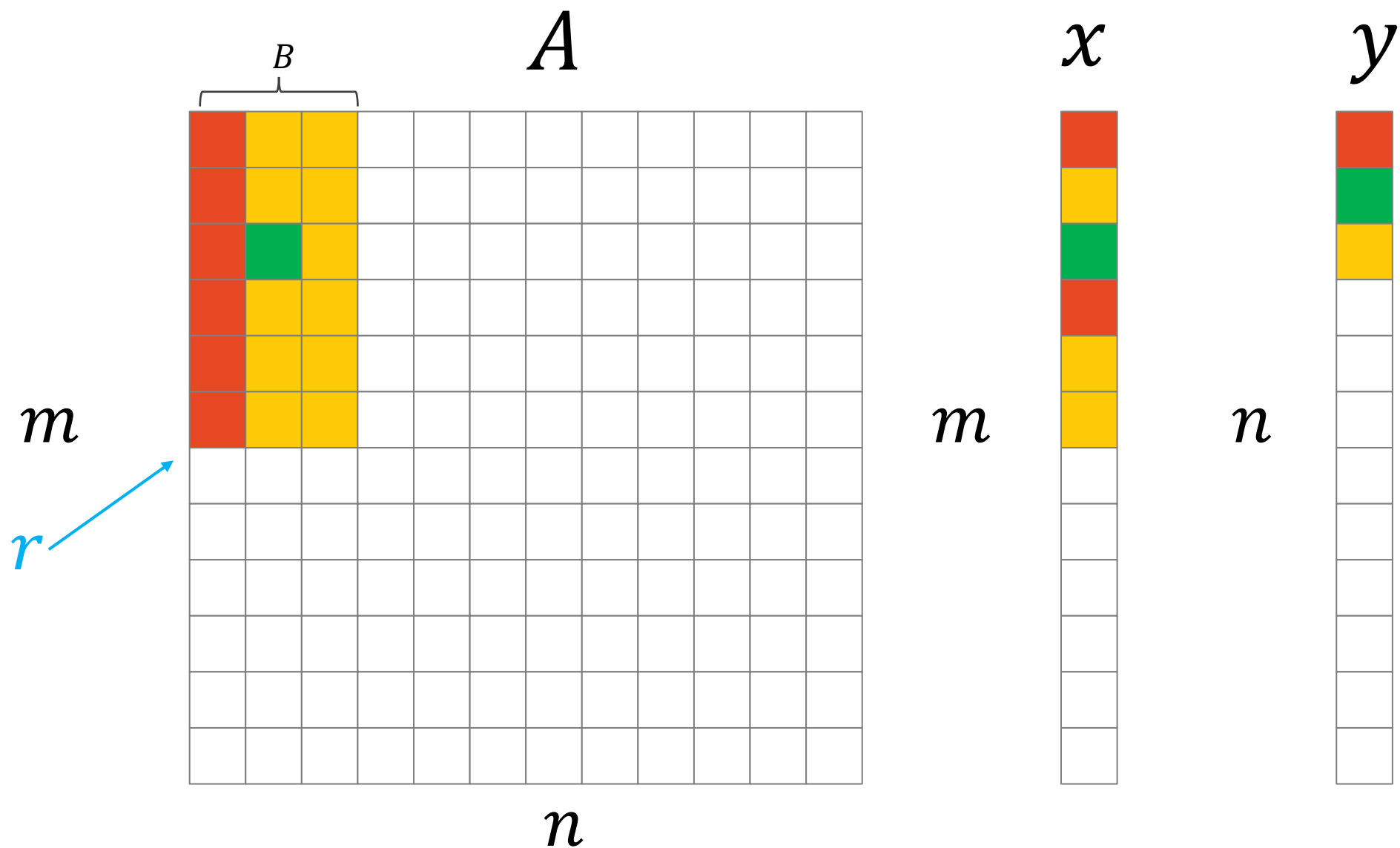
Transpose Matrix Vector Product

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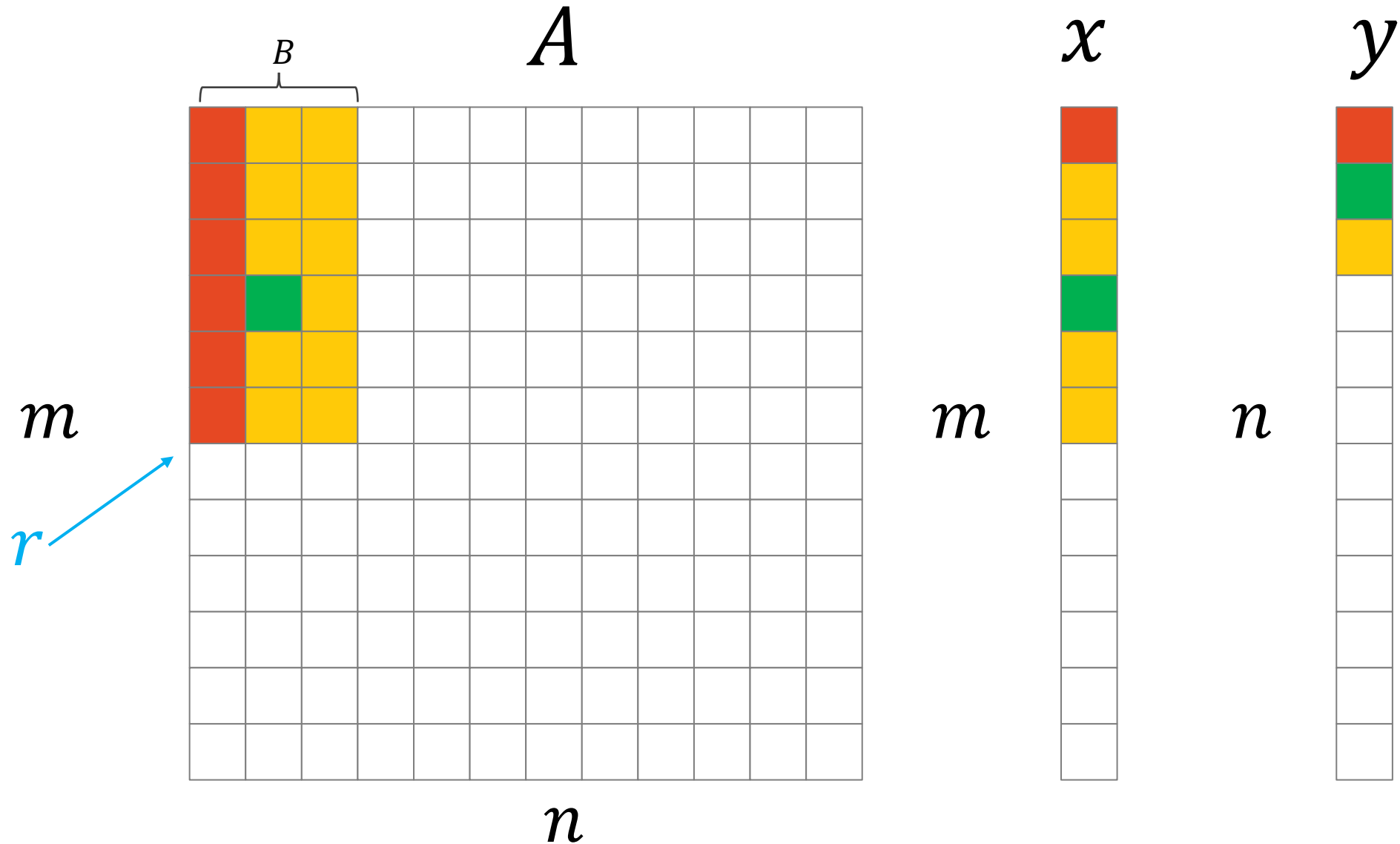
Transpose Matrix Vector Product

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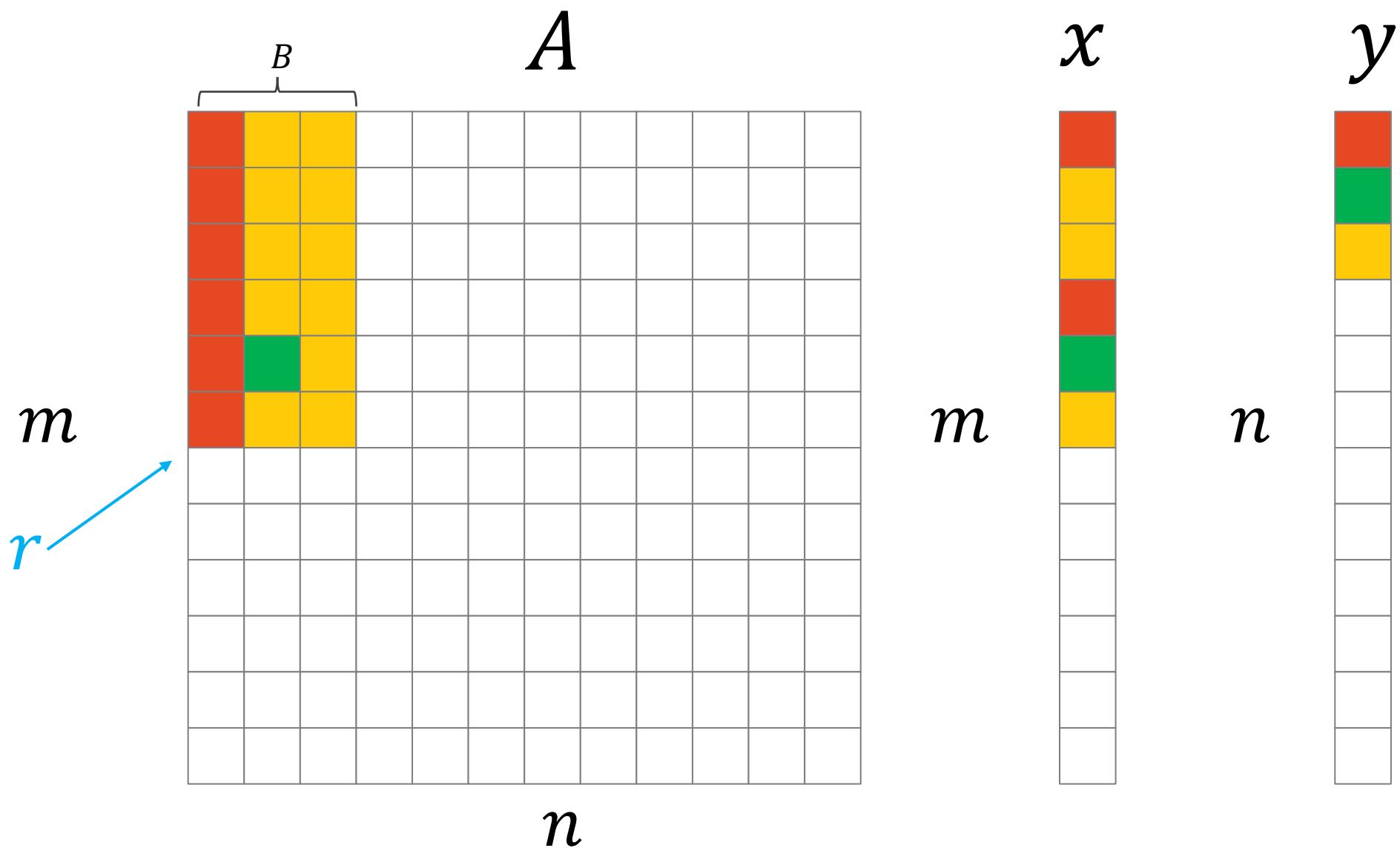
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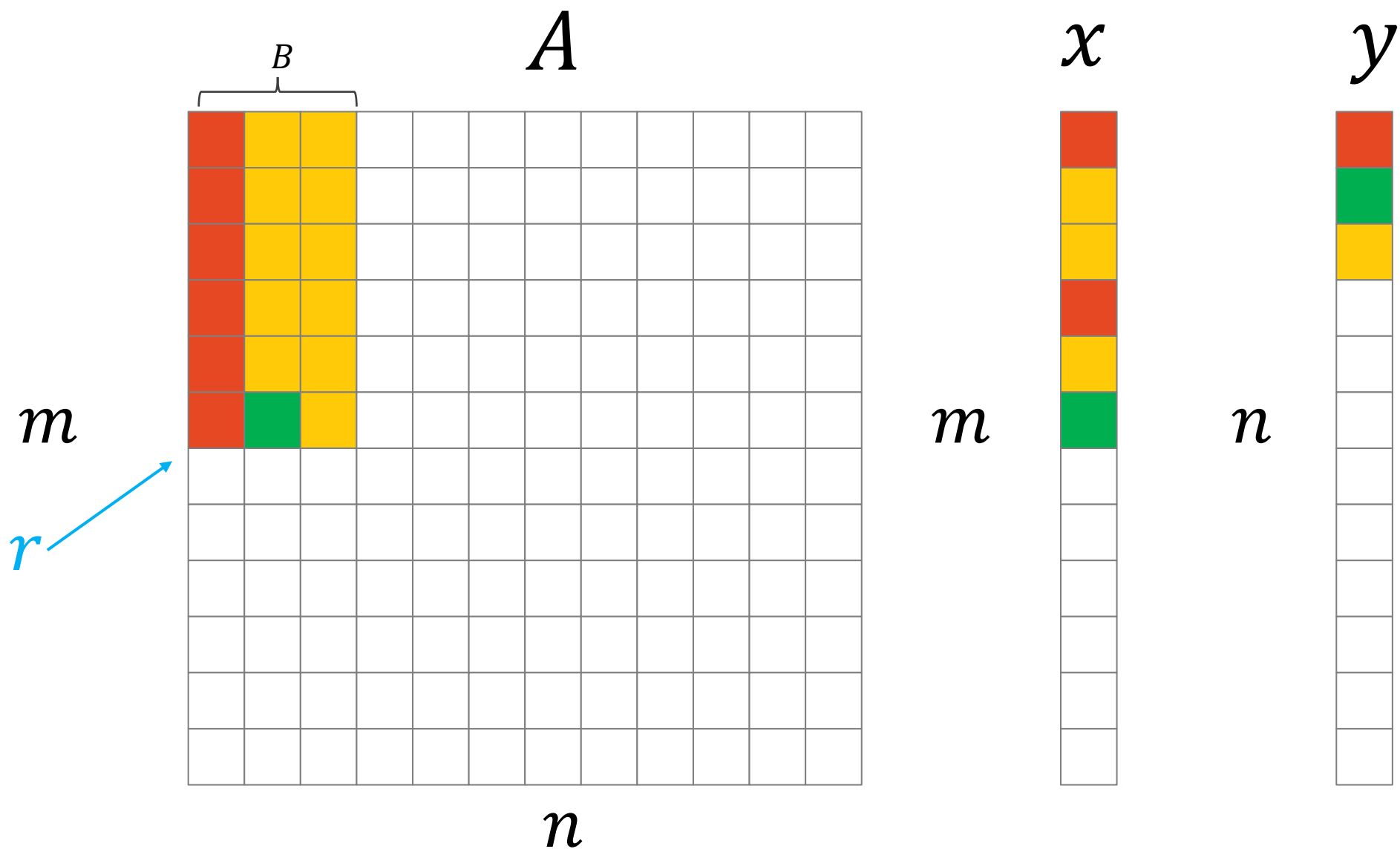
Transpose Matrix Vector Product

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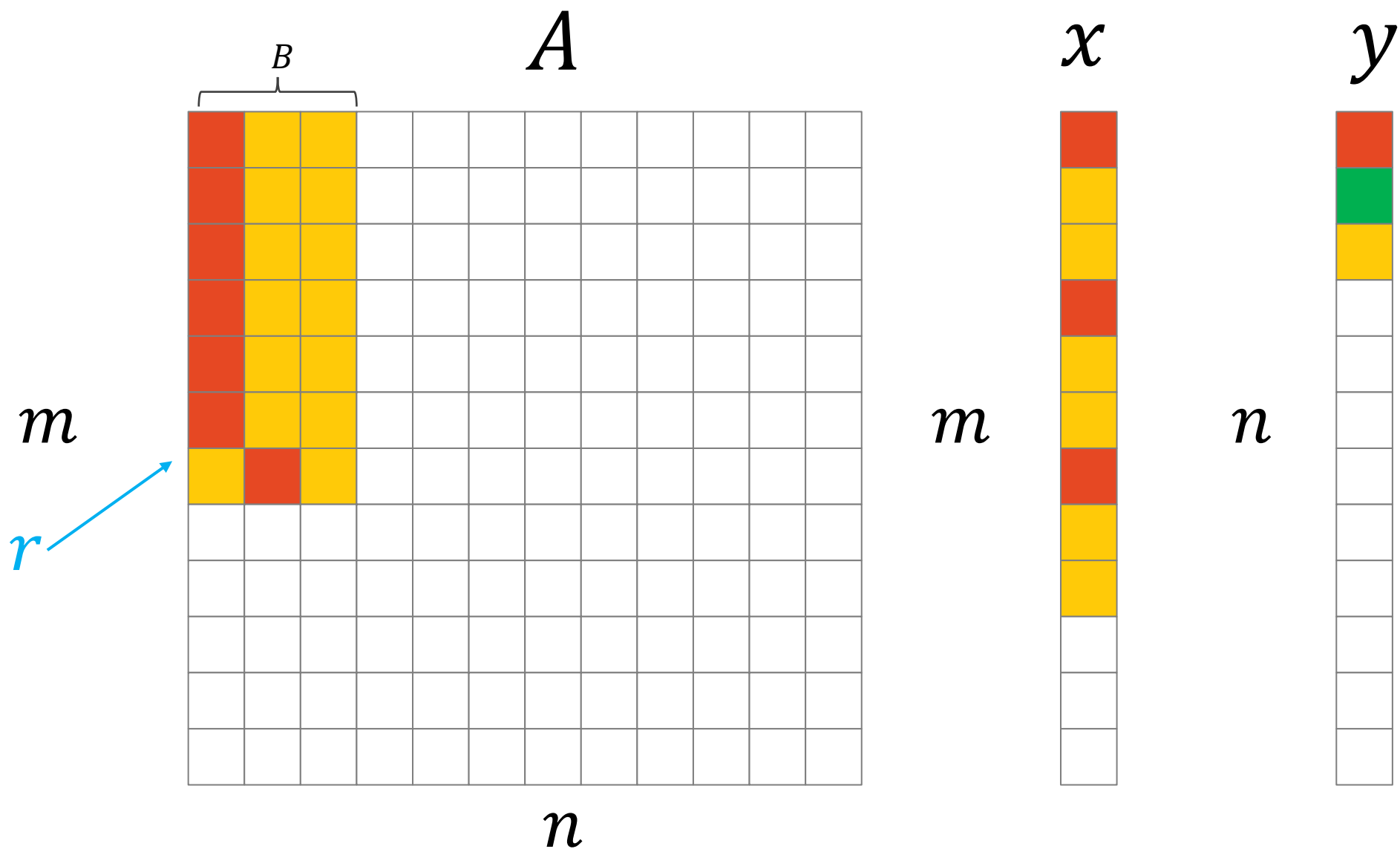
Transpose Matrix Vector Product

$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$



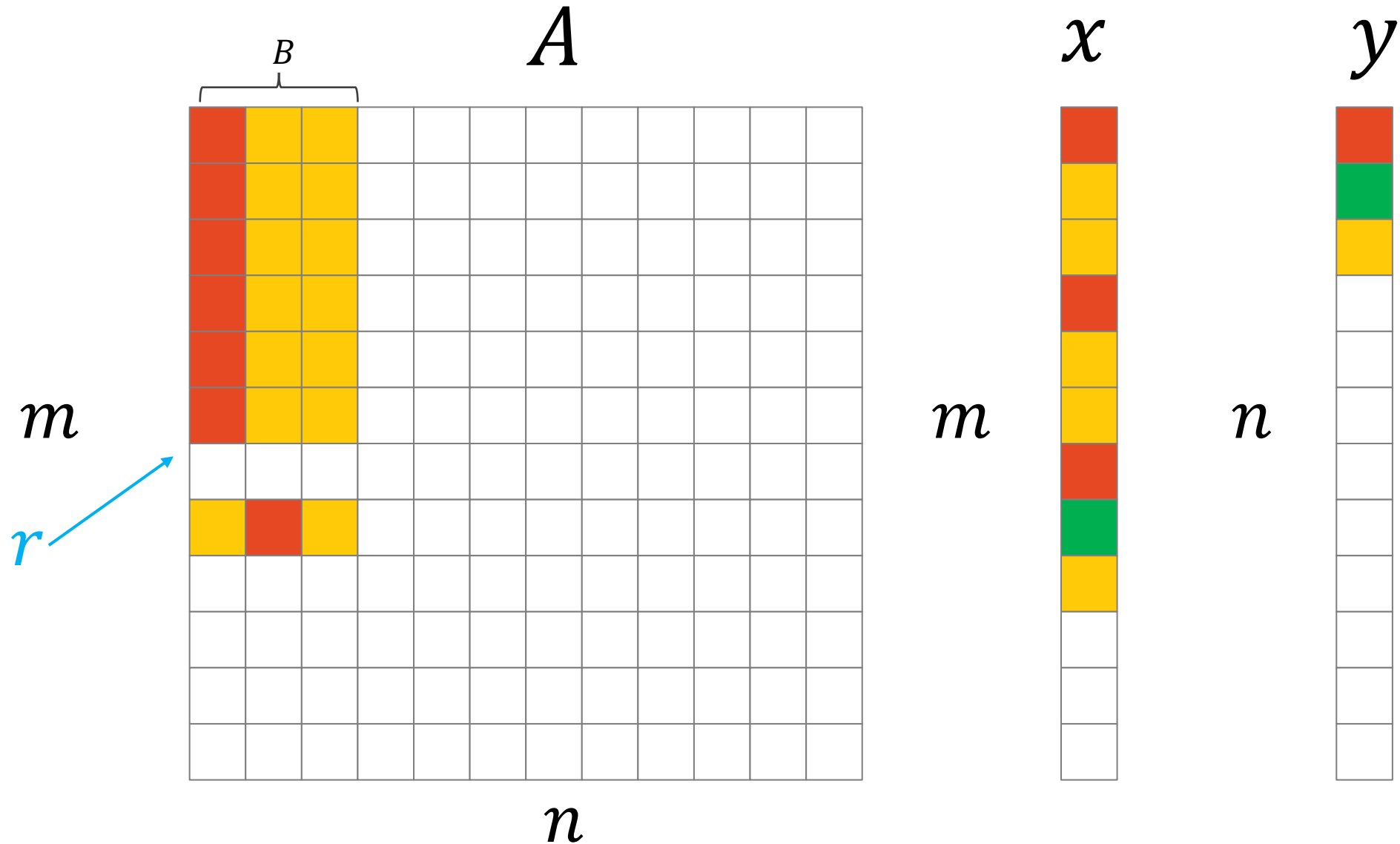
Transpose Matrix Vector Product

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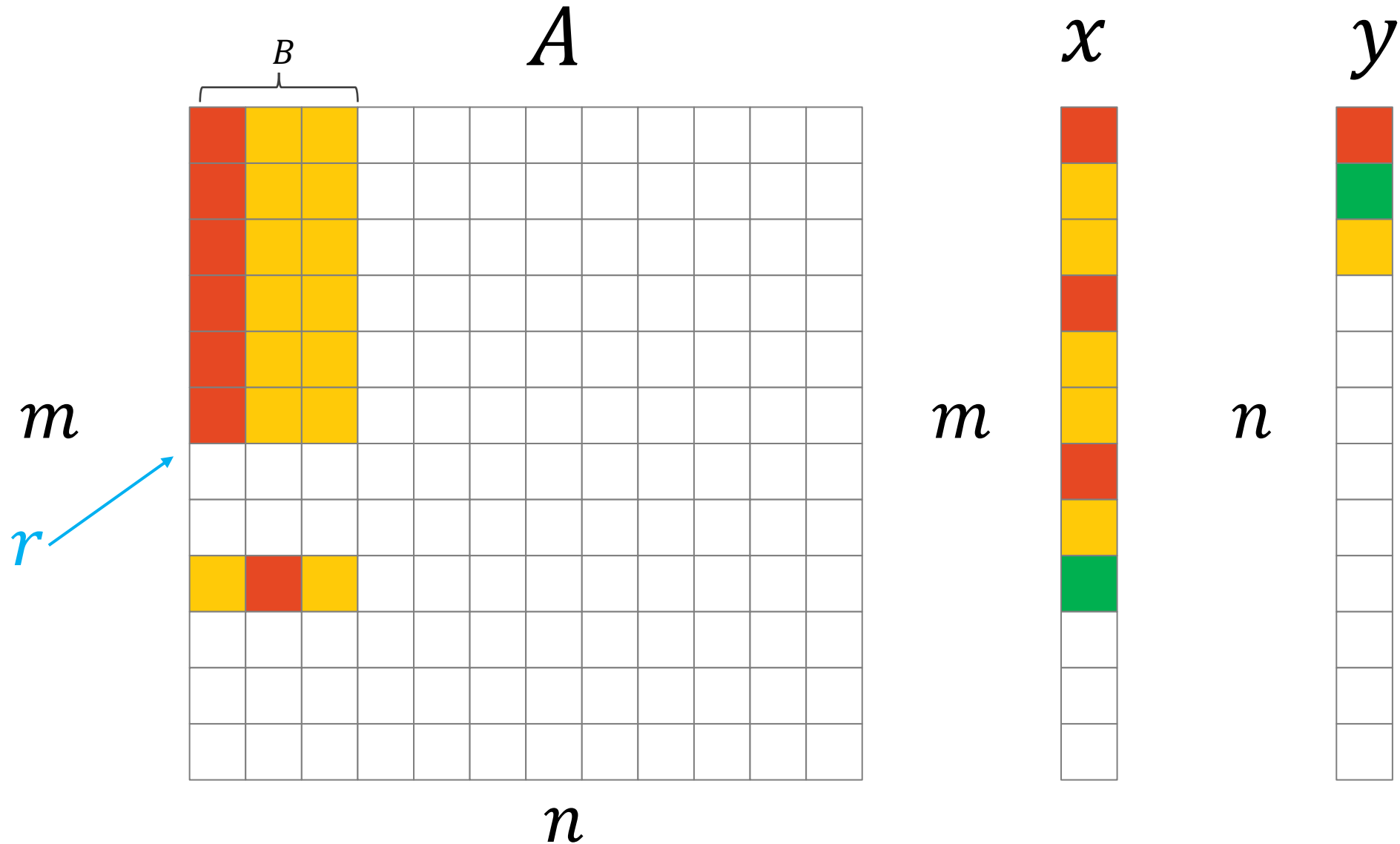
Transpose Matrix Vector Product

$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$



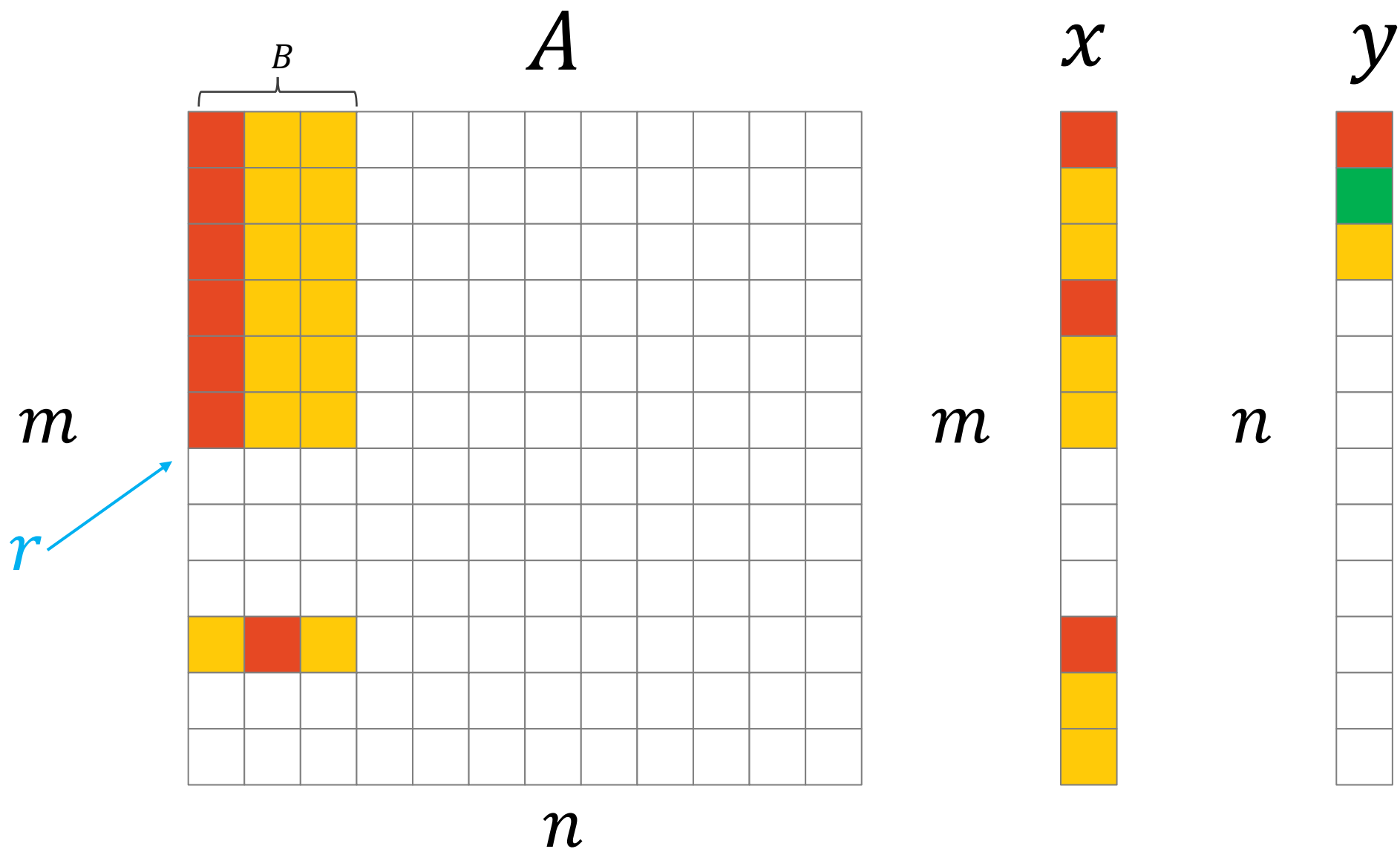
Transpose Matrix Vector Product

$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$



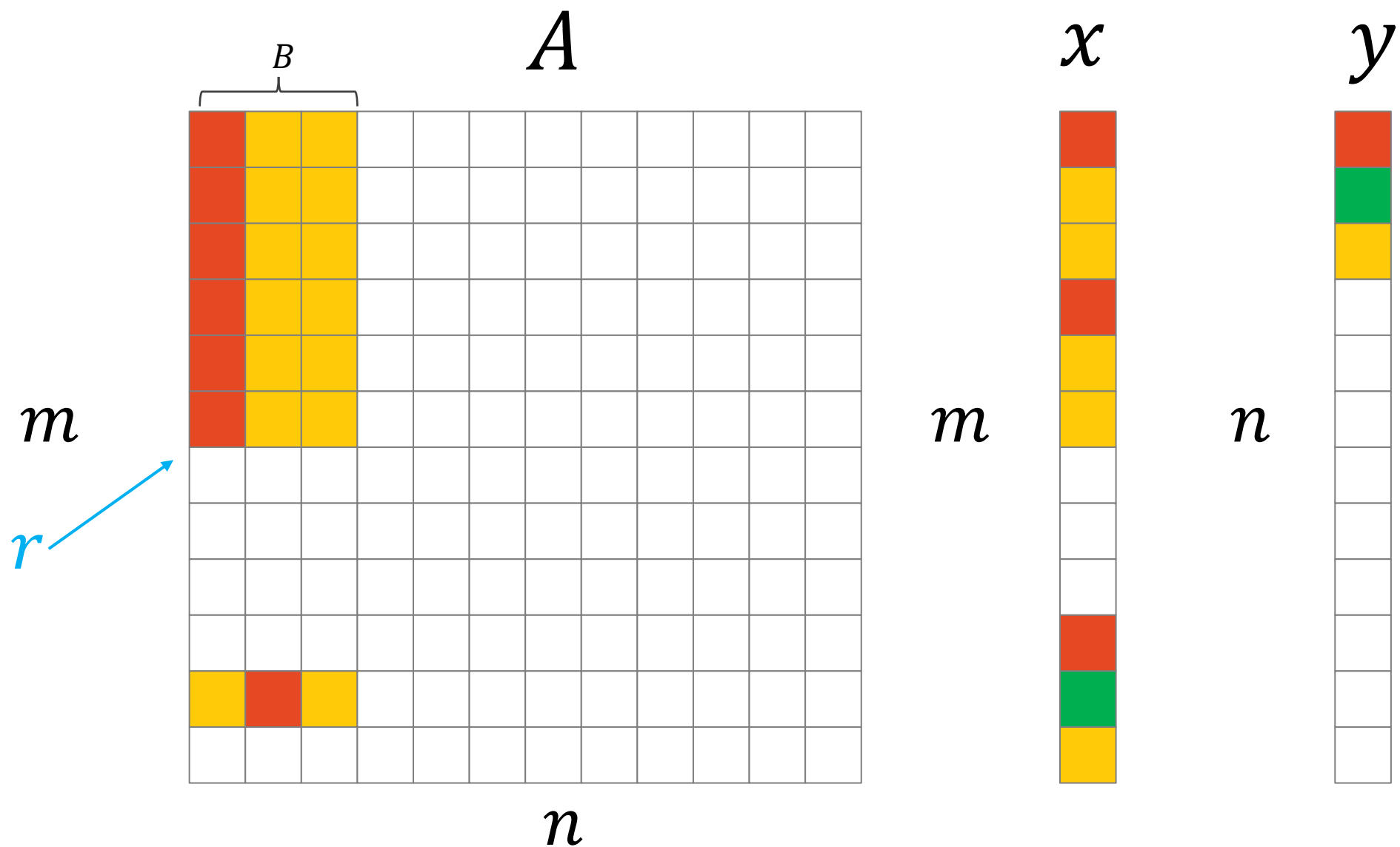
Transpose Matrix Vector Product

$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$



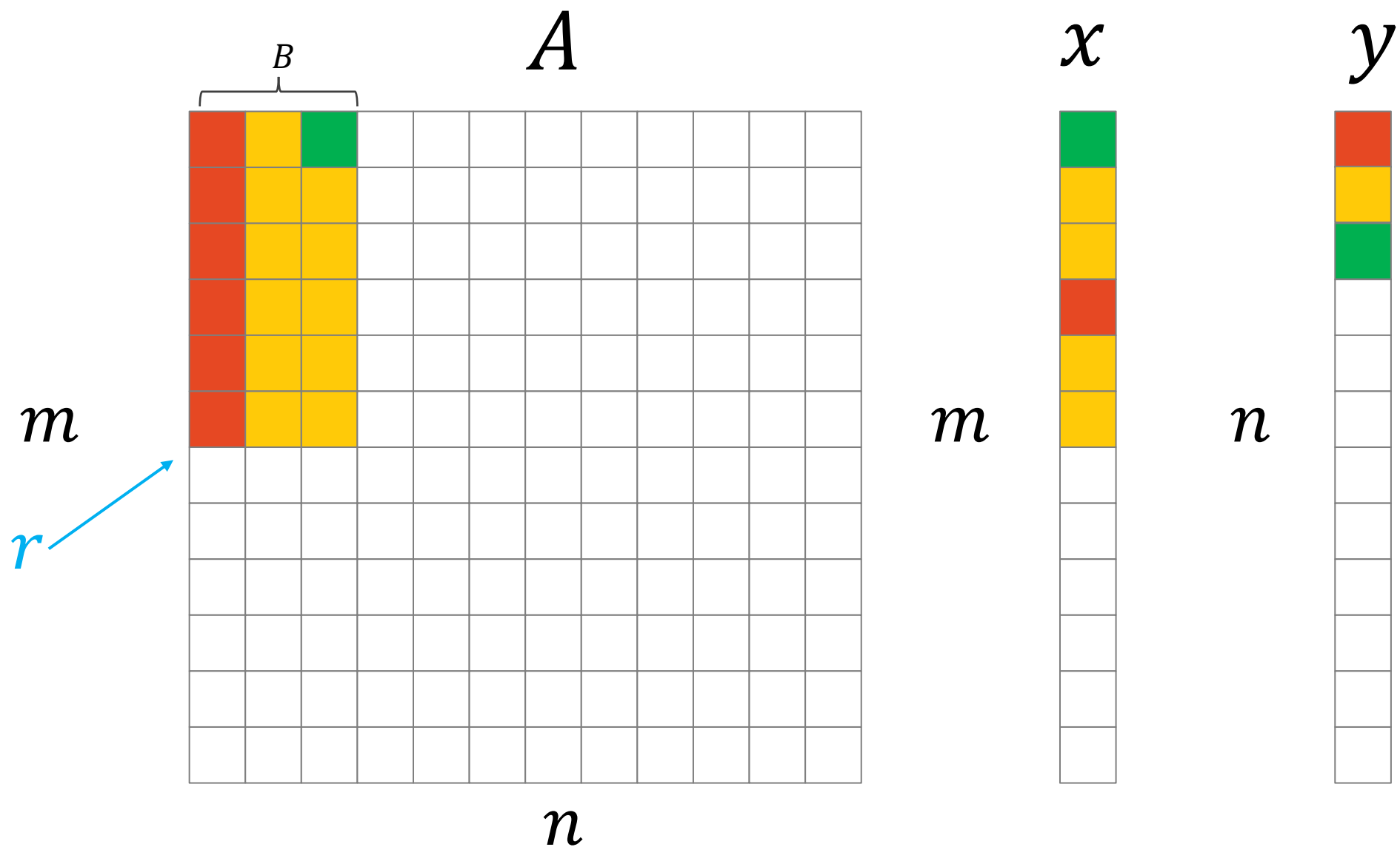
Transpose Matrix Vector Product

$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$



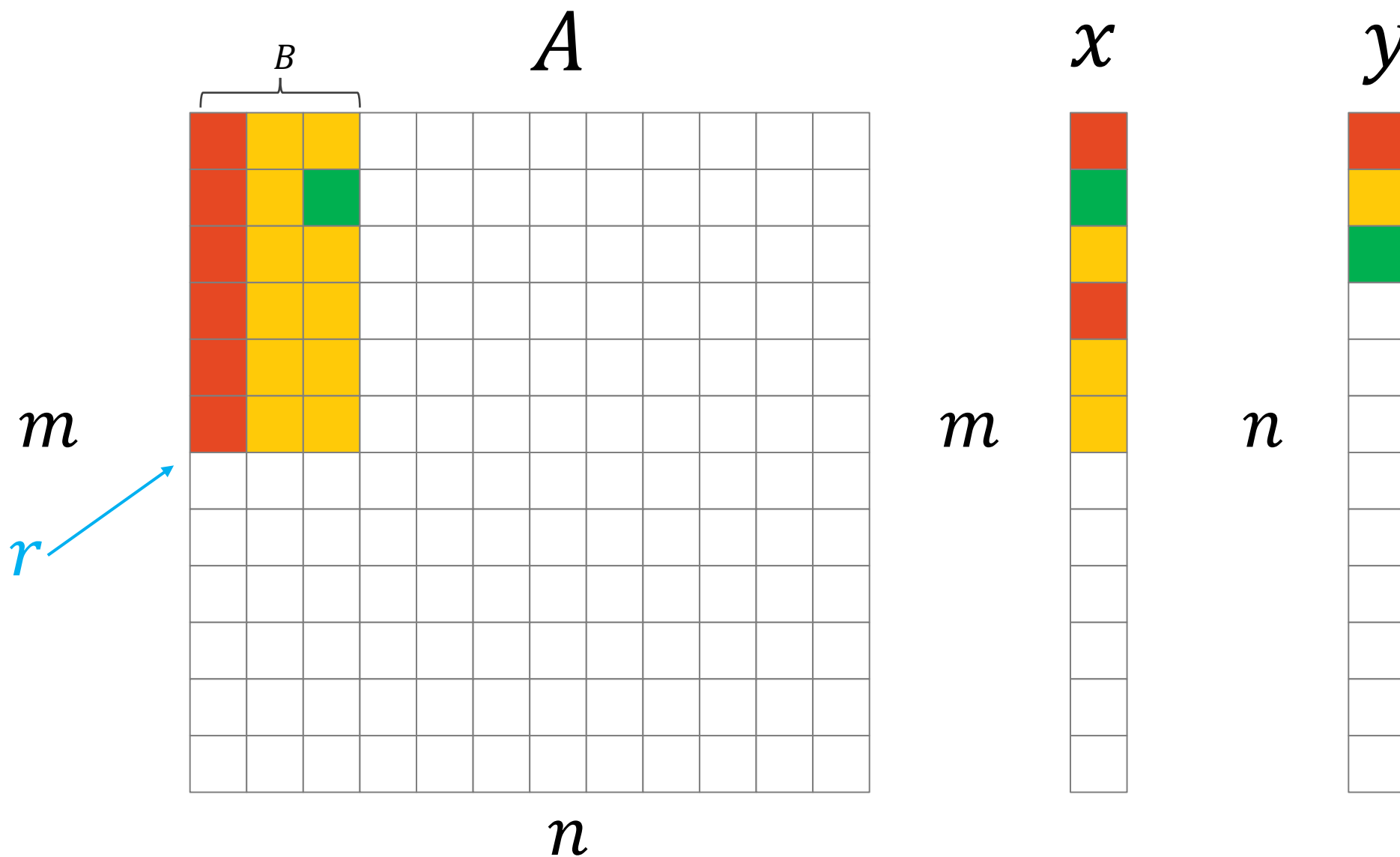
Transpose Matrix Vector Product

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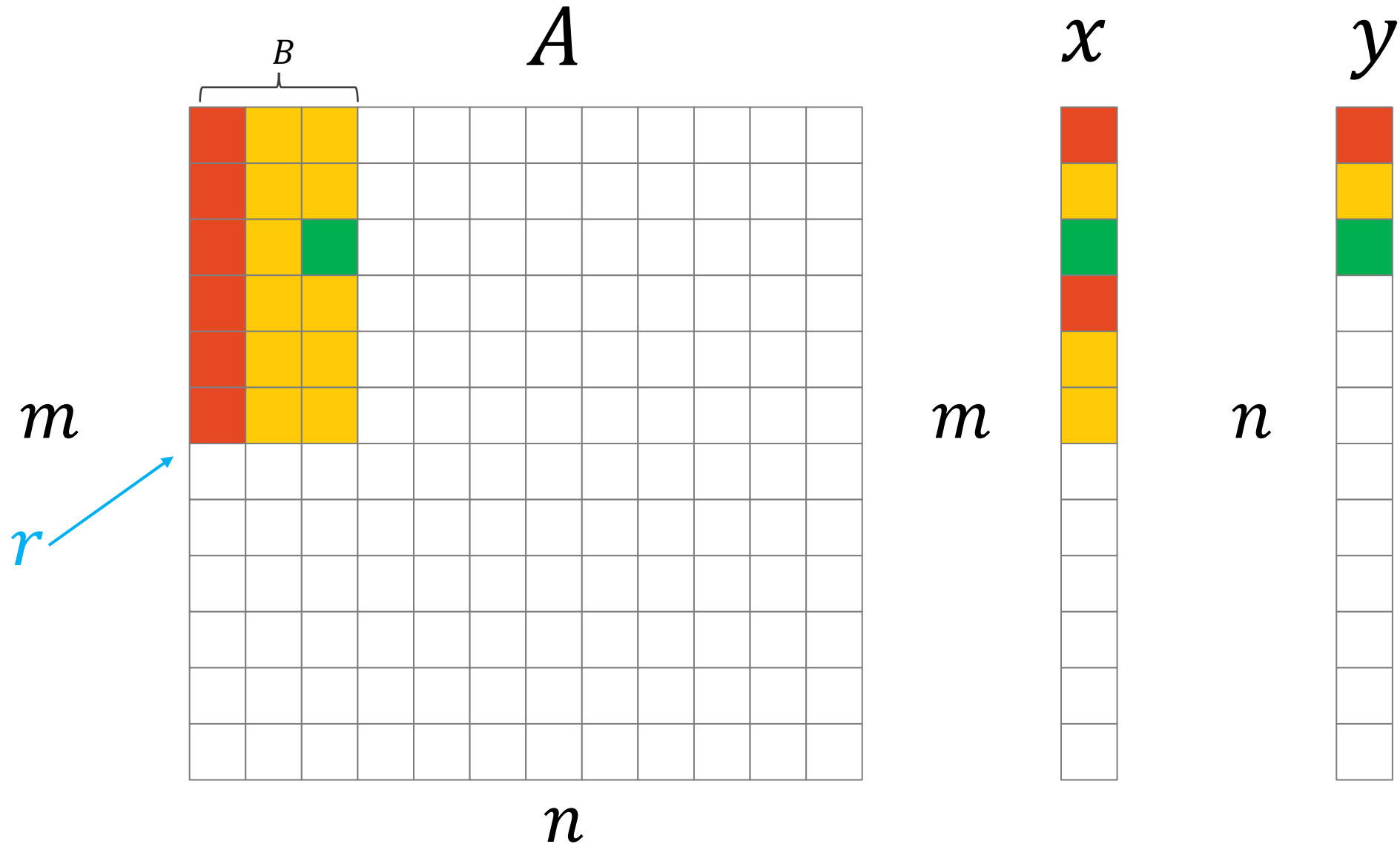
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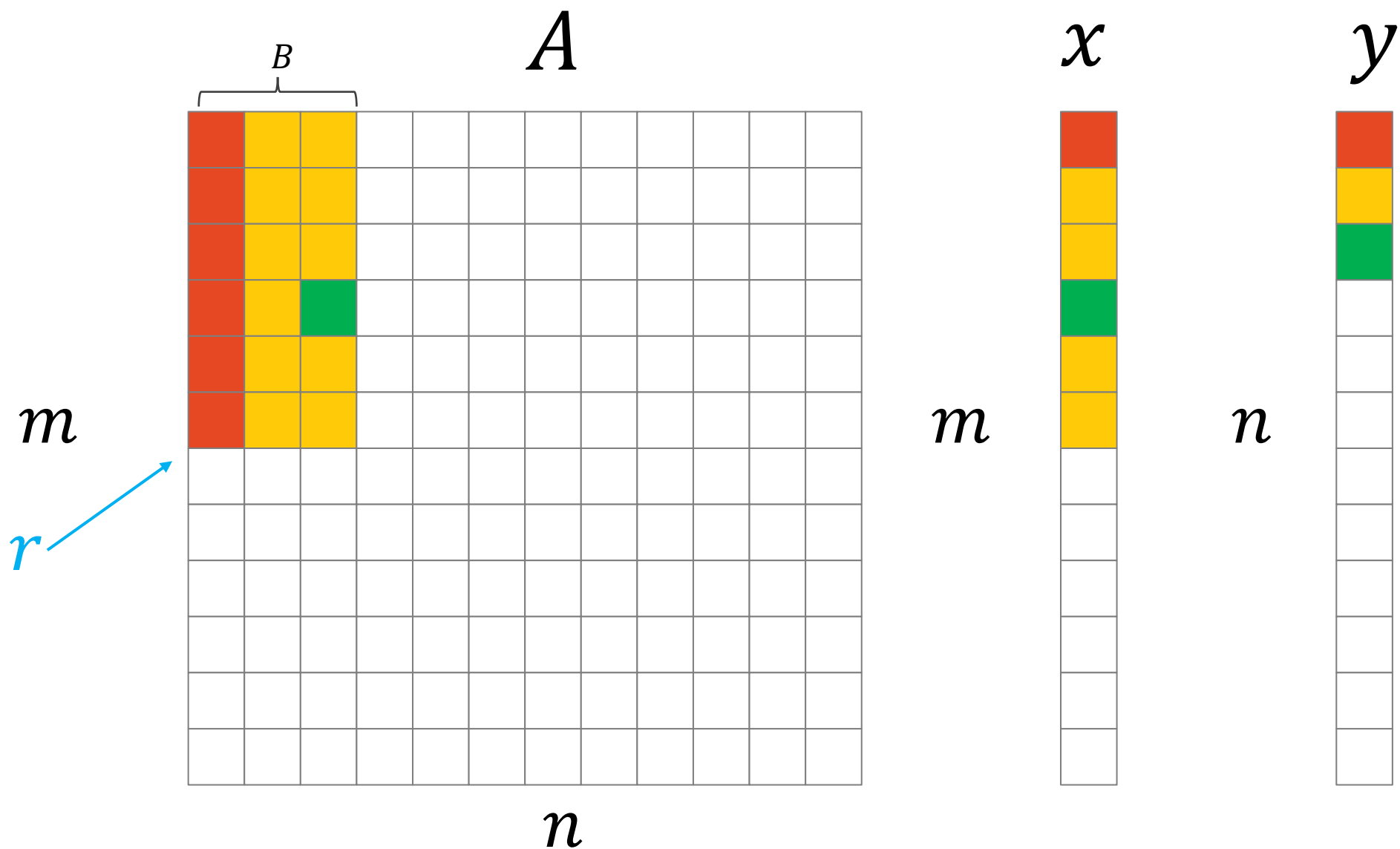
Transpose Matrix Vector Product

$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$



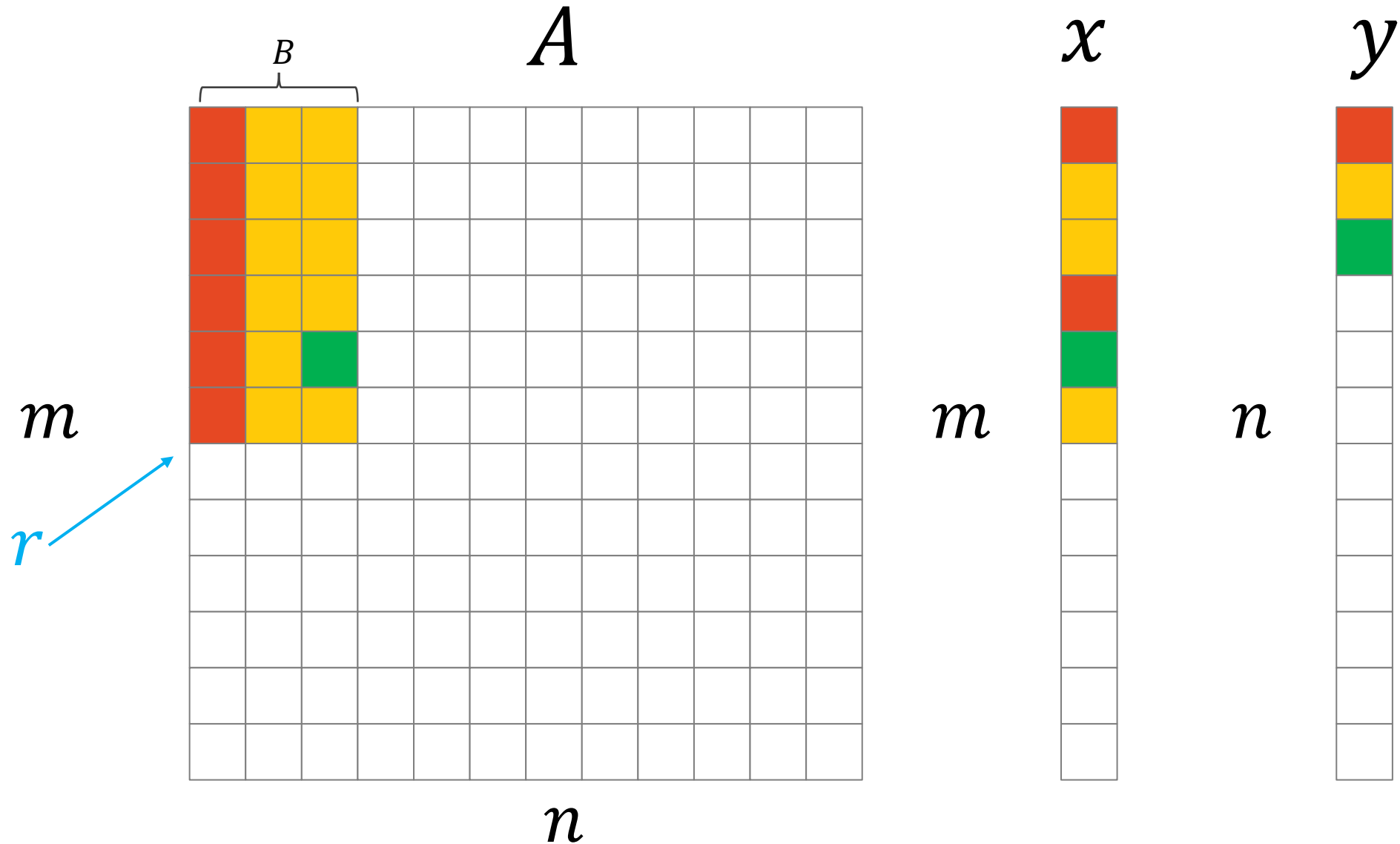
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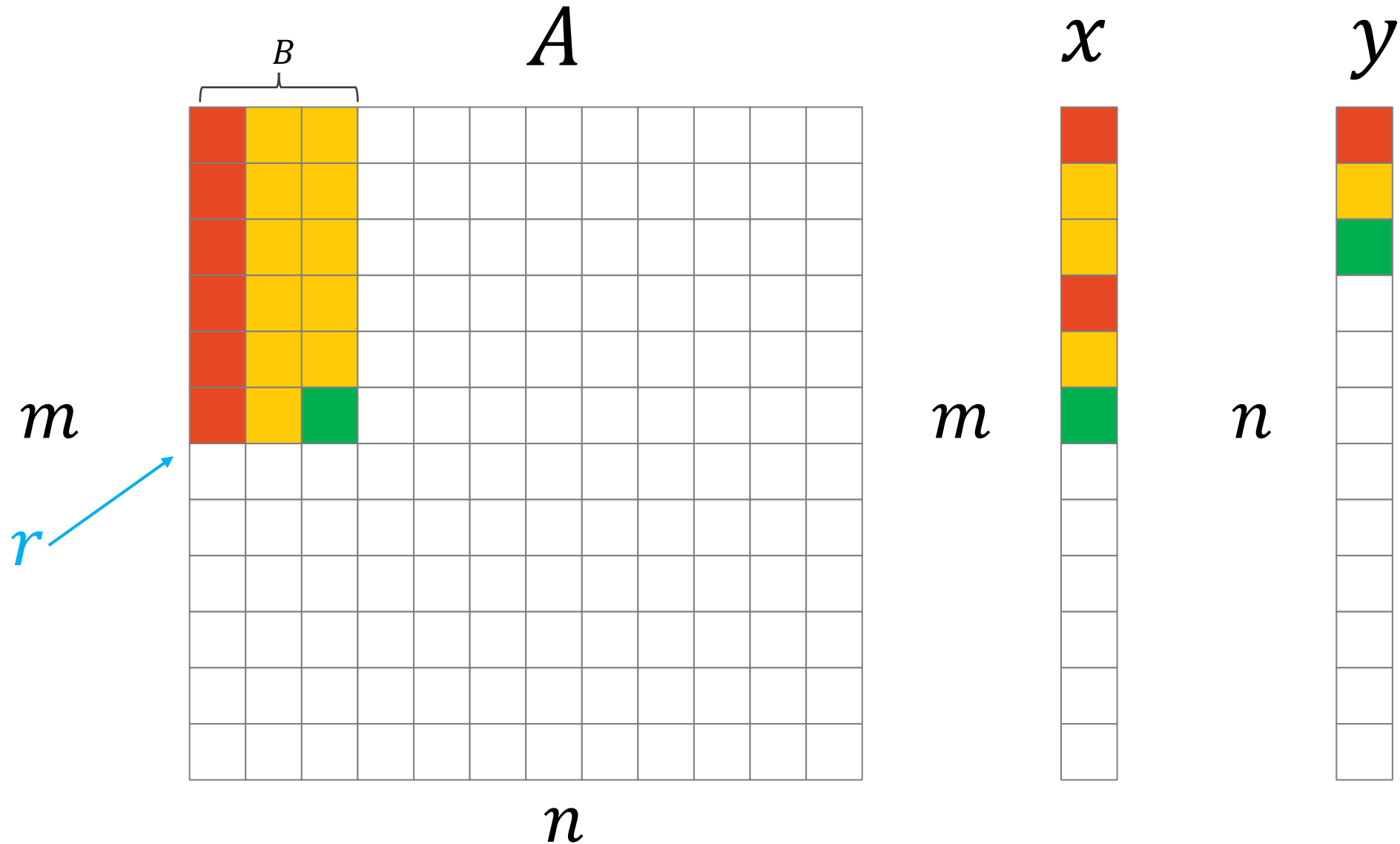
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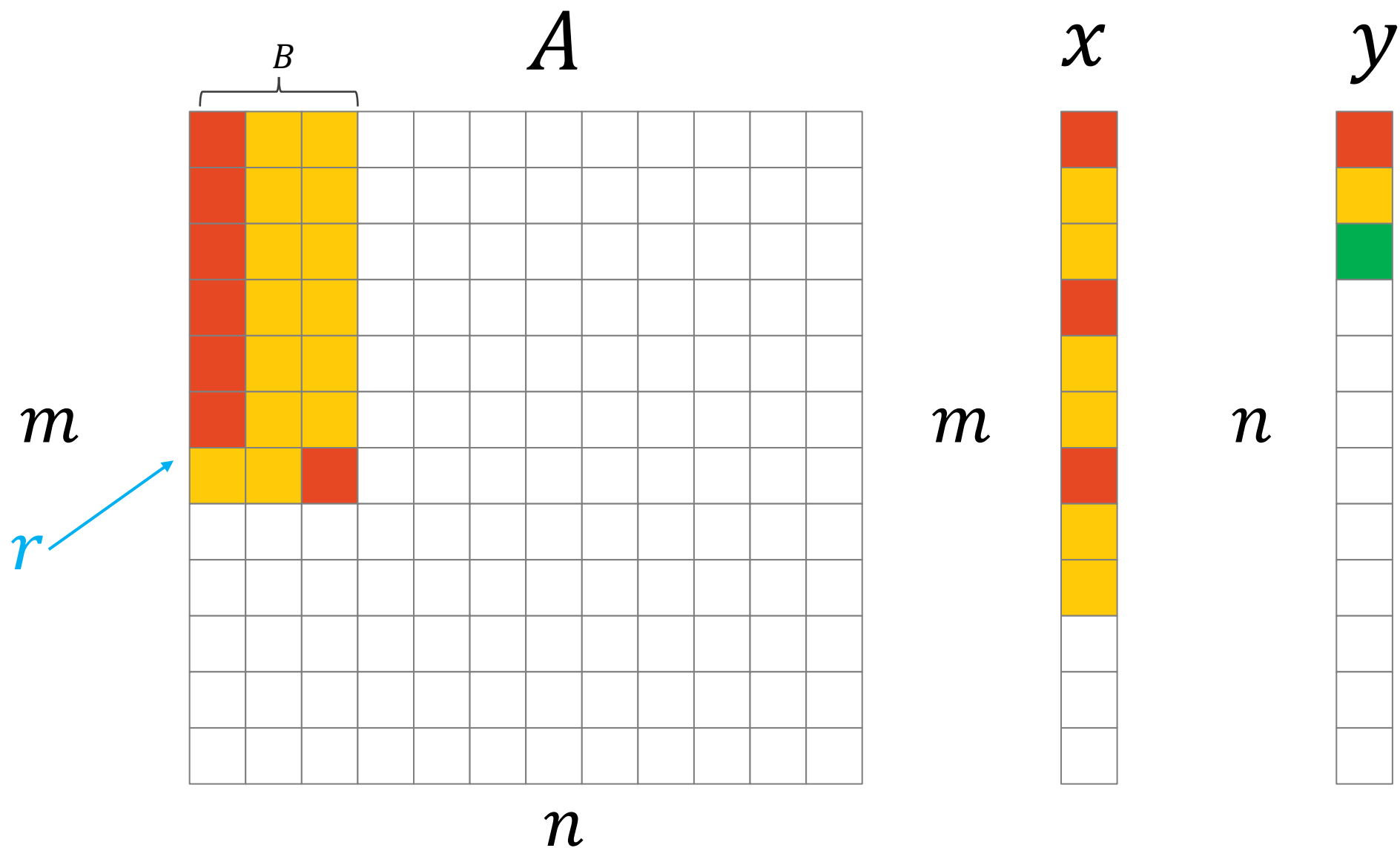
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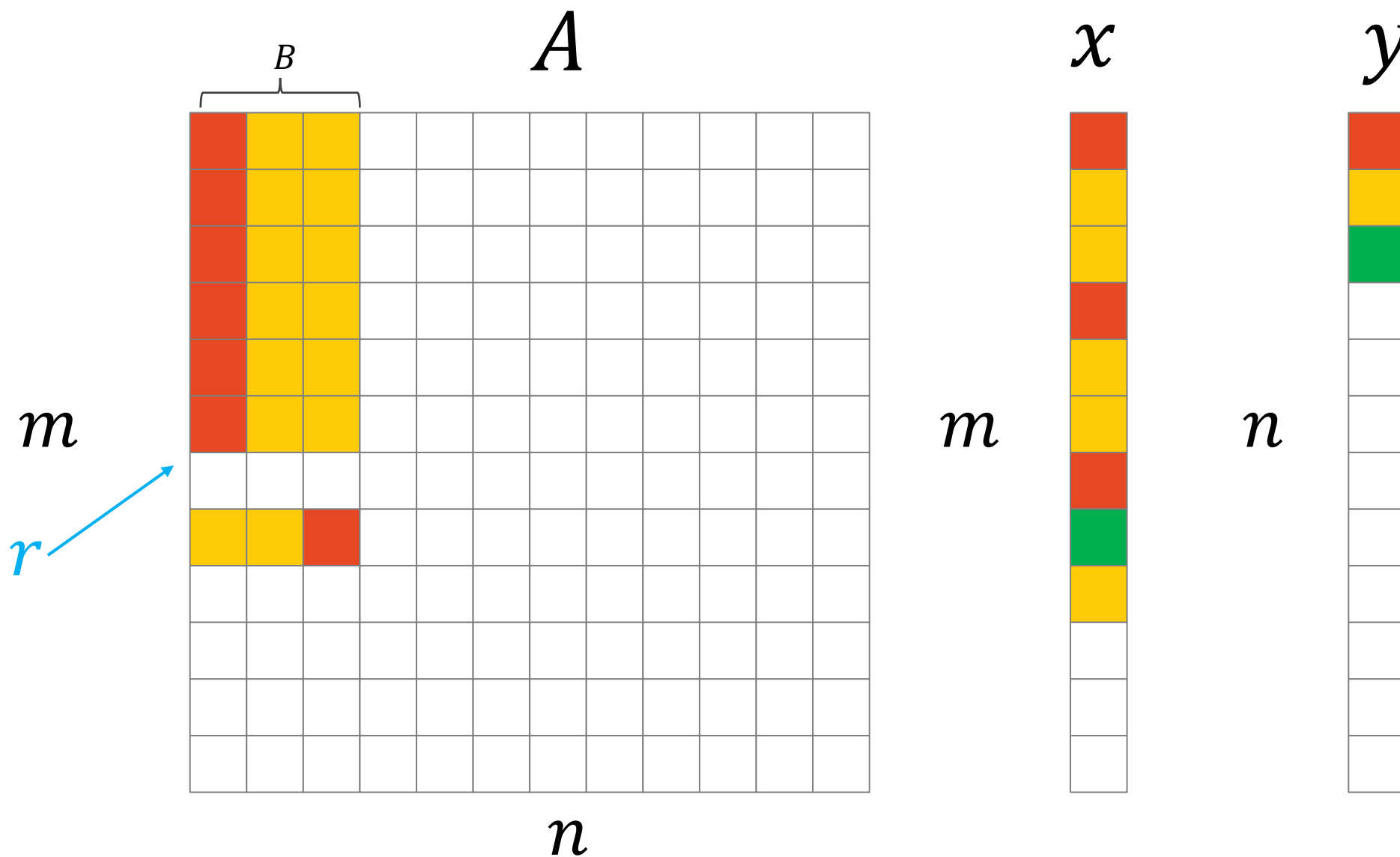
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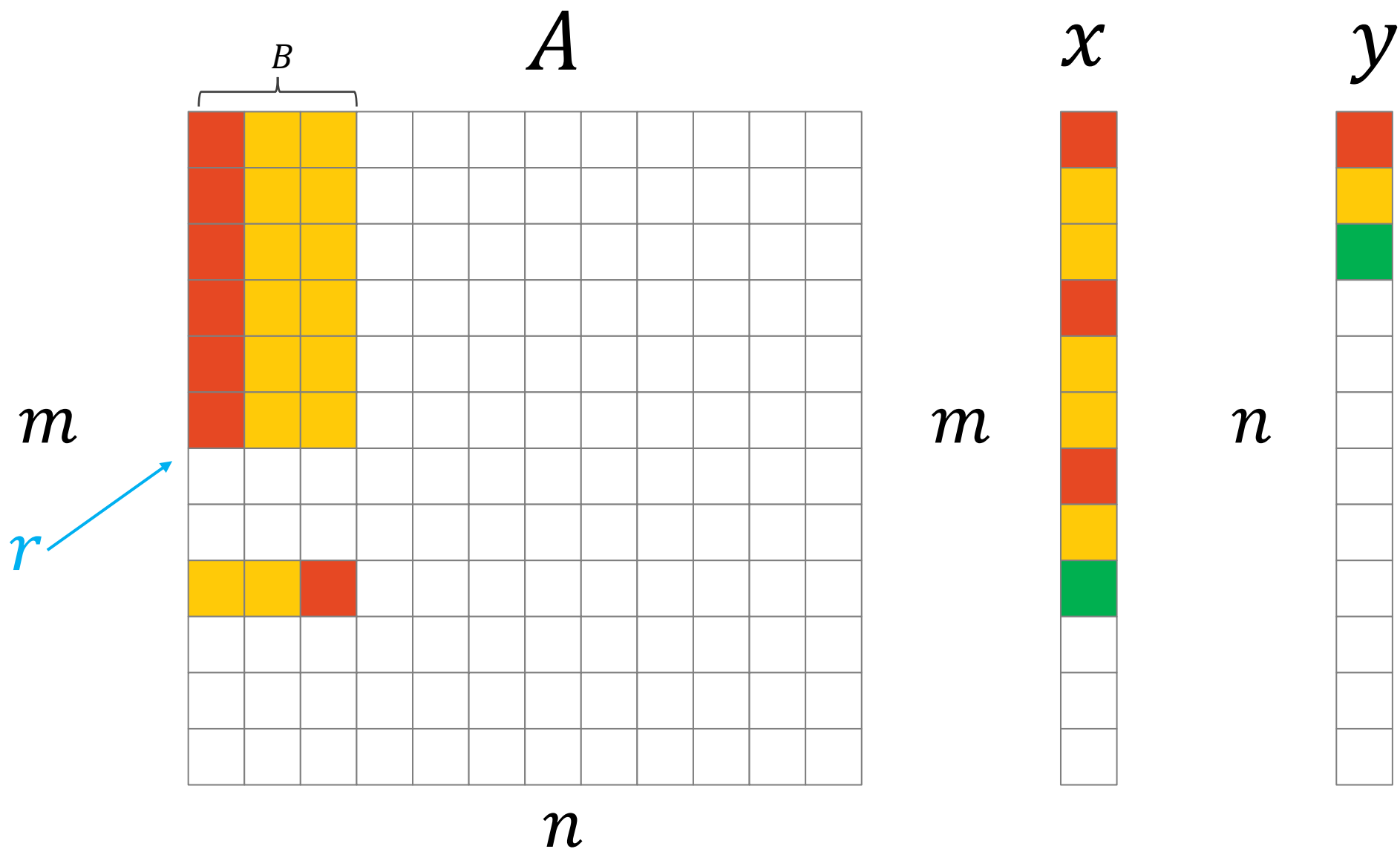
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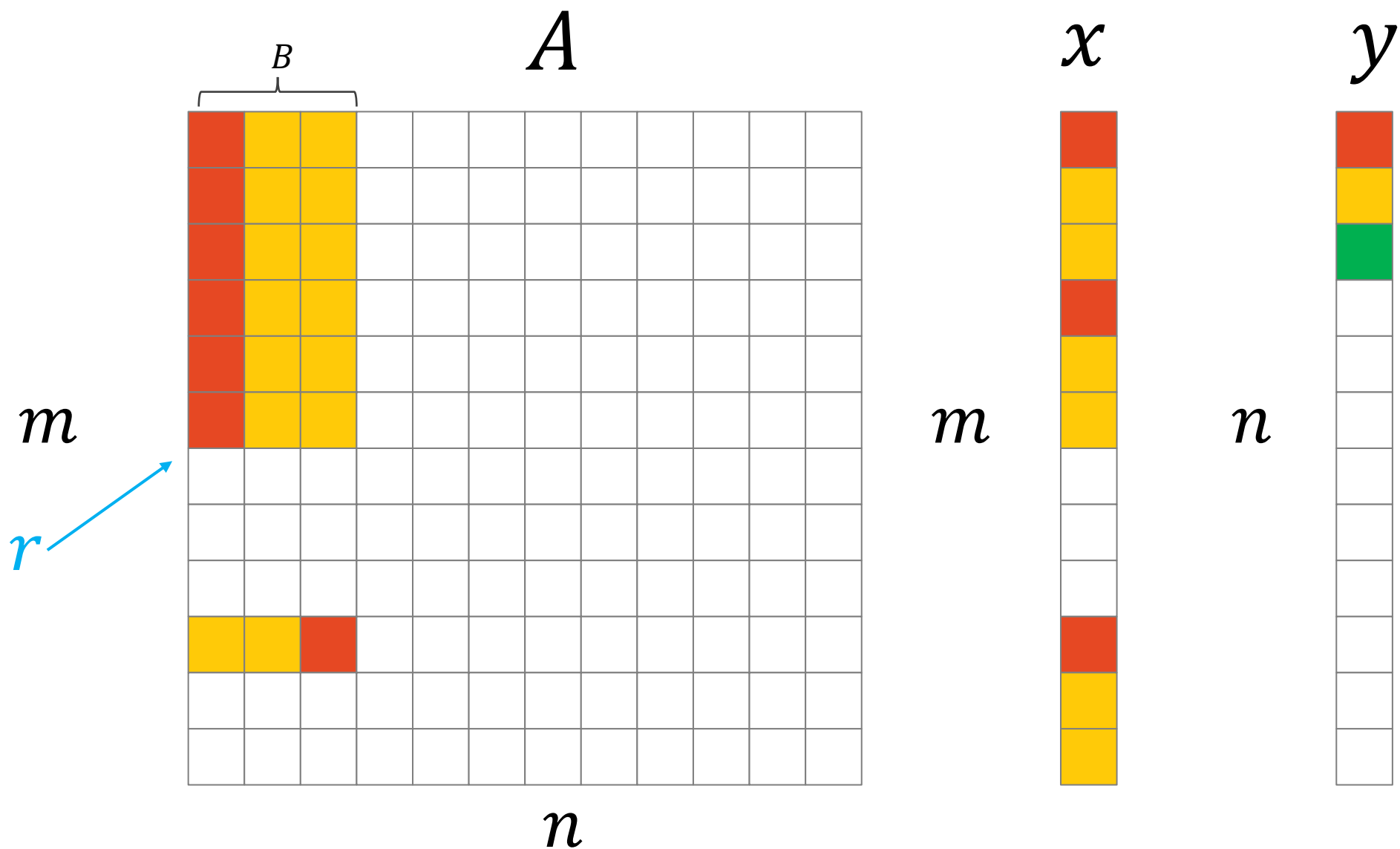
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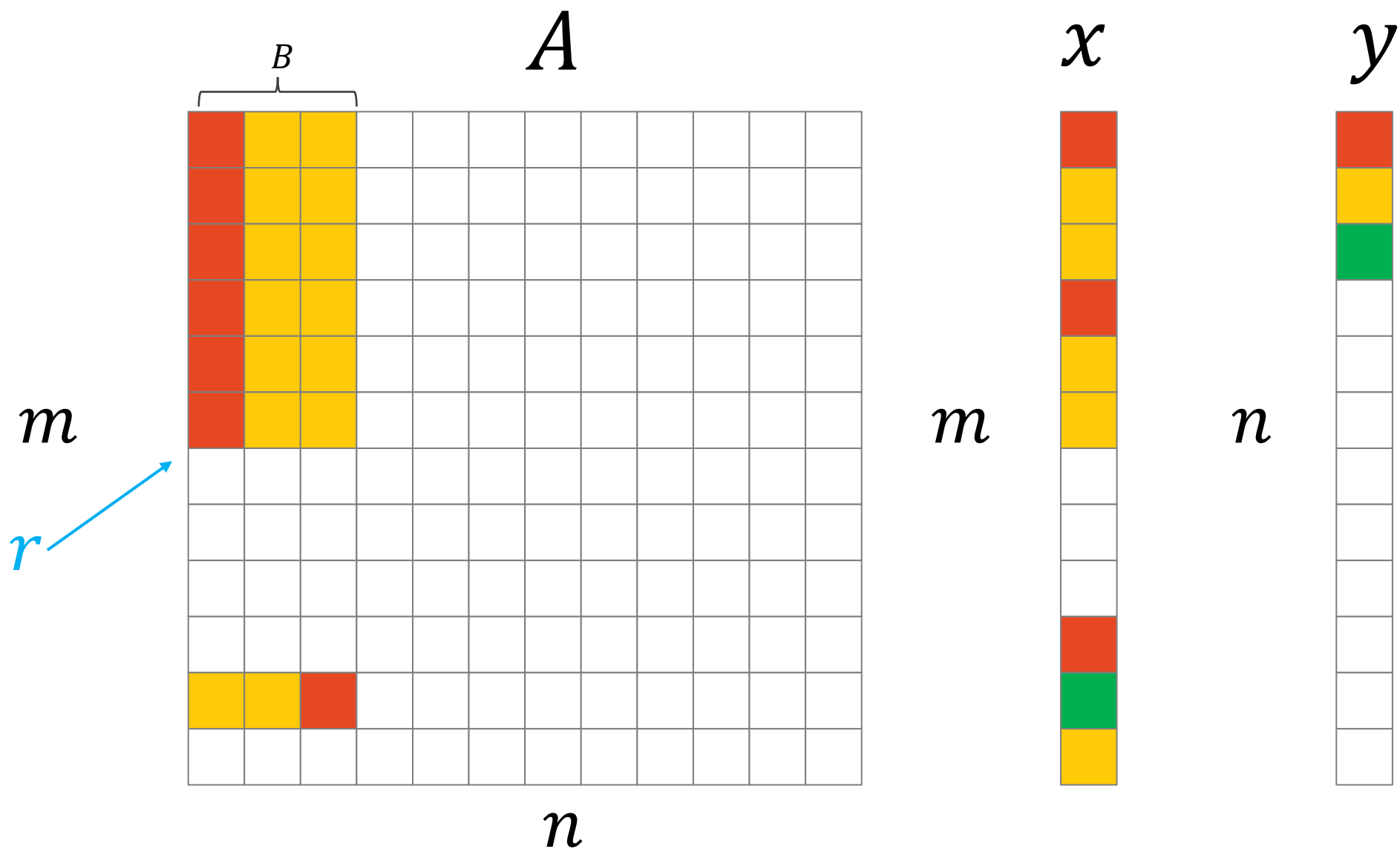
Transpose Matrix Vector Product

$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$



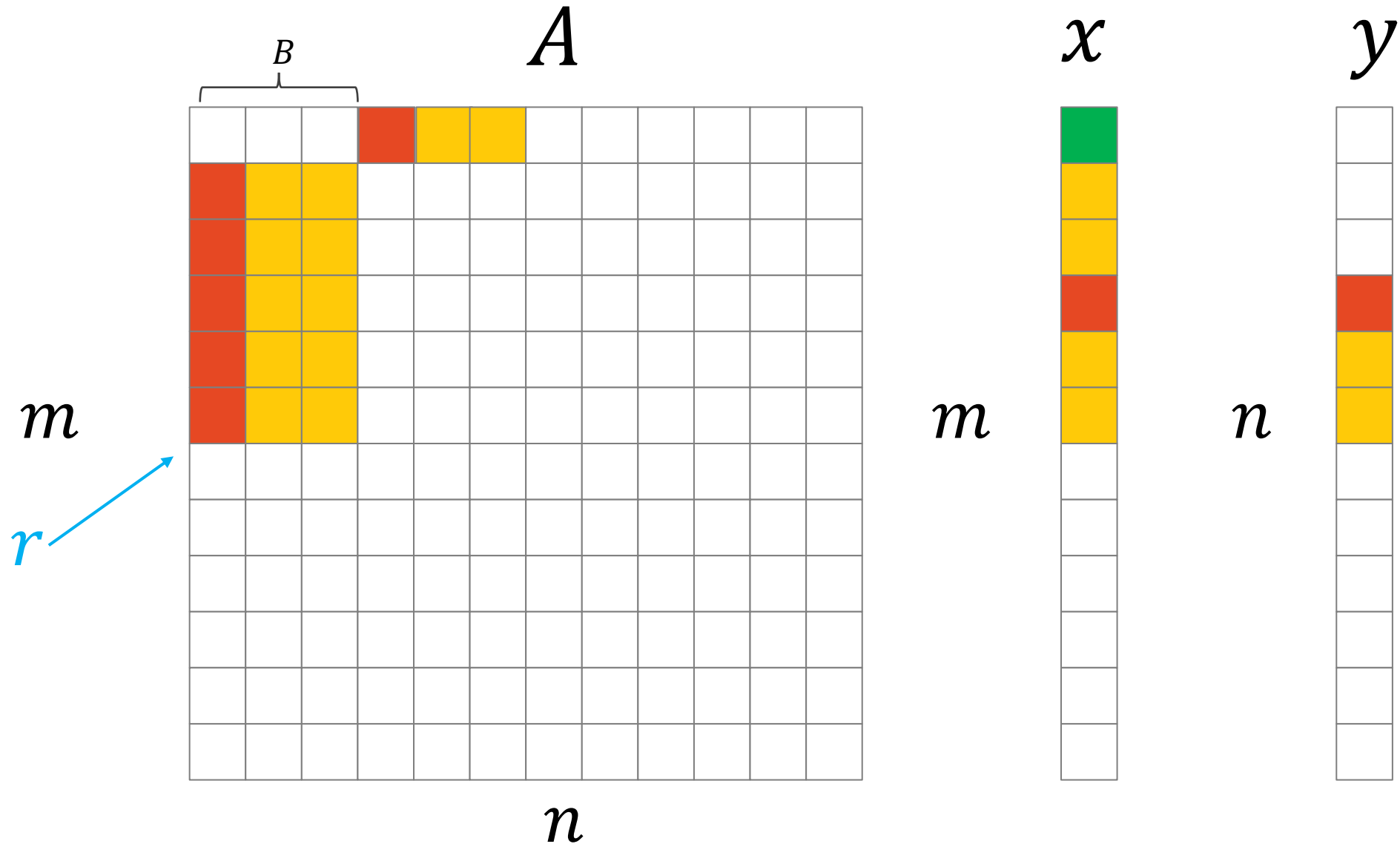
Transpose Matrix Vector Product

$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$



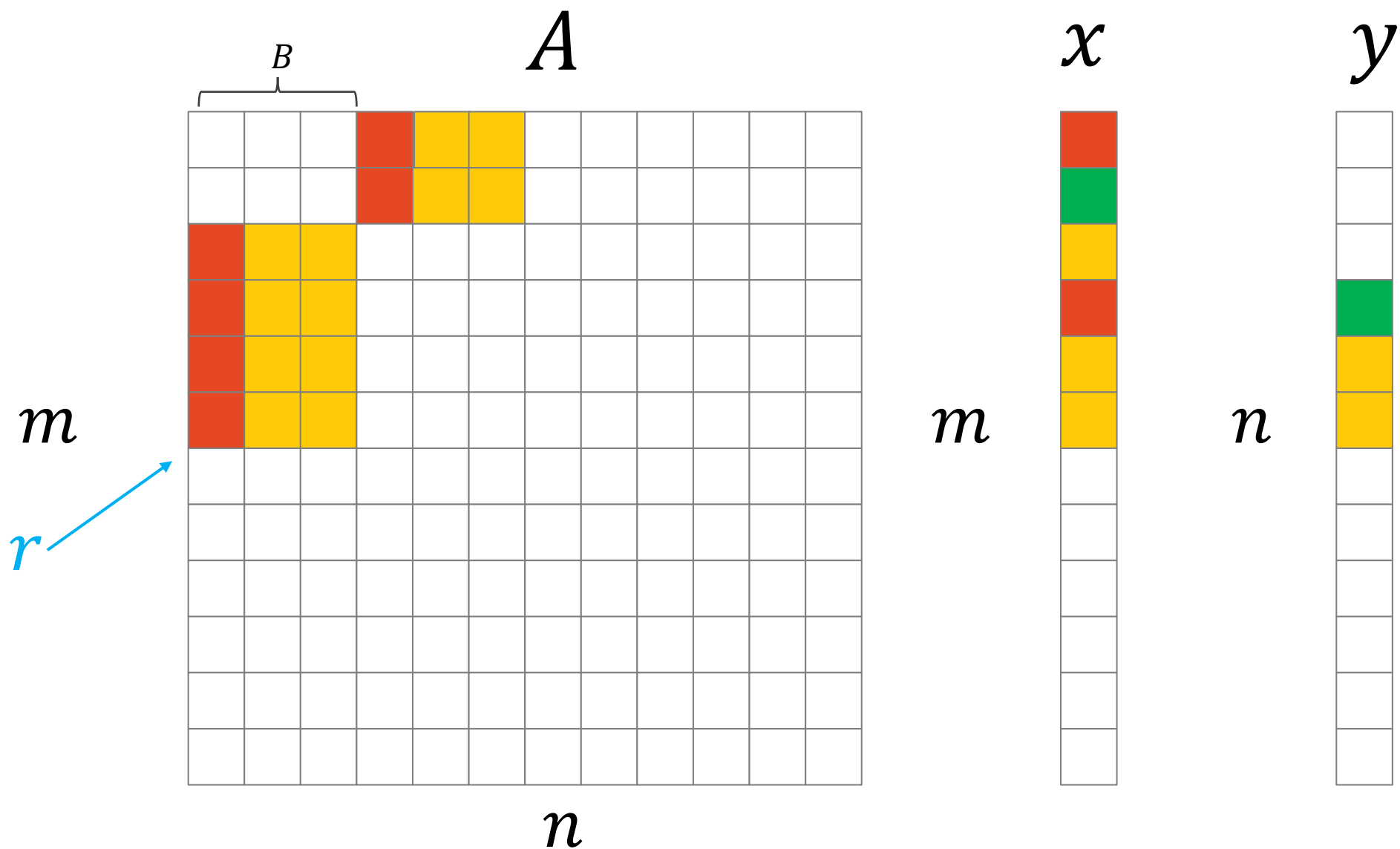
Transpose Matrix Vector Product

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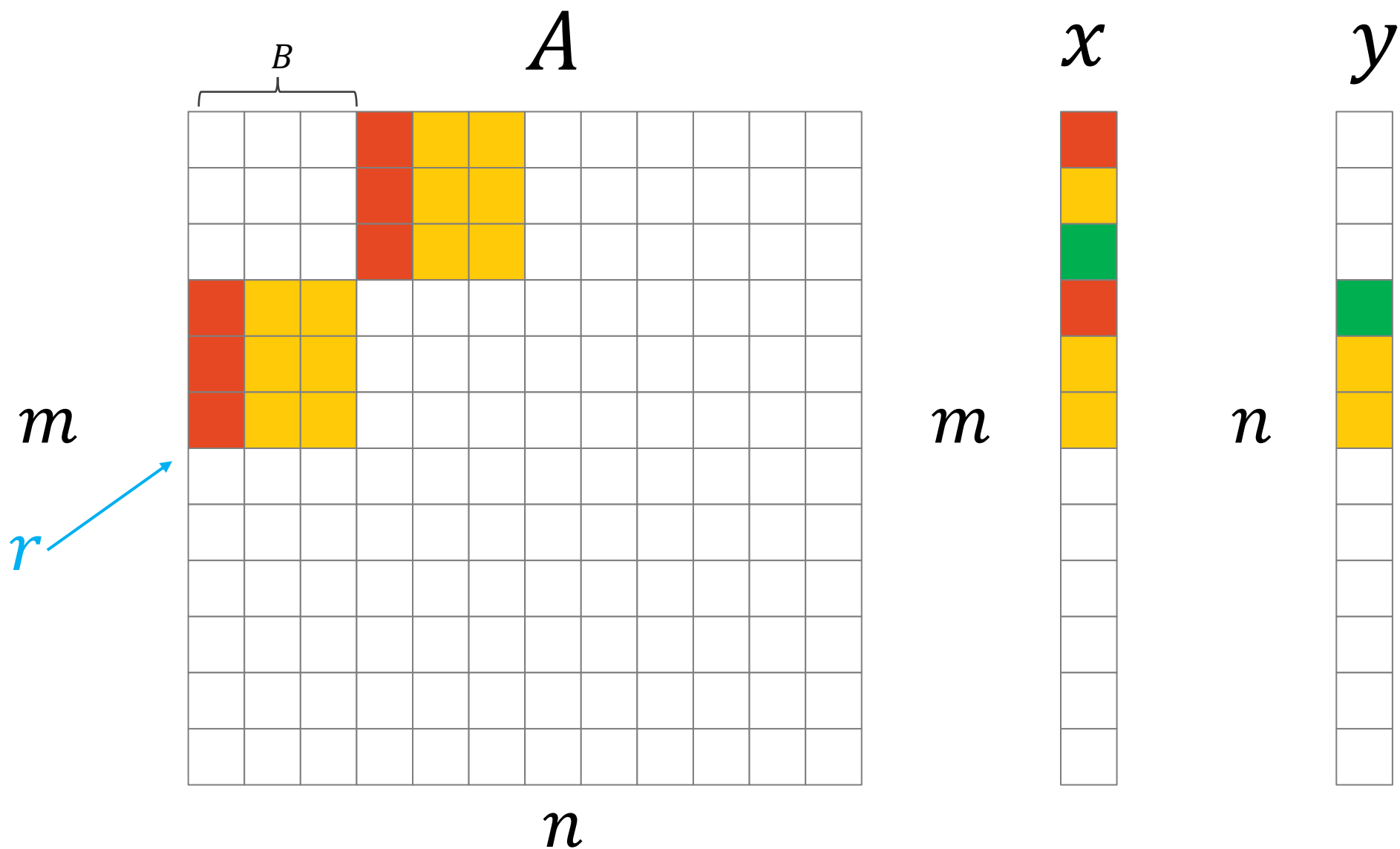
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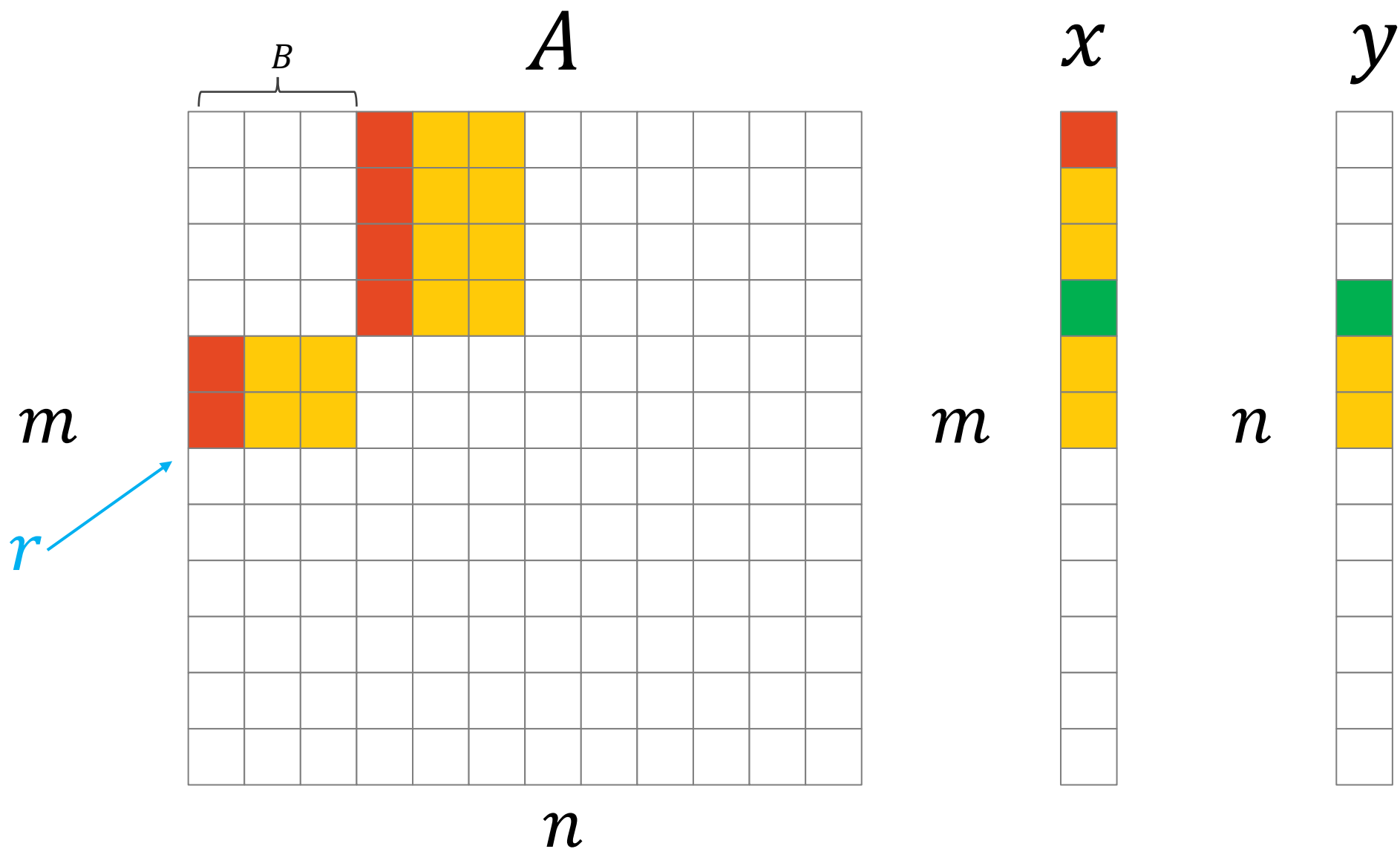
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$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$

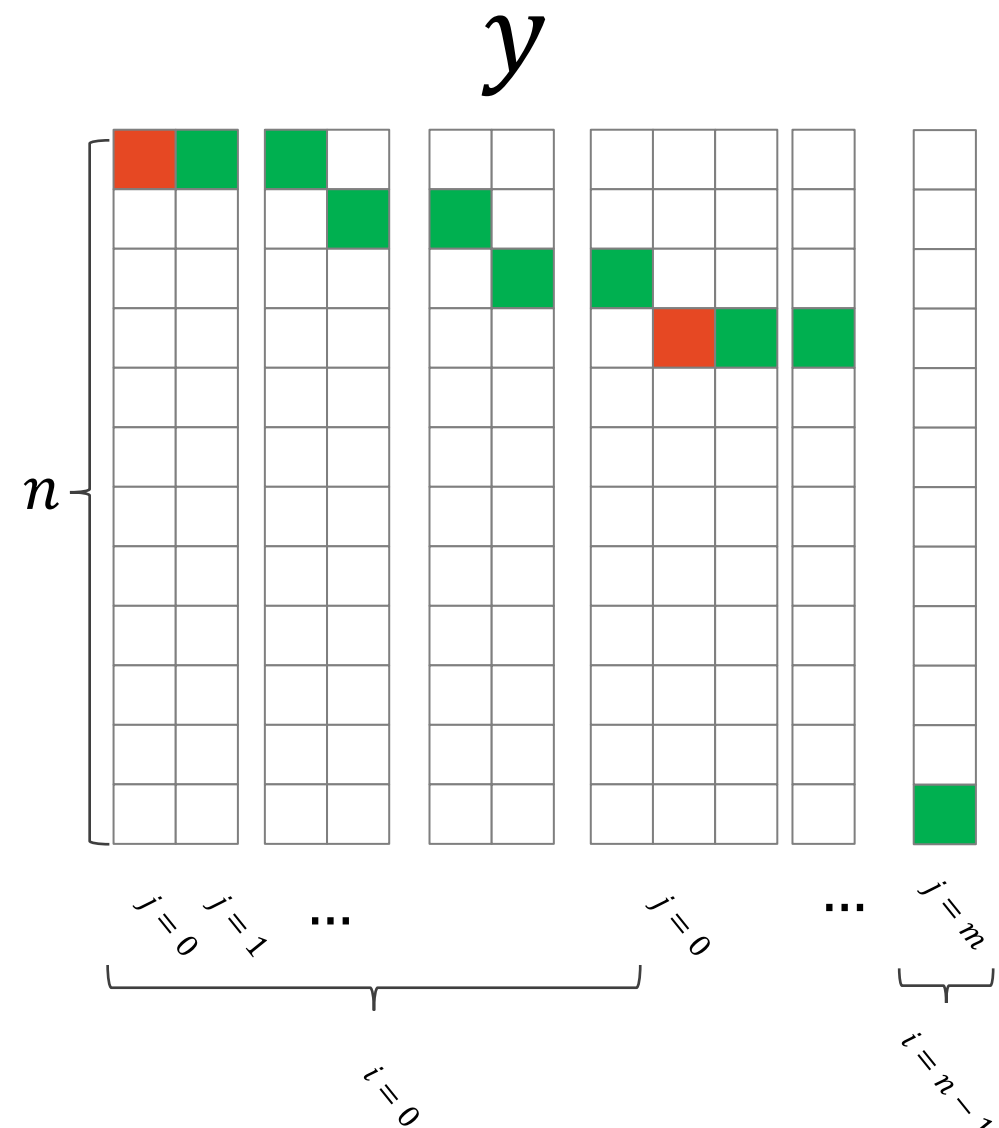
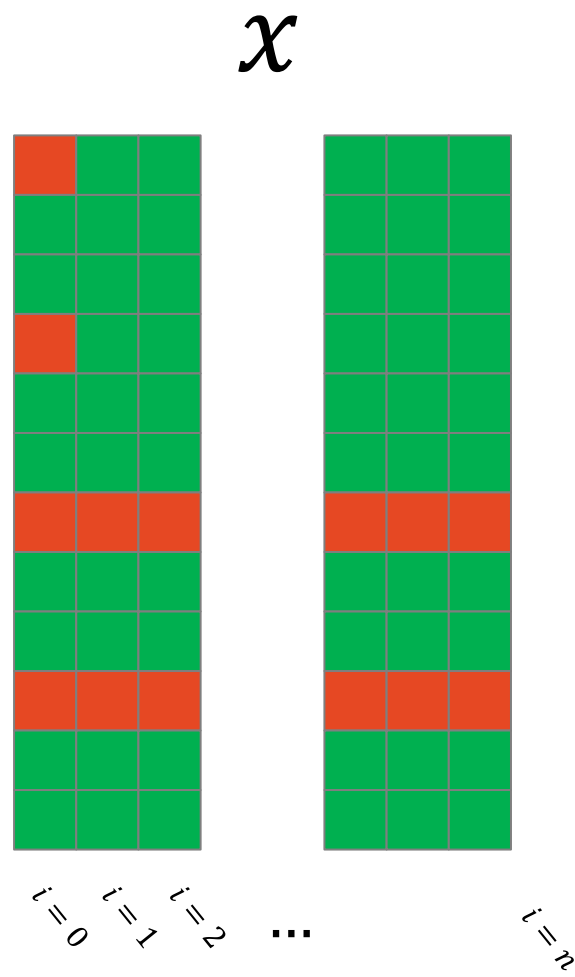
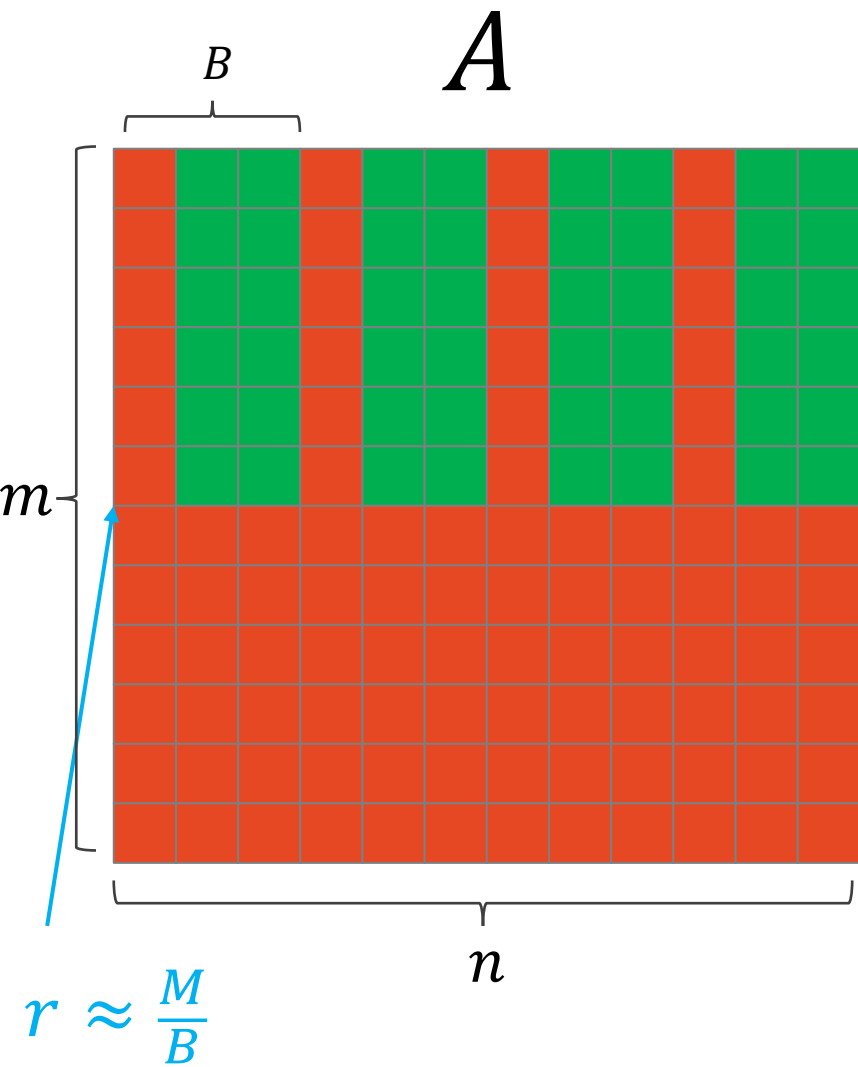


Transpose Matrix Vector Product

$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$



Hits and Misses



Analysis

$$mn + m + n \gg M \text{ and } \min(m, n) \gg M/2$$

```
for(int i=0; i<n; i++){  
    y[i] = 0;  
    for(int j=0; j<m; j++){  
        y[i] = y[i] + A[j][i]*x[j];  
    }  
}
```

$$Q(n, m; B, M) \approx 2mn - r(B-1)\frac{n}{B} - rn - (m-r)(B-1)\frac{n}{B}$$

All misses for A and x

All access for A and x

Hits in A

Hits in upper
part of x

Hits in lower
part of x

$$r \approx \frac{M}{B}$$

$$Q(n, m; B, M) \approx 2mn - \left(\frac{M}{B-1} + m\right)(B-1)\frac{n}{B}$$